

GE Healthcare

LightSpeed 5.X Installation Manual

(Book 1 of 2)



OPERATING DOCUMENTATION



2349182-100
Rev 18

Book 1 of 2: Mechanical Installation

Pages 1 - 154

Effectivity

The information in this manual applies to the following LightSpeed 5.X CT Systems:

- LightSpeed Pro16
- LightSpeed RT
- LightSpeed 16
- LightSpeed Ultra
- LightSpeed Plus

IMPORTANT PRECAUTIONS

LANGUAGE

Предупреждение

(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告

(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

- Tento provozní návod existuje pouze v anglickém jazyce.
- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL

(DA)

- Denne servicemanual findes kun på engelsk.
- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for tekniker, operatøren eller patienten.

WAARSCHUWING

(NL)

- Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING

(EN)

- This Service Manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.

HOIATUS

(ET)

- Käesolev teenindusjuhend on saadaval ainult inglise keeles.
- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS

(FI)

- Tämä huolto-ohje on saatavilla vain englanniksi.
- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION

(FR)

- Ce manuel de service n'est disponible qu'en anglais.
- Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

(DE)

- Diese Serviceanleitung existiert nur in Englischer Sprache.
- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, Mechanische oder Sonstige gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

- Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο.
- Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

(HU)

- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.
- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

(IS)

- Þessi þjónustuhandbók er eingöngu fáanleg á ensku.
- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA

(IT)

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告

(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다 .
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우 , 번역 서비스를 제공하는 것은 고객의 책임입니다 .
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 .
- 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .

BRDINJUMS

(LV)

- Šī apkalpes rokasgrāmata ir pieejama tikai anglu valodā.
- Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis anglu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS

(LT)

- Šis eksploatavimo vadovas yra prieinamas tik anglų kalba.
- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė.
- Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote.
- Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.

ADVARSEL

(NO)

- Denne servicehåndboken finnes bare på engelsk.
- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE

(PL)

- Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.
- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.

ATENÇÃO

(PT)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente consertar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode pôr em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.

ATENȚIE

(RO)

- Acest manual de service este disponibil numai în limba engleză.
- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!

(RU)

- Данное руководство по обслуживанию предлагается только на английском языке.
- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.

UPOZORNENIE

(SK)

- Tento návod na obsluhu je k dispozícii len v angličtine.
- Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.

ATENCION

(ES)

- Este manual de servicio sólo existe en inglés.
- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

- Den här servicehandboken finns bara tillgänglig på engelska.
- Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DIKKAT

(TR)

- Bu servis kılavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "Damage in Shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

Rev. June 13, 2006

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Healthcare personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment, if not properly used, may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, GE Healthcare Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, GE Healthcare Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS



CAUTION
Risk of
Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.



ATTENTION
Danger
d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers: please contact the GE Healthcare Sales or Service representatives.

GE personnel: please use the GE Healthcare iTrak/PQR Process to report all omissions, errors, and defects in this publication.

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Revision History

Revision	Date	Reason for change
0	06/12/03	New product release.
1	07/02/03	General Updates.
2	09/05/03	Integrated H-Power 60 information. General Updates.
3	10/31/03	Updates for LS 5.X RT
4	12/16/03	CQA 10213170: New Gantry Display Service Mounting Location information (Figure A-14, and accompanying text), in Appendix A. CQA 13008123: Updates to Table/Gantry Interference Test Requirements, in Chapter 5. CtCge 85679: Corrected the keyboard and mouse connections in Chapter 2, Figure 2-3.
5	01/29/04	Updated PDU front block diagrams. Chapter 8, Sec. 4.3: Added a step for replacing table caution labels for LS16, Ultra & Plus.
6	4/09/04	Updates for LightSpeed 5.X RT M4.
7	7/30/04	PQR 13012134 Chapter 1, Sec 3.5 Install Alignment Bars: Corrected the Alignment Bars location in the figure. Chapter 2, Sec 8.0 System Ground Connections: Corrected figure 2-31.
8	12/07/04	PQR 13026492 Chapter 1: Updated Floor Anchor information Added Chapter 8 - System-Level Safety Tests.
9	6/1/05	PQR 13038013: In Chapter 1: - Added note for RT base vs. HP base alignment bars. - Revised diagram for HP base alignment bar. - Revised diagram for Table Base Gantry/Table Level Locations A and B in 4.7 - Level the Table. - Added diagram for alternate anchoring holes on gantries in 4.13 - Gantry & Table Redundant Anchor Holes. PQR 13038082: In Chapter 5: deleted procedures for monitor setup for CRT and LCD, referring instead to product documentation. PQR 13038084: In Chapter 5: revised procedures for Autovoice/Intercom Check. PQR 13038088: In Chapter 6: deleted manual scan for 20cm QA Phantom Image Series Image Performance Verification. PQR 13038090: In Chapter 6: deleted "CT number adjustment" from flowchart in Introduction. PQR 13038092: In Chapter 6: deleted manual scan parameters for System Functional Test. YMSct03605: In Chapter 5: added the intercom adjustment method for console intercom pots and for gantry intercom pots in Section 4.2.3 Autovoice Volume. Chapter 5: - Corrected expected results in several instances for RT interference tests. - Grouped all tests for "RT" and all tests for "Pro16, LS16, Ultra & Plus" into product-specific headings. Chapter 6: Revised directions for reference procedures in image quality calibration process.

Revision	Date	Reason for change
10	9/27/05	PQR 13044488: Updated Top Cover info in Appendix A. PSR 13050792: Added Section 4.4 to Chapter 5 Chapter 5: - Added Sections 4.2, 4.5, 4.6, 4.7, 4.8 and 4.9 - Updated Section 4.1 Chapter 9: Updated Section 3.0 (Form 4879)
11	2/24/06	PQRs 13054383, 13064122: Updated Chapter 5, Sect. 4.10.3 - Tests for RT Systems Only. PSR 13059056: Updated front cover dolly instructions for RT, in Appendix A, Section 3.0. Chapter 3: Added Section 4.0 - Axial Head Holder Shim Installation Added Appendix D - Gantry Bearing Gap Inspection Chapter 8 - System-Level Safety Tests: Updated procedure. Chapter 5: Updated screens & instructions in "Reconfig the OC" for 06MW3.X SW including: - Figure 5-12: Preferences Setup Screen - Figure 5-14: Network Settings Screen
12	4/14/06	Chapter 5: Updated Section 4.10.3 - Tests for RT Systems Only
13	10/10/06	Chapter 1: Updated Section 4.2 - Position the Gantry, for Limited Access Sites. Chapter 3: Corrected anchor Torque value. Appendix A: Added Section 9.0 - Gantry Auxiliary (Mini) Dolly Installation w/Wood Block Added Appendix D- Regulatory Clearance Quick Reference Guide Chapter 8: Updated Patient Touch Leakage Test procedure
14	10/18/06	Corrected Errors in Appendix D .
15	4/2/07	PQRs 13084736, 13093391, 13095452: Updated Table 6-5 and section 6.4.3.1 Series Means (AvXo-AvXc) Failure Recovery specifications for LS 16 and Pro16. PSR 13093409: Updated Check/Set Date and Time instructions and moved to section 2.9. Updated instructions and figures in the Reconfig the OC section and moved to section 2.8. Chapter 1: Added new Section 1 and Section 5; performed text updates to Sections 2, 4, and 9; added new images to Sections 4 and 9. Chapter 2: Performed text updates and added images to Section 4. Chapter 3: Performed text updates to Sections 1, 2, and 4. Added images to Section 1. Chapter 4: Added Section 6. Chapter 5: Performed text edits to Section 4. Chapter 8: Renamed Chapter and added Chassis Leakage Test section. Chapter 9: Performed text edits to section 6. Added Chassis Leakage Test Completion Form. Updated the Important Precautions section. Updated front and rear covers.

Revision	Date	Reason for change
16	9/07/07	Chapters 1-6: References to form 4879/Chapter 9 updated/removed. Chapter 1: Updated Seismic mounting information; added Bearing Gap Inspection (PQR 13096524/FCTge30480); Section 4.1.4: added cross reference to Appendix C: Regulatory Information Chapter 2: Section 5.8 added (previously Section 4.7 of Chapter 5. Section 5.9 (new) added. Chapter 4: Section 6, FE Workflow updated and Install Checklist and eForms information added (sec 7 & 8). Chapter 5: Section 4.7 revised. Chapter 7: Replaced Section 11 with disclaimer, per George Farrington and Dave Palermo. Chapter 8: Typos in step 13c of Patient Touch Leakage Test and Table 8.1 corrected. References to Chassis Ground Leakage Test form updated in procedure. Both procedures streamlined and updated per feedback from GF; formatting made more consistent for both. Chapter 9: Removed this chapter ("Forms" chapter). Forms now reside on the Service CD. Removed Appendix D (<i>Bearing Gap Inspection</i>), incorporating the procedure into the manual. Removed Appendix C (<i>Option Cables</i>) and Appendix F (<i>Additional Characterization Procedures</i>). PQR 13111971: Updating Regulatory information (Appendix E, now Appendix C)
17	9/12/07	Chapter 3: - Section 3.3: updated the name and direction number of the UPS Installation manual. - Section 5.0: updated information to reflect that the Mechanical Installation Checklist is now part of GE e4879 and must be accessed online. Chapter 4: Section 2.1.1: Corrected <i>Regulatory Minimum Working Clearance by Major Subsystem</i> tables to match what is in the LS 5.X Pre-Installation Manual. Added tables with information for the UPS and A1 Disconnect.
18	3/28/08	PQR 13145875, 13147124: Chapter 5, Section 4.9, Mechanical Characterization, Section 4.10, Interference Test (RT Gantry): updated information based on changes with software version 07MW11.10. Customer Configuration Preference for Language requirements: Updated Chapter 5, Section 2.8.2, Step 5c Procedure, and Section 4.5 Warning Labels. Also updated GE form e4879 to record information.

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Preface

Publication Conventions

Please become familiar with the conventions used within this publication before proceeding.

Section 1.0

Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles are used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger, and warning identify important safety information. Text (Hazard) styles are applied to the paragraph contents applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information normally includes:

- Type of potential hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided below. They include paragraph prefixes and modified text styles.

 CAUTION Pinch Points Loss of Data Sharp Objects	Caution indicates a hazard exists that can or <u>could cause minor injury</u> to self or others if the instructions are ignored. They include, for example: • Loss of critical patient data • Crush or pinch points • Sharp objects
 DANGER EXCESSIVE VOLTAGE CRUSH POINT	DANGER INDICATES A HAZARD EXISTS THAT <u>WILL CAUSE SEVERE PERSONAL INJURY</u> OR DEATH IF THE INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE: • ELECTROCUTION • CRUSHING • RADIATION
 WARNING ROTATING EQUIPMENT BARE WIRES	WARNING INDICATES A HAZARD EXISTS THAT <u>COULD OR CAN CAUSE SERIOUS PERSONAL INJURY</u> OR DEATH IF THE INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE: • Potential for shock • Exposed wires • Failure to tag and lockout system power, which could allow for un-command motion.



NOTICE
Equipment
Damage
Possible

Notice indicates a hazard is present that can cause property damage, but has absolutely no personal injury risk. They can include:

- Disk drive crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information is always preceded by the exclamation point  contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) are used to make the reader aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make the reader aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0 Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection. It conveys information that should be considered important to the reader.

Example: Makes the reader aware that the paragraph(s) that follow are examples of information stated previously.

Comment: Represents "additional" information that may or may not be relevant.

2.2 Page Layout

Publication Part Number & Revision Number

Publication Title

The current section and its title are always shown in the footer of the left (even) page.

An exclamation point in a triangle is used to indicate important information to the user.

Paragraphs preceded by Alphanumeric (e.g. numbers) characters is information that must be followed in a specific order.

The current chapter and its title are always shown in the footer of the right (odd) page.

Paragraphs preceded by symbols is (e.g. bullets) is information that has no specific order.

Headers and footers in this publication are designed to allow the reader to quickly identify the location within the document. The document's part number and revision number appear in every header on every page. Odd numbered page footers indicate the current chapter, its title, and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Character Styles

Within this publication different character styles indicate computer input and output text. Character (input, output, and variable) styles are applied to the text within a paragraph to indicate directions. Computer screen output and input is also formatted using mono-spaced (fixed width) fonts.

Example: This paragraph denotes computer screen fixed output. It's output is fixed from the sense that it does not vary from application to application. It's the most commonly used style to indicate filenames, paths, and text.

Example: Variable Output	<i>This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput></i>
Example: Fixed Input	This paragraph denotes fixed input. It's typed input that does not vary from application to application. The user is required to supply fixed text as input.
Example: Variable Input	<i>This paragraph denotes computer input that can vary from application to application. The user is required to supply variable text as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.</i>

2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles indicate actions requiring the reader to press either a hard or soft button, switch, or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer-generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: Hard Keys	A power switch <u>ON/OFF</u> or a keyboard key like <u>ENTER</u> is indicated by applying a character style using both over and under-lined bold text. This is a hard key.
Example: Soft Keys	The computer <u>MENU</u> button that requires the reader to click with the mouse or touch with the hand uses over and under-lined regular text. This is a soft key.

Chapter 1

Position Subsystems

Potential for Data Loss and/or Equipment Damage.

To prevent potential data loss and equipment damage, please do the following:

- When instructed, record data collected from the procedures in this chapter on GE Form e4879. For more information about this form, see [Section 8.0](#) of Chapter 4.
- Only use the Installation manual that arrives with your system. Any other revisions of this manual may not exactly match your system.

Section 1.0

Installer/FE Notices

1.1 Shipping, Warehouse, and Transportation Warning

- This gantry is designed to be moved using the shipping dollies and should not be lifted or moved using a lift truck.
- Do not hoist gantry or table using dollies.

1.2 International Shipments

- Dollies must be used to remove the gantry from the shipping skid and to transport the gantry to the customer's site.
- If lifting is required, refer to the Pre-Installation Manual for instructions.

1.3 On Site Warning (Mechanical Installation Team)

This system requires a gantry bearing gap inspection *before* electrical calibration is started. See [Gantry Bearing Gap Inspection, on page 39](#).

Section 2.0

Install Table/Gantry Introduction

This chapter describes how to mount, position, and level the CT Scanner subsystems.

Note: Before you start the installation, make sure the site preparation complies with conditions and instructions found in the pre-installation manual. Failure to comply will result in excessive installation delay and potential increased, unrecoverable installation costs. This product is designed to meet specific mechanical installation standards, which should be reviewed prior to installing this system.

2.1 Skill Set Required



NOTICE

All Mechanical Installers shall meet all minimum requirements.

Only people experienced and trained in the installation process of a GE CT Scanner should perform the work described in this document. A person doing the installation should have:

- Installed a GOC 1 or GOC 2 console.
- Completed three LightSpeed Ultra 4 or 8 or 16 CT installations as a helper, prior to leading an installation. Installation leader must have completed project management, leadership training, and completed all required training prior to becoming a project leader.
- Reviewed the LightSpeed Installation Video, Safety Video, Installation Training CD, and successfully passed all related tests. (Refer to [Table 1-1.](#))

Table 1-1 Installation and Training Media

MEDIA	PART NUMBER
LightSpeed Installation CD	2289471-100
LightSpeed Installation Video	2289468-100
LightSpeed Installation Training CD	2289779-100
GE Safety CD	2290198-100
GE Safety Video	2290199-100

- A working understanding of the mechanical installation process prior to beginning.
- Completed environmental health and safety (EHS) training, understand and apply lockout/tagout processes, completed Hazcom training, completed Eye & Personal Protection Safety, and have taken Blood Borne Pathogens training.
- All mechanical vendor training and records must be verifiable and available to GE Healthcare.
- Mechanical vendors shall be responsible for training all employees involved in the installation of GE Healthcare CT products.
- It is the mechanical vendors responsibility to require and monitor that all of the above safety requirements are maintained and met on installation sites.
- Mechanical vendor installation teams must have basic measurement skills, system layout skills, and the ability to read and understand blue prints. Must have professional communication skills needed to communicate with customer any changes or additional requirements needed to complete the installation project.

2.2 Floor and Room Preparation

2.2.1 Preparation

Consult your local GE Sales and service representative about your specific needs. It is the purchaser's (buyer) responsibility to provide an approved support structure and an approved method of mounting. General Electric is not responsible for any failure of the support structure or method of anchoring.

The LightSpeed 5.X system has a total floor load of approximately 6660 lbs (3021 kg) (**Pro 16**) or 6500 lbs (2950 kg) (**LS 16, Ultra, Plus**). About 5050 lbs (2291 kg) (**Pro 16**) or 5175 lbs (2350 kg) (**LS 16, Ultra, Plus**), including patient (450 lbs (204 kg)), is concentrated in the table-gantry assembly. Refer to the *Pre-installation Manual* for more information.

2.2.2 Flooring

Do not place the scanner on any resilient flooring. Resilient tile or carpeting may slowly yield over a period of time and disturb the alignment of the table to the gantry. Refer to the floor template to determine locations where resilient flooring material should be removed.

Limitations include:

- No part of the floor surface within the table, gantry, or the two interface areas between the table and gantry, should be higher than the support areas for the table and gantry.
- The floor structure must withstand the occupied weight of table and gantry, as well as the individual contact area loading of these components.
- The method and placement of anchors, or through bolts, must not reduce the structural strength of the floor.

If you have to remove the gantry covers in order to move the gantry into the room, refer to [Appendix A](#) for the cover removal procedure. Please read the caution statement on [page 117](#) before removing the gantry covers.

2.3 Overview

Procedures in this chapter provide detailed instructions to position, level, and anchor the gantry and table securely for operation. The LightSpeed 5.X system uses adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads located on the gantry base. The table has five (5) pads used for leveling it.

The process you will be following is:

- 1.) Use the room-layout template to determine the general position of the gantry and table.
- 2.) Move the gantry into position.
- 3.) Level the gantry.
- 4.) Use the three alignment bars to position the table relative to the gantry.
- 5.) Level the table to the gantry, and anchor the system.

Use the template to position the system; however, use the gantry and table to locate and drill the anchor holes. Drill the anchor holes with the system in place. Refer to [Section 4.0, on page 33](#) and [Figure 1-20, on page 45](#), for an example of this procedure. This CT system installation procedure requires the items listed in [Section 2.5, on page 28](#) and [Section 2.6, on page 28](#).

2.4 Pre-Installation Template

Always use the room-layout template (in two pieces), during installation (see [Table 1-2](#), for part number). The gantry and table will not be properly aligned if existing holes are used. The template shows the location of the gantry and table anchor holes.

This template is shipped with the system. It may also be ordered via the Web, from Coakley-Tech.

Table 1-2 LightSpeed 5.X Room Layout Templates

System	Part Number
Pro16	2363851
LS16, Ultra & Plus	2363851
RT	2389013

2.5 Installation Alignment Tool Kit

The installation personnel provide the installation tool kit, P/N 2110953. The tool kit contains the following special tools (See [Figure 1-1](#)). **This installation alignment toolkit requires yearly calibration and must have a current calibration sticker.**

- **Guide Pin (Bolt) Tool** included in the tool kit (not needed).
- **Long Alignment Bar** (in two pieces), P/N 2106369: use to position the table, relative to gantry
- **Two Short Alignment Bars**, P/N 2106368: use to position the table, relative to the gantry
- **Adjuster Tool**, P/N 2107863: use to turn the adjusters, which in turn level the system.
- **½ in. Drill Bushing**, P/N 2106205: use to center and guide the bit when drilling the anchor holes
- **Vacuum Attachment (funnel)**, P/N 2110944: use to vacuum debris out of the anchor holes
- **Spanner Wrench**, P/N 2110003: if necessary, use to turn the adjusters after anchoring system
- **Anchor Seating Tool**, P/N 2247965: used to properly seat anchors

Figure 1-1 Installation Alignment Toolkit Accessories

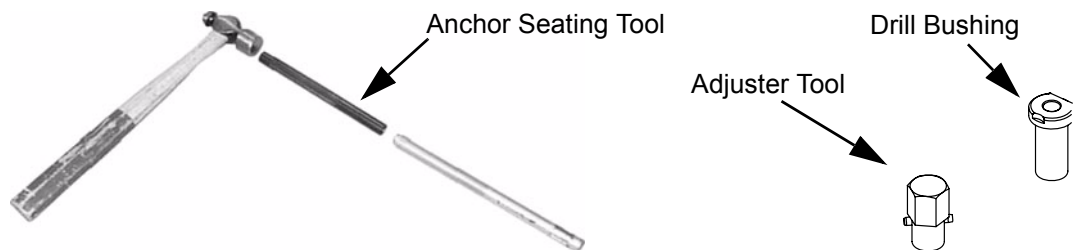


Figure 1-2 Gantry/Table Alignment Toolkit



2.6 Required Common Tools and Supplies

The following tools and supplies are required for installation of the scanner. Please refer to [Appendix B, on page 137](#), for pictorial descriptions of the tools and supplies.

WRENCHES

- Standard and Metric combination wrench sets
- Standard and Metric hex key (Allen wrench) sets
- ½ in. and 3/8 in. Drive torque wrench: 0-100 N-m (0-100 ft.-lb.) Must be calibrated yearly.

SOCKETS AND EXTENSIONS

- 3/8" and ½" Drive ratchet wrenches
- 1" Drive 3" and two 6" long extensions
- 3/8" Drive 12" long extension
- Standard & Metric 3/8" drive socket sets
- ¾" Deep well socket 3/8"
- 1", 1-1/8", 1-¼" and 1-½" Sockets for ½" drive
- 3/8" Drive universal joint
- Metric hex bit set ¼" or 3/8" drive, including:
 - 14 mm Hex bit 3/8" or ½" drive (14 mm ball hex helpful)
 - 10 mm Hex bit 3/8" drive

SCREW DRIVERS

- Phillips screwdriver set (small, medium, and large)
- Straight blade screwdriver set (small, medium, and large)

DRILL BITS

- Complete set of standard (U.S.) drill bits
- Metric tap set
- ½ in. Masonry bit, min. 8 in. long USA – 18 in. optional (for rear table hole)
- 3 in. (76 mm) Hole saw with 1/4 in. (6 mm) masonry bit (to remove flooring)

POWER TOOLS

- 3/8 in. or ½ in. Drill, cordless or electric
- Reciprocating saw (Sawzall or equivalent) and assorted blades.
- Hammer drill & bit (8 in. min, 12 in. max)
- Sears 17740 Shop Vacuum or equivalent, with "HEPA" or dry wall dust filter (Sears part number 17918) or equivalent
- 25 ft. Extension power cords

HAND TOOLS

- Ball-peen hammer (1lb or 2lb)
- Tongue and groove pliers (large)
- Diagonal cutting pliers, large (to cut 1/0 ground)
- Framing square (e.g., Empire 16 in. x 24 in. aluminum square)
- Diagonal cutting pliers, small
- Large pry bar
- 4 ft., 2 ft. and 9 in. Torpedo levels (see [Table 1-3: Recommended Levels](#))
- Laser level (see [Table 1-3: Recommended Levels](#))

Table 1-3 Recommended Levels

9 in.	Johnson Magnetic Level, model 7500M* Johnson Magnetic Level, model 4500 Stanley Magnetic Level
2 ft.	Johnson Professional Box Beam Level, model 9624* Empire Titan Professional Box Beam Level, model 900 series
4 ft. (nominal)	48 in. Johnson Professional Box Beam Level* 42 in. Stanley Contractor Grade Level 48 in. Empire Titan Professional Box Beam Level, model 900 series 48 in. Stabila Aluminum Box Beam Level, Kit 24816
Laser**	Sears Laser Level Tool (\$40)

* Preferred levels

** Laser level is required for table cradle alignment. Using this tool reduces the alignment procedure by 1 man-hour. New tables have a cradle center line to be used with the laser level.

ELECTRICAL TOOLS

- DVM
- Continuity tester
- Dale 600 Leakage Meter
- Temperature/humidity tool: Oregon Scientific Wireless Weather Station Model BAR608HGA

PERSONAL SAFETY EQUIPMENT

- Safety shoes*
- Safety glasses*
- Gloves
- LOTO kit (supplied)*
- Hearing protection*
- 6 ft. and 8 ft. Step ladder

* These PPE items are absolutely required for every installation job, with NO exceptions.

Note: A box labeled Installation Support Kit is shipped with each system. It contains paint, masking tape, cleaners, towels, and other materials needed to install this CT scanner.

Section 3.0

Delivery and Inventory Procedure

3.1 Delivery Procedure

3.1.1 System Transportation - Temperature Extremes

When transporting the CT system, ensure that the system is not exposed to temperatures or humidity outside the following specifications.

Temperature: -18° to +49° C (0° to +120° F)

Humidity: 20% to 80%

NOTICE Component freezing occurs if CT system is exposed to temperatures below -18° C (0° F) for a period longer than two days.

Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature prior to installation.

3.1.2 Working with the Mover

Follow the instructions provided by your installation specialist regarding working with equipment movers. Help direct movers on where to place equipment, and which items you need first.

Generally, movers should move all equipment into the customer room. Door removal and other site changes to move equipment should be done only as directed by the install specialist.

For component sizes and weights, refer to the *LightSpeed 5.X Pre-Installation* manual.

3.1.3 Floor Protection

It is suggested that the movers use floor protection. Most equipment movers can provide floor protection during the equipment delivery. Installers should provide floor protection for the room.

3.1.4 Equipment Delivery Route

Prior to equipment delivery, review the delivery route with the movers. Refer to the installation specialist for any additional delivery instructions.

3.1.5 Removing Gantry Dollies and Covers

Refer to [Appendix A](#) for the dolly and cover removal procedures. Please read the caution statement on [page 117](#) before removing the gantry covers.

3.1.6 Check for Shipping Damage

Check for the following damage, which may have occurred during shipping:

- Equipment damage: paint scrapes and cover damage.
- Damage to hospital property: floors, door frames and walls.

If damage is found—or items are missing in shipment—notify the appropriate service personnel:

- For item(s) missing in shipment or short shipped, contact the install specialist to have the item(s) shipped.
- Report damaged items to:

- (800) 548-3366, Option 8

For further details, see [“DAMAGE IN TRANSPORTATION,” on page 8.](#)

3.1.7 A1 Breaker

NOTICE The site **MUST** have a main disconnect with Lockout/Tagout capability.
Lock-out and tag-out the A1 breaker now.



Figure 1-3 Sample A1 Breaker



3.1.8 Installation Support Kits

An Installation Support Kit is shipped with every system. Locate this box now and open it. Use all included materials during the installation process. Leave these items ON SITE for future service needs.

Purchase cleaning supplies locally.

Section 4.0

Install and Level Table/Gantry

4.1 Lay the Floor Template

4.1.1 Tools Required

- Standard Install Tool Kit
- Install Support Kit
- GE Site Print
- Installation Manual
- Floor Template for your system
- 4 ft. Level
- Chalk line
- China marker

4.1.2 Time & Personnel

- 1.0 Hour labor on site
- 2 Engineers

4.1.3 Safety

CAUTION Potential for Injury

The LightSpeed gantry presents a variety of mechanical and electrical hazards. Use appropriate safety procedures when working on the CT system.

4.1.4 Establish the Room Layout

BEFORE YOU BEGIN

Place the main unit of the temperature and humidity tool in the control room and the remote unit in the scan room. Monitor environmental conditions at the installation site by recording the temperature and humidity of each room at the end of each day.

Use the GE site print developed for your site to establish the room layout, observing all operating and service clearances shown on the print. Likewise, before continuing, confirm that the site meets the following requirements:

- The site print is on-site and matches the actual dimensions of the room.
- The placement of equipment on-site matches the placement of equipment on the print.
- The site meets all regulatory requirements. Installation cannot proceed if the site does not comply with all regulatory requirements. (Refer to [Appendix C: Regulatory Clearance Quick Reference Guide](#).)

Note: If the site does not meet ALL of above requirements, contact your management and your Project Manager of Installation.

Use the supplied template to locate the anchor holes. Make sure they clear structural interferences in the floor.

Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

PROCEDURE

NOTICE



The floor must meet the minimum floor specification in order to begin this procedure. Refer to the *Pre-Installation Manual* for details.

- 1.) Lay out the two (2) pieces of the floor template (**Pro16, LS16, Ultra and Plus:** P/N 2363851; **RT:** P/N 2389013).

- 2.) Start with the gantry template—align per the GE print.

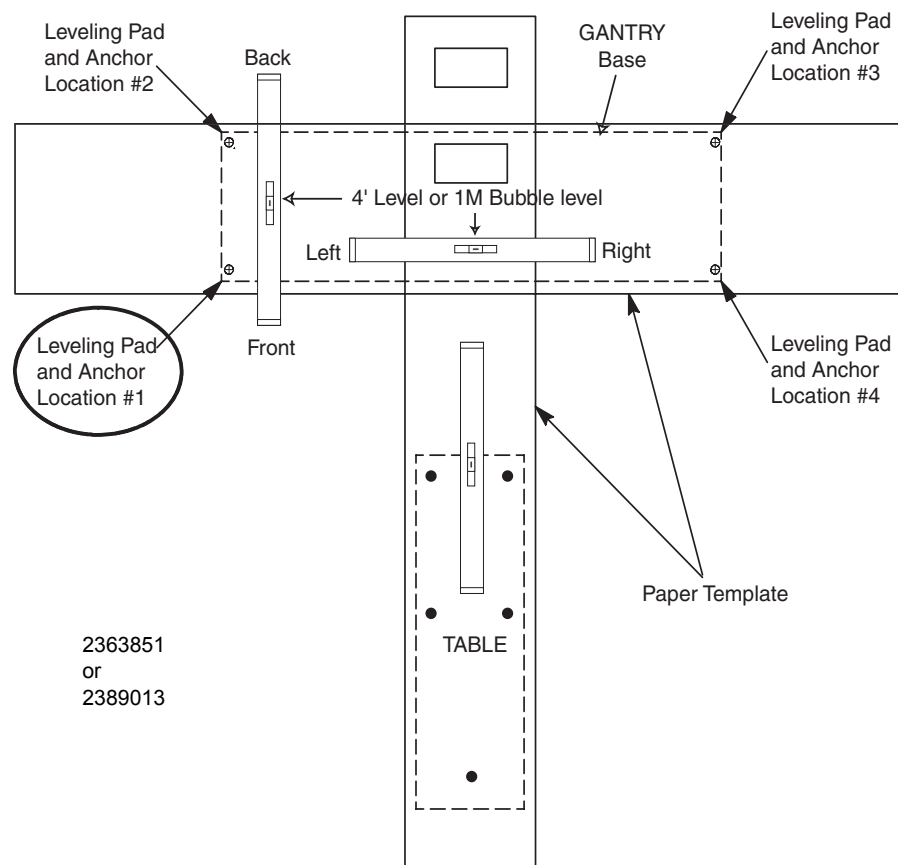
Note: If the floor has existing anchor penetration, the new anchors must be set at least 102 mm (4 in.) from the nearest penetration.

- 3.) Place the table template over the top of the gantry template. Align the scan and table center-lines and secure the templates to the floor. Make sure there are no potential clearance issues.

Note: LightSpeed 5.X service clearance areas are different than other LightSpeed systems.

- 4.) Now, check the level of the floor (See [Figure 1-4](#)) across the templates.

Figure 1-4 Hole Locations



- 5.) Make a mark (e.g., use a center punch) at each of the gantry's mounting hole locations (there are four (4) of these).
- 6.) Using a center punch, mark the three (3) table mounting hole and two (2) leveler locations.

NOTICE



Positioning requires cutting 7 holes in the floor.

Before you drill or cut any flooring, make sure the appropriate hospital personnel have approved the location of the table/gantry.

- 7.) Cut tiles (or other resilient flooring) around all holes punched in the template for the gantry and table. Use a utility knife with a heat gun, a 3 in. hole saw with a 1/4 in. masonry bit or other adequate tools to cut the flooring.
- 8.) Some sites may require sealing the floor penetrations after the flooring is removed.

- Note:
- Use RTV or other sealant to seal the floor covering, as necessary.
 - All documentation in this manual is based on mounting the table/gantry on a concrete floor only.

4.2 Position the Gantry

4.2.1 Gantry Prep - For Access Greater Than 711.2 mm (28 in.)

Remove all the transportation packaging, except for dollies, from the gantry.

Note: Some sites require floor protection. Locate and install any required floor protection now.

4.2.2 Gantry Prep - For Access Less Than 711.2 mm (28 in.)

Note: Perform the following procedure ONLY if your system is limited-access compliant. Do not install if your system was built before January 1, 2007.

- 1.) Remove all the transportation packaging, except for dollies, from the gantry.

Note: Some sites require floor protection. Locate and install any required floor protection now.

- 2.) Remove the gantry dolly from the left side of the gantry so that the gantry can be positioned closer to the left side wall:

Note: Use Floor Protection for this process.

- a.) Lower the gantry to the floor so that the gantry is resting on the floor.
- b.) Remove the three (3) M14 hex bolts that secure the gantry to the dolly.
- c.) Replace the removed dolly with the shipped black gantry-positioning dolly, and reinstall the three (3) M14 hex bolts.
- d.) Raise the gantry so that it is once again off of the floor.

4.2.3 Gantry Positioning - All Sites

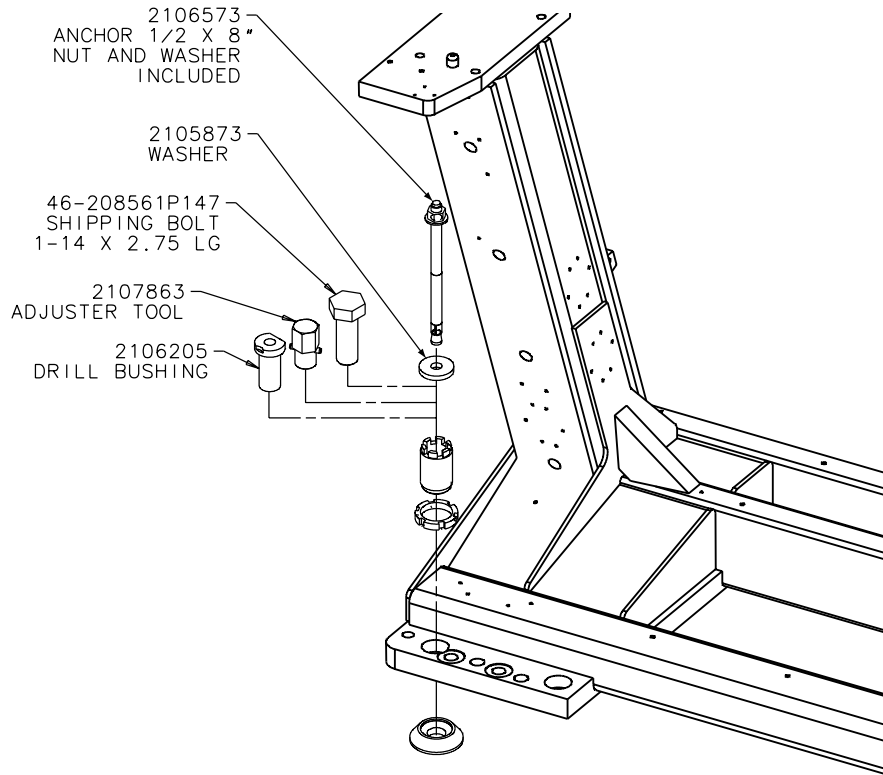
- 1.) Position the gantry over the template appropriately.

Figure 1-5 Gantry and Table Paper Templates



- a.) Locate the four (4) leveling pads, and position each of them beneath its associated adjuster.
- b.) Use the dollies to evenly lower the gantry, until it is just off of the floor (approximately 12.7 mm or ½ in.). Use a ½ in. ratchet to raise and lower the dollies.
- c.) Carefully rotate the gantry into the correct position over the template.

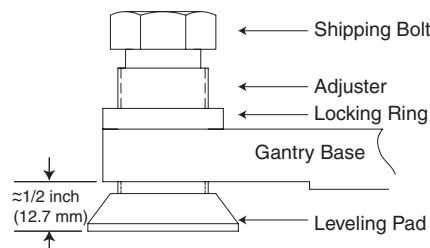
Figure 1-6 Gantry Base Installation Hardware (RT varies)



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm). Void between adjuster and anchor must be filled according to local building codes for seismic application.

- 2.) Remove the paper templates from the floor and discard properly.
- 3.) Loosen the locking rings and shipping bolts so you can fine-tune the leveling pads to compensate for slight variations in the floor surface.

Figure 1-7 Gantry Base Leveling Pads (Starting Positions)



- 4.) Position the gantry so that the adjusters are centered over their respective holes scribed earlier into the floor.
- 5.) Using a 1/2 in. ratchet, gently lower the gantry until it rests on the floor, over the marked areas.
- 6.) Using a 14 mm hex socket, remove the dollies from the gantry by removing the three dolly bolts found at both ends of the gantry ([Figure 1-8](#)).

IMPORTANT:

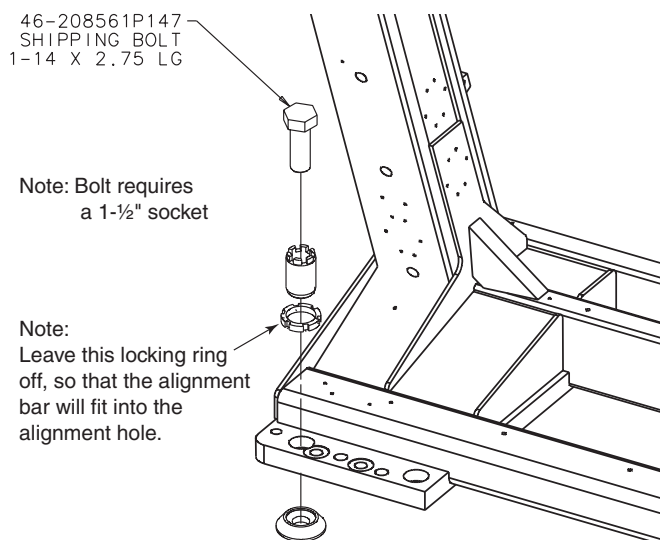
Make certain to route the gantry power cord under the two rear gantry rails, before removing the gantry shipping dollies.

Figure 1-8 Gantry Dolly Bolts



7.) Remove the (4) gantry shipping bolts, using a 1½ in. socket.

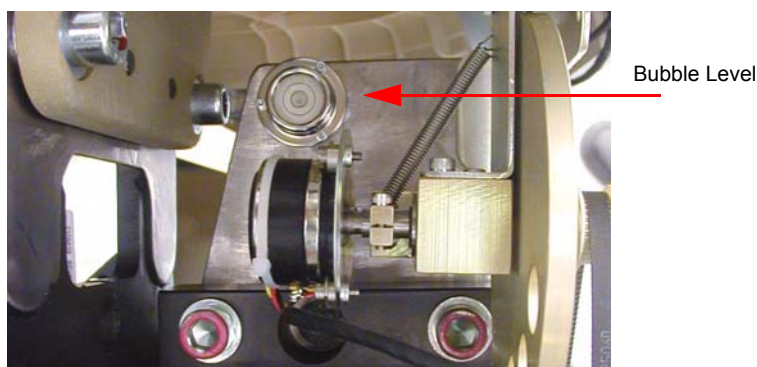
Figure 1-9 Gantry Shipping Bolts



4.3 Level the Gantry

The gantry uses two (2) bubble levels that are permanently mounted to machined surfaces on the stationary base to tell when it is level.

Figure 1-10 Gantry Bubble Level



The bubble levels are located on both ends of the gantry stationary base, near a point where the rotating structure pivots mount to the base structure. (See [Figure 1-10](#).) The gantry is properly leveled when the bubble is centered. (See [Figure 1-14](#), on page 41.)

- 1.) **Remove adjuster #1 lock ring** and only loosen all the other (#2, #3 and #4) adjuster lock rings. Do this using spanner wrench 2110003 (the spanner wrench can be found in the alignment toolkit). Check to make sure auxiliary levelers do not cause interference.
- 2.) Systematically turn each of the gantry's adjusters (locations 1, 2, 3, and 4, in [Figure 1-11](#) or [Figure 1-12](#)) until both bubble levels are centered left to right, and front to back.
 - Begin by turning each adjuster no more than 1 turn at a time.
 - Use the adjuster tool, 1-1/8 in. socket, and the ½ in. drive ratchet to turn each adjuster. (Refer to [Figure 1-8](#), on page 38.)

SYSTEMATIC PROCEDURE FOR LEVELING GANTRY FOLLOWS:

- 1.) Level the left side from front to back by turning adjusters #1 and #2.
- 2.) Level the right side from front to back by turning adjusters #3 and #4.
- 3.) Level the side (right or left) that is higher, with respect to the other side. Turn both adjusters on a side equally until that side is level. The side should now also be level.

Figure 1-11 Gantry Base “Adjuster” Locations (except RT) — Top View

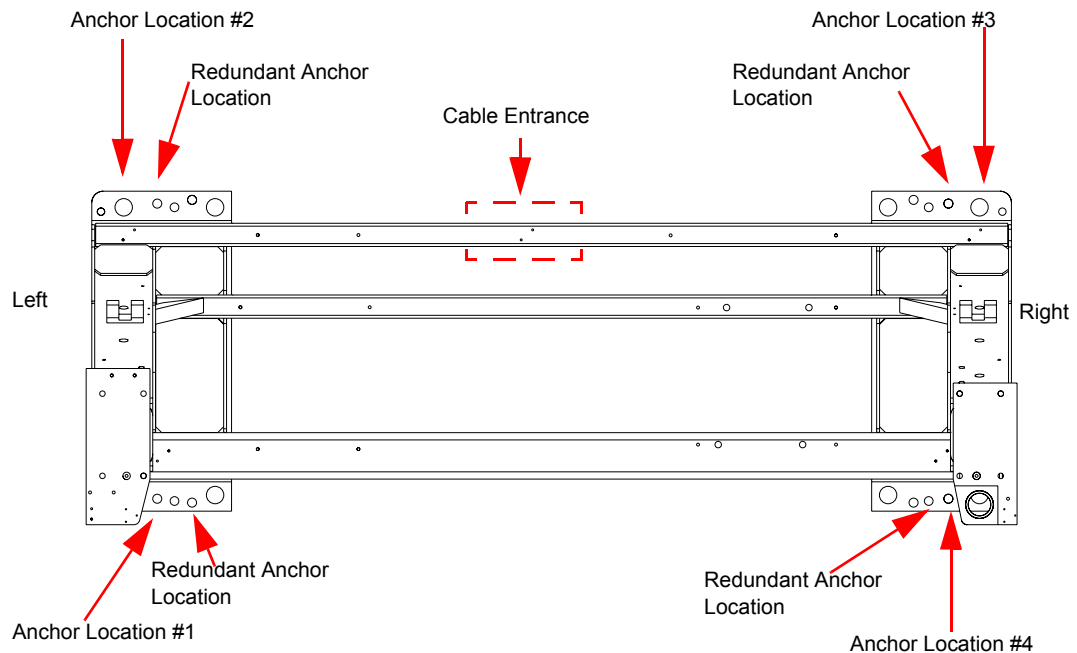
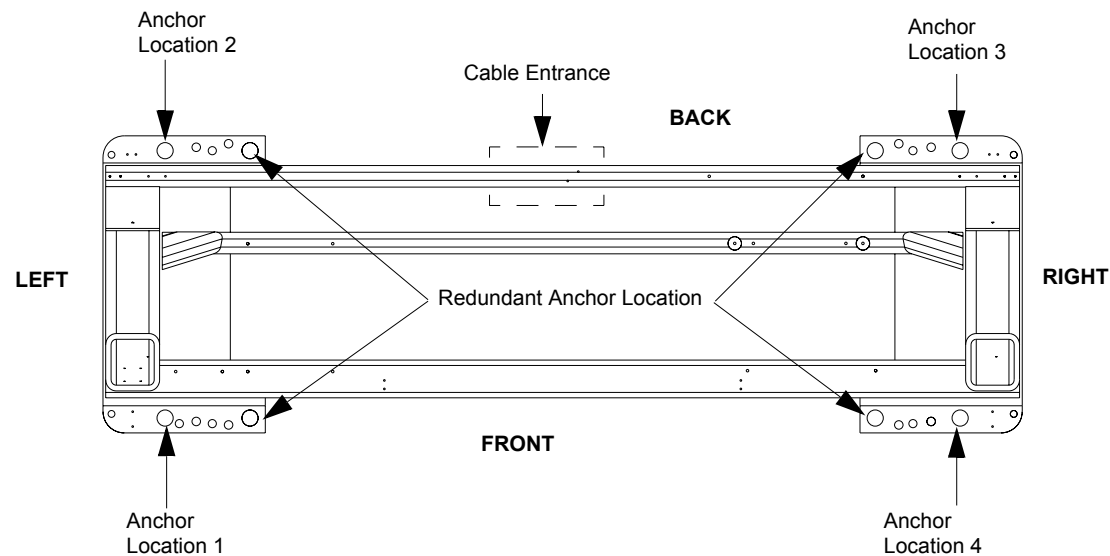
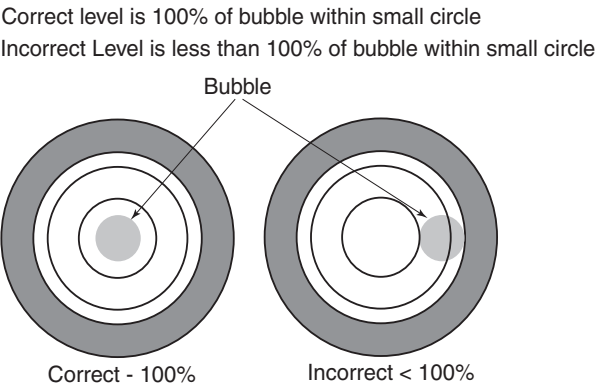


Figure 1-12 RT Gantry Base “Adjuster” Locations — Top View



- 3.) When the bubble levels are centered (Figure 1-13), each of the four (4) leveling pads should be carrying a portion of the gantry weight. Distribution of the gantry weight prevents the base frame from rocking during normal operation. **DO NOT leave any adjuster un-loaded or floating.**

Figure 1-13 Bubble Level Centering



4.4 Gantry Bearing Gap Inspection

All CT systems require a Gantry Bearing Gap inspection before starting electrical calibration.

All international gantries are shipped in a wooden shipping crate that should not be removed until it arrives at the installation site. This shipping container is designed to reduce the risk of shipping damage.

4.4.1 Personnel Requirements

REQUIRED PERSONS	PRELIMINARY REQS	PROCEDURE	FINALIZATION
2 (mechanical suppliers or installation team)	15 min	15 min	5 min

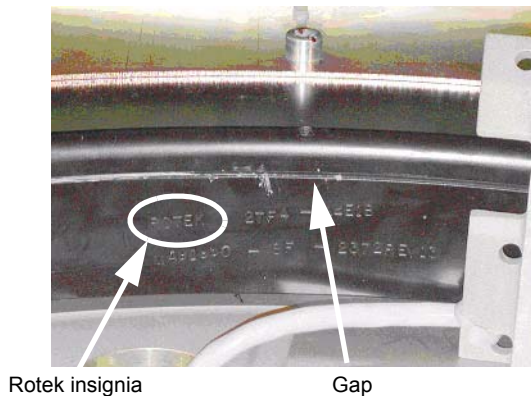
4.4.2 Tools and Test Equipment

- Standard tool kit
- Inspection document
- 2.5 mm Allen wrench
- Rear cover dollies (2)
- Flashlight

4.4.3 Damage Indicators

On the inside edge of the black-colored bearing assembly, a mark similar to that shown in [Figure 1-14](#) will be seen, if this is a Rotek bearing.

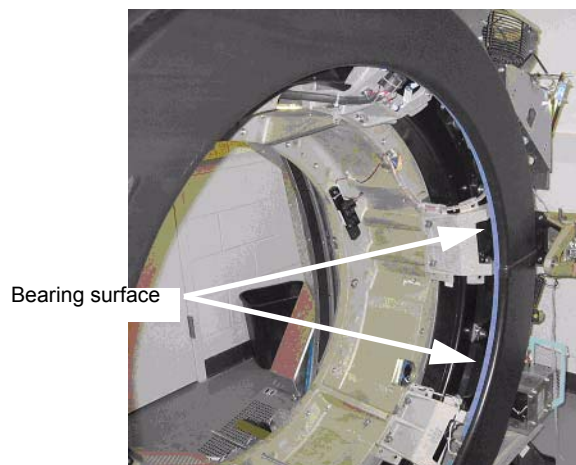
Figure 1-14 Gantry Bearing - Rotek Label



The mark has a serial number in the same format as:
ROTEK 2TF4-44E1B-MA91960-8F-2372-REV13.

The gap to inspect is shown in [Figure 1-15](#) next to the serial number.

Figure 1-15 Gantry Bearing



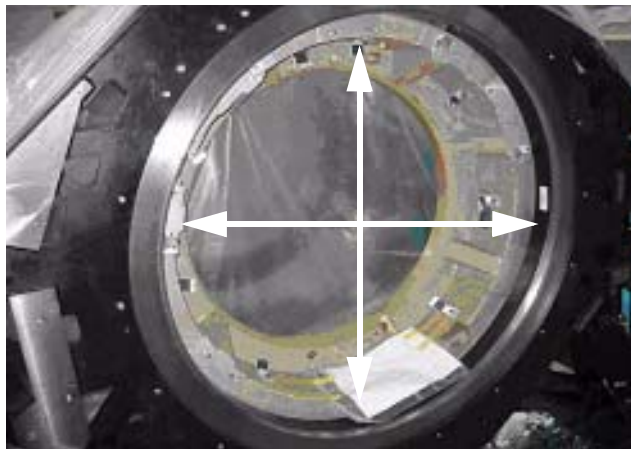
On most systems, a change in the bearing gap does not cause the gantry to make unusual sounds, unless the gap is severe. If the gantry is badly damaged and the gap is severe, it can cause operation issues. Some systems are shipped with shock indicators that must be returned to Milwaukee.

A severe failure may be seen during installation as a problem rotating the gantry.

4.4.4 Procedure

- 1.) Remove the scan window following the procedure in Appendix A, [6.1, on page 133](#).
- 2.) Remove the top and rear gantry covers, following the procedures in Appendix A, [Section 2.0, on page 119](#) and [Section 4.0, on page 129](#)
- 3.) Use a 2.5 mm hex wrench as a tool to measure the gap at the positions shown in [Figure 0-1](#). The location of the gantry components does not matter. Measure four (4) locations 90 degrees apart from each other.

Figure 0-1 Inspection Locations



- 4.) If the 2.5 mm hex wrench easily fits without effort in the gap, the gap is out of spec. [Figure 1-16](#) shows a gap that is too large. [Figure 1-17](#) shows a gap that is good. Notice that the hex wrench does not fit in the gap in [Figure 1-17](#), but does in [Figure 1-16](#)

Note: Do not use force when putting the wrench in the gap. Either it slips in or it doesn't.

Figure 1-16 Gap Too Large



Figure 1-17 Gap Is Good



- 5.) Replace the top and rear gantry covers, following the procedures in Appendix A, [Section 2.0, on page 119](#) and [Section 4.0, on page 129](#).
- 6.) Replace the scan window, following the procedures in Appendix A, [Section 6.0, on page 133](#).

4.4.5 Finalization - Mechanical Installers

If the Bearing Gap Inspection passes, complete the signoff on the GE Form e4879, Installation Data verification form, that this inspection was completed.

If the Bearing Gap Inspection fails, contact your site FE.

4.4.6 FE Service Action Required

If the Bearing Gap Inspection fails, the mechanical installer notifies the site FE that the inspection failed.

The site FE should:

- 1.) Open a bearing inspection dispatch.
- 2.) Follow the inspection procedure described in this section.
- 3.) Record the bearing inspection results.

If no damage is found, close this dispatch and continue with the electrical calibration procedures.

If the system is damaged, go to the Equipment Delivery Quality web site and follow their instructions.

To enter a damaged in shipping claim, go to this web site:

<http://egems.med.ge.com/edq/home.jsp>

4.4.7 FE Inspection Completion

- 1.) After the Gantry Bearing Inspection passes, complete the opened service dispatch with the following information:
 - Gantry Serial Number
 - Gantry Type
 - System ID
 - Site Name
 - Installation date
 - Was the Gantry transported to the site in the shipping crate? (Yes/No)
 - Was the Gantry lifted or hoisted, were riggers used, or was the Gantry delivered via flatbed wrecker? (Yes/No)
 - Number of locations that fail the gap inspection if any: _____
- 2.) Close the service dispatch.

Should any follow-up be required after this inspection, the site engineer will be contacted directly by CT Engineering.

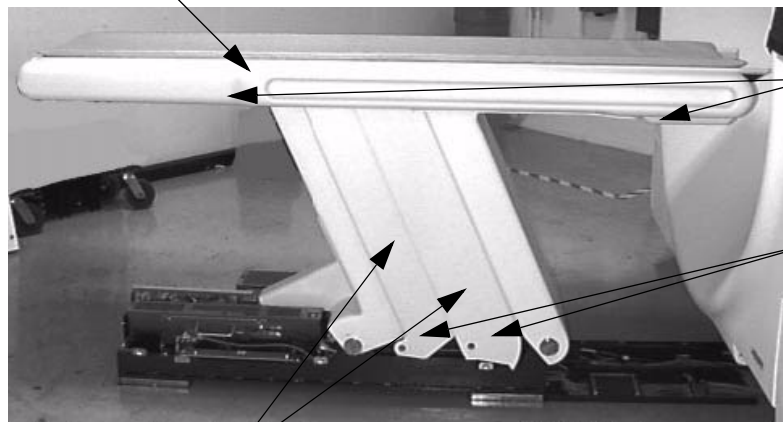
4.5 Position the Table

- 1.) Remove all the transportation packaging and boxes, except dollies, from the table.
- 2.) Wheel the table to its approximate position relative to the gantry, using the marks made earlier.
 - a.) Locate the table leveling pads and position them against the base of the table, using the adjusters with a 1½ in. socket and ½ in. ratchet.
 - b.) Use the dollies to evenly lower the table until it rests on the leveling pads using an ½ in. ratchet.
- 3.) **Loosen adjuster shipping bolts and locking rings “one turn” before you lower the table to the floor.** Loosen the locking rings and shipping bolts so you can **fine tune** the leveling pads to compensate for slight variations in the floor surface. Use a 1½ in. socket with 6 in. extension to loosen the shipping bolts.
- 4.) Remove the table left and right upper side covers (refer to [Figure 1-18](#)).

Note: The captive screws have a tendency to fall out occasionally.

Figure 1-18 Table Cover Removal

Remove left and right side covers.



Loosen captive “slotted” screws.

Remove 3mm hex screws.

Remove the two panels, on both sides.

- 5.) Remove the four side panels using a 5 mm hex key. Without disconnecting the ground straps at the z-channel, carefully lay the side panels on top of the padded cradle.

NOTICE

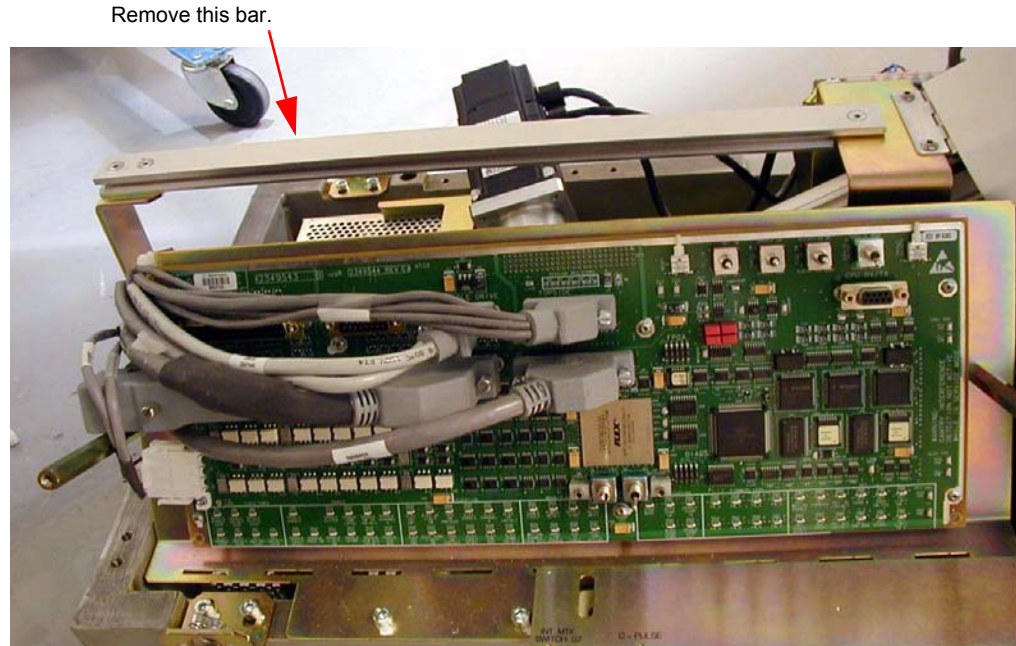


Potential for Equipment Damage.

Take extreme care to prevent damaging the MTCB printed wire assembly. Use some packing material to protect these components.

- 6.) Remove the table base cover center support bar. (See [Figure 1-19](#).)

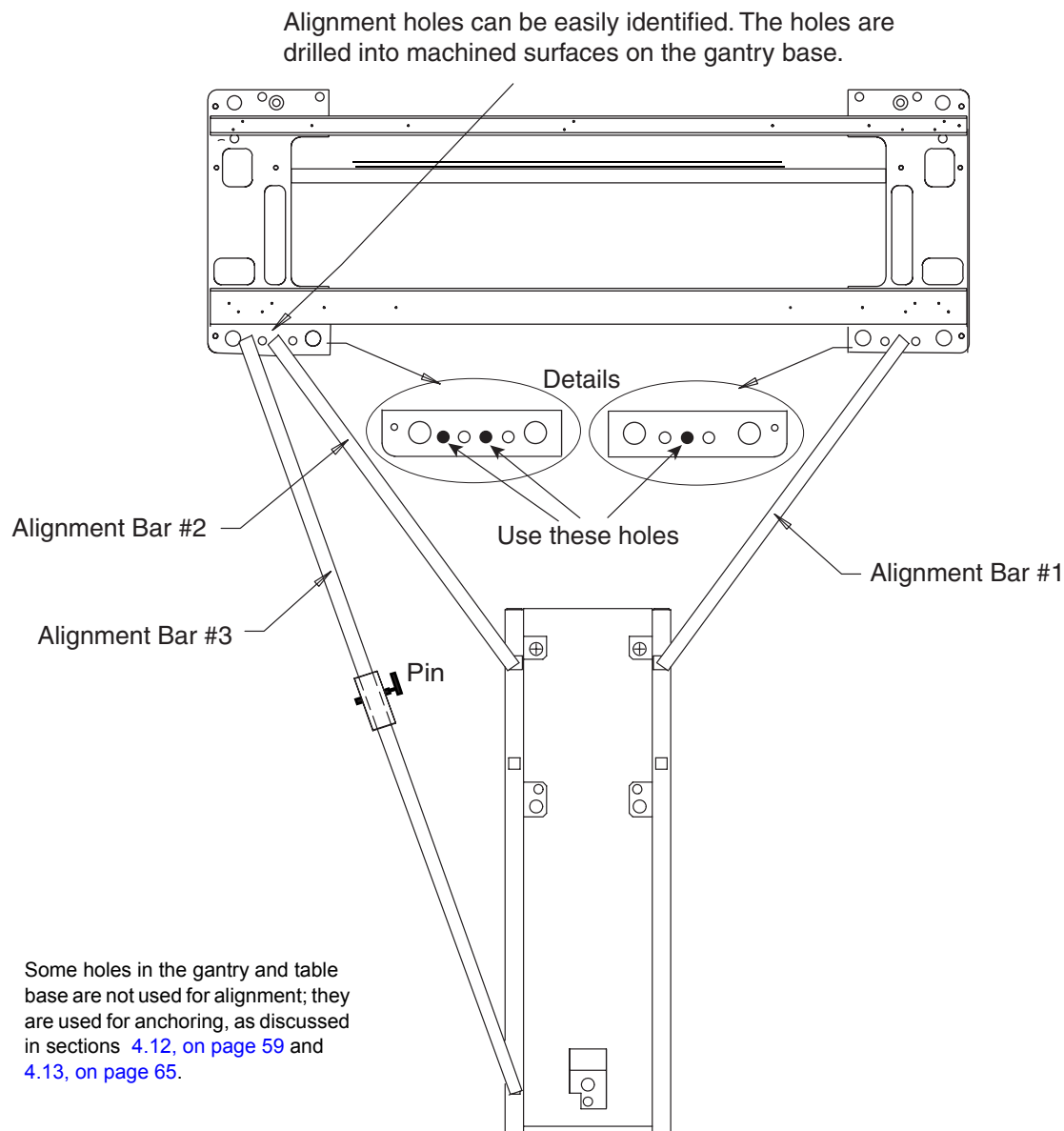
Figure 1-19 Center Support Bar



4.6 Install Alignment Bars

- 1.) Assemble the two (2) halves of the long gantry/table alignment bar (#3):
Adjuster locking ring at location #1 (see [Figure 1-20](#)) must be removed to properly position the alignment bars. Also, observe the following:
 - **Bars should be at room temperature ($70^{\circ}\text{F} \pm 1\text{ degree}$ or $21^{\circ}\text{C} \pm 1\text{ degree}$).**
 - **The alignment bar must be calibrated. They must have a valid calibration sticker to be accurate.**
 - Do not use or trust bars that have a missing locking pin or loose end cams.
- a.) Insert the pin and rotate the handle to tighten the joint. If the joint remains loose, tighten the tip of the expanding pin to increase its diameter and try again.
- b.) Check the serial numbers on both halves of the alignment bar; make sure the serial numbers match.

Figure 1-20 Gantry & Table Alignment Bar & Hole Locations for HP60/80/100



- 2.) The short alignment bars (#1 and #2) are identical and interchangeable. Refer to [Figure 1-20](#):
 - a.) Install the short alignment bars (#1 and #2) first. Bars should fit snugly.
 - b.) Pivot the far end of the table until the long alignment bar (#3) fits into place.
 - c.) Adjust the position of the table until the three alignment bars fit into the $\frac{3}{4}$ in. alignment holes in the gantry base and the $\frac{1}{2}$ in. alignment holes in the table base.
 - d.) Fully seat all three bars on their mating surfaces.

Note: The LightSpeed RT base is different from the HP60/80/100 base. For the LightSpeed RT base, note the hole pattern shown below in [Figure 1-21](#). The RT has the same configuration for alignment bars and anchor positions, but it has extra (smaller) holes on the outside edge for the plates on the base. These extra holes are not used for installation.

Figure 1-21 Close-up of RT alignment bar locations



- 3.) After installing the alignment bars, lower the table to the floor and remove the transportation dollies ([Figure 1-22](#)).

Figure 1-22 Table transportation dollies



- a.) Temporarily remove alignment bar #1, to remove the front table dolly.
- b.) Replace alignment bar #1 after you remove the front dolly.
- 4.) Remove the five adjuster shipping bolts and loosen the lock rings.
- 5.) Use the 1½ in. socket and 6 in. extension to remove the shipping bolts.

4.7 Level the Table

4.7.1 Before You Begin

- Only adjusters should be used to level the table.
- Use the adjuster tool, 1¹/₈ in. socket, and ½ in. drive ratchet to turn each adjuster.
- Place a mark on the adjuster to aid in counting adjuster rotation.

4.7.2 Procedure

- 1.) Loosen and unload adjusters #5 and #6. (Refer to [Figure 1-23](#).) Do not remove the shipping

bolts.

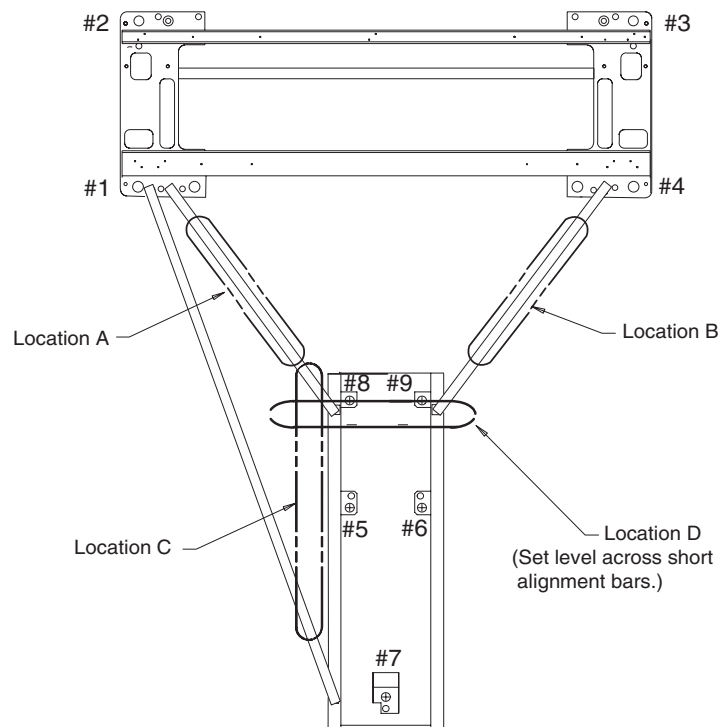
- 2.) Using a 4' level, measure the levelness of the table at locations A, B and C.
 - Place the level on the machined surfaces of the alignment bars.
 - Do NOT rest the level on the radiused surface on alignment bar #3. (See [Figure 1-20.](#))Because the leveling pads rest against the bottom of the base, you need to raise the table from its lowest position to get it leveled.
- 3.) Map out the adjustments needed to level all three locations.
- 4.) Remove the shipping bolts from adjusters #7, #8 and #9 only.
- 5.) Systematically level locations A, B, and C (refer to [Figure 1-23](#)) by turning each adjuster. We suggest a maximum of one (1) turn at a time.

NOTICE

Always use the adjuster tool to raise or lower the table.

- Use adjuster #8 to level Location A.
- Use adjuster #9 to level Location B.
- Use adjuster #7 to level Location C.
- Use adjusters #8 and #9 to level Location D.

Figure 1-23 Table Base Gantry/Table Level Locations



- 6.) Continue to make adjustments until Locations A, B, and C are centered at the same time.
- 7.) Check location D. Fine tune only.
- 8.) Screw down the adjusters at table Locations #5 and #6, until each rests tightly against its leveling pad. Tighten each adjuster a little more than hand tight, but take care not to unload any of the other table adjusters. It is important that locations #5 through #9 ([Figure 1-23](#)) all bear part of the table's load.

- 9.) The table base covers need a minimum of 27 mm $\pm$ 3 mm between the bottom of the table base and the floor for the covers to fit correctly. Check the distance between the table base and the floor, to ensure the covers will fit properly. See [Figure 1-24](#) and [Figure 1-25](#).

Figure 1-24 Typical Table Base Clearance

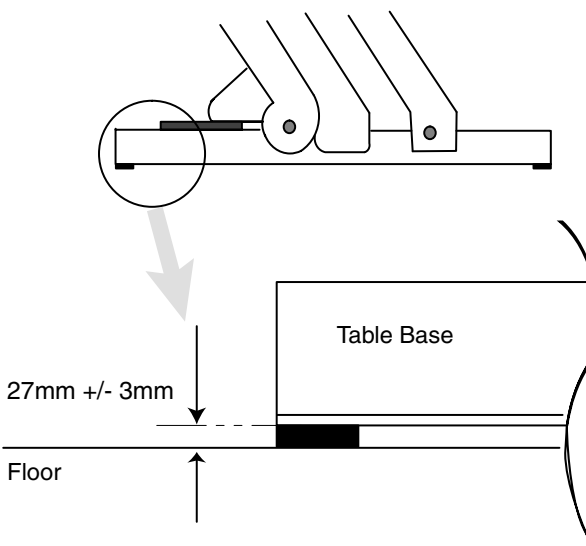


Figure 1-25 Excessive Table Base Cover Clearance



Note: If excessive clearance cannot be avoided, fill the gap. For example, you can use 0.5" round pipe insulation to fill the gap.

4.8 Table-Gantry Mechanical Alignment

4.8.1 Required Tools

- 2 ft. Carpenter's Square
- 2 ft. Level
- Masking Tape
- Flat-blade Screwdriver
- Craftsman Laser Trac™ Level with Carrying Case and Laser Enhancing Glasses
Sears item #00948247000
Mfr. model #48247
- 4 ft. Level
- 9 in. Level
- Metric Tape Measure
- 1ft. Pry Bar

4.8.2 Alignment Conditions

- Complete all sections of this procedure before drilling any anchor holes.
- The alignment tool is used to align and level the table and gantry.
- The alignment tool MUST have a current calibration sticker.

4.8.3 Alignment Specification

- Mechanical alignment must be within ± 3 mm
- Table cradle travel must be parallel
- Table must be perpendicular to the gantry

4.8.4 Procedure

NOTICE Use caution while removing the gantry scan window.



- 1.) Remove the gantry scan window. (Refer to [Figure 1-26.](#))

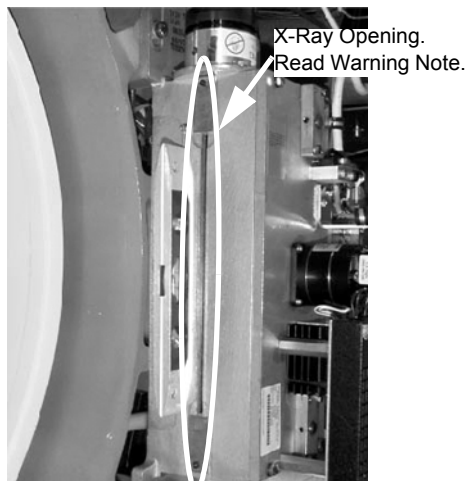
Figure 1-26 Removing Gantry scan (mylar) window



- 2.) Do not remove the alignment bars unless told to do so in the procedure.

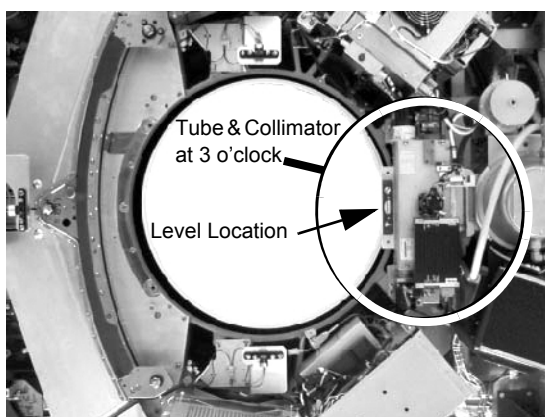
Note: Do not place any object near or into the x-ray opening. This slot must not be nicked or marred (refer to [Figure 1-27](#)).

Figure 1-27 X-Ray Opening



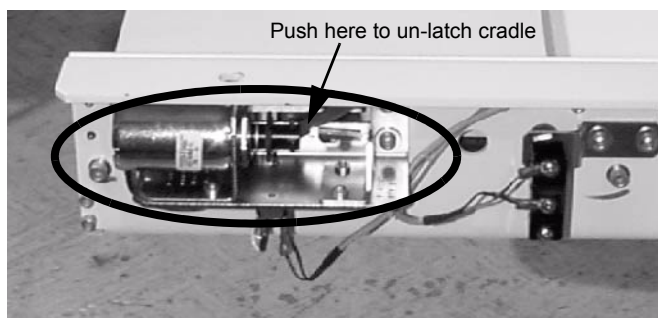
- 3.) Rotate the gantry by hand until the collimator face plate is at 3 o'clock (refer to [Figure 1-28](#)). Use a 9-in. level to verify the plate is vertical. With the power off, the gantry movement is tight.

Figure 1-28 Three o'clock position



- 4.) Remove the cradle shipping bolt and the cradle right side cover. The cradle shipping bolt is located under the table/cradle assembly at the foot-end of the table. It has a RED TAG attached to it.
- 5.) Push the plunger on the right rear of the table to unlatch the cradle (refer to [Figure 1-29](#)).

Figure 1-29 Unlatch Cradle



4.8.4.1 Cradle Center Procedure

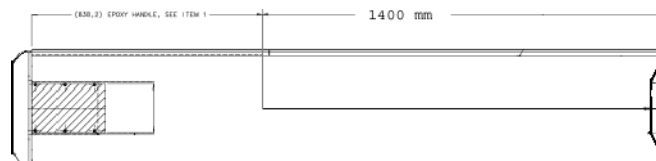
- 1.) Center the cradle on the drive rollers assembly by pushing the cradle into the gantry to its maximum position (1400 mm) and back to just before the latch position six times. The cradle should be centered as shown in [Figure 1-30](#).

Figure 1-30 Centered Cradle



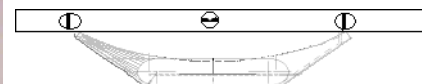
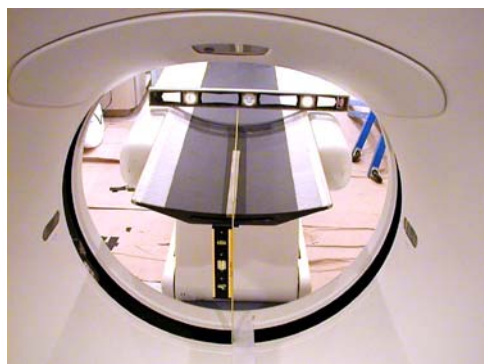
- 2.) Adjust the cradle as needed, using the six bolts located on the back of the cradle ([Figure 1-31](#)).

Figure 1-31 Use bolts to adjust cradle



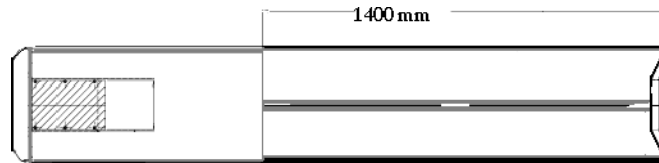
- 3.) Unlatch and move the cradle toward the gantry a few inches until the front of the cradle is near the gantry opening.
- 4.) Use a 2 ft. level to check the cradle level from side to side. Level the cradle using the table front level adjusters #8 and #9. (See [Figure 1-32](#).)

Figure 1-32 Place level across cradle



- 5.) Place the level in the center of the cradle and check that it is level (see [Figure 1-33](#)). The width and the length of the cradle must be level when finished. Re-tighten all adjuster lock rings.

Figure 1-33 Place level along the length of the cradle



- 6.) Find the table center line marked on the front edge of the cradle. If your system has no center line mark, complete the following procedure to make a cradle center line.
- 7.) Using a metric tape measure or other suitable measuring device:
- a.) Using supplied masking tape, place a 380 mm (15 in.) strip on the front of the cradle. See [Figure 1-34](#).

Figure 1-34 Masking tape on the front of the cradle

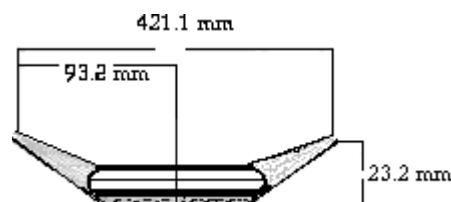


- b.) Determine the cradle center-line by measuring the cradle width 421.1mm (16.6 in.) (210.5 mm (8.3 in.) center). Use a square or straight edge to determine the center.

Note:

The white edges on the cradle are not always parallel and a second measurement is needed. See [Figure 1-35](#).

Figure 1-35 Front of Cradle



- c.) Find the center of the inside slot. Width of the inside slot is 186.4 mm (7.3 in.) (93.2 mm 3.7 in.) center). Use a square or straight edge to determine the center. See [Figure 1-36](#).

Figure 1-36 Determine slot center



- d.) Use these two dimensions to determine the table center line by connecting the two marks. Measure back 300 mm (11.8 in.), then measure across the cradle and place a second

cradle center mark on the tape. Use a square or straight edge to connect the two marks.
See [Figure 1-37](#).

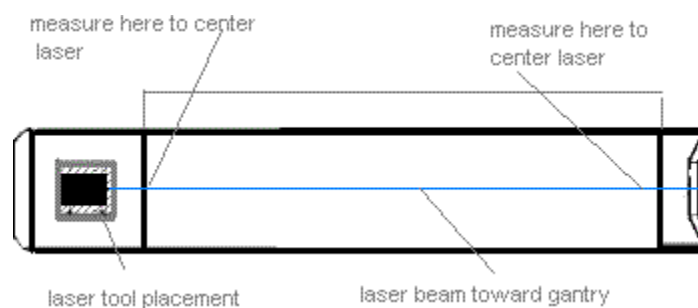
Figure 1-37 Use a square to connect marks



4.8.4.2 Laser Cradle Center Procedure

- 1.) Assemble the laser tool
 - a.) Mount the laser to the magnetic base unit
 - b.) Note the CL alignment pin on the magnetic base unit
- 2.) Level the laser tool
 - a.) Place the laser tool on the rear of the cradle with the laser pointing toward the gantry. See [Figure 1-38](#).

Figure 1-38 Position Laser Tool



- b.) Using the three adjustment screws level the laser. See [Figure 1-39](#).

Figure 1-39 Laser Level Adjustment Screws



- 3.) Center the laser on the cradle

WARNING



POSSIBLE EYE INJURY.
LASER BEAMS CAN CAUSE PERMANENT EYE DAMAGE.
WHEN OPERATING THE LASER LEVEL, DO NOT STARE AT THE LASER BEAM.

- a.) Turn on the laser. See [Figure 1-40](#).

Figure 1-40 Turn on the Laser



- b.) Place a metric tape across the cradle in front of the laser. See [Figure 1-41](#).

Figure 1-41 Place Metric Tape in Front of Laser



- c.) Adjust the laser tool so that the beam is centered at ~ (210mm)
d.) Move the metric tape to the front edge of the cradle and adjust beam center (210mm)

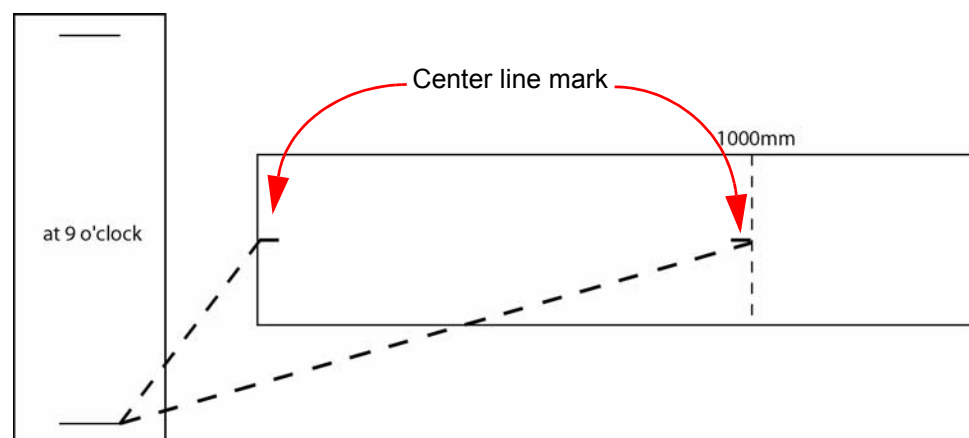
- 4.) Using a metric tape carefully measure from cradle center line to collimator faceplate. The laser beam should pin point the exact measurement. Record this measurement as "A" on the worksheet.
- 5.) Carefully rotate the gantry by hand until the tube and collimator is in the nine o'clock position.
- 6.) Use a bubble level to precisely position the collimator faceplate at the nine o'clock position.
- 7.) Carefully measure from cradle center mark to collimator faceplate.
- 8.) Record this measurement as B on the worksheet.
- 9.) Perform the equation in the SPEC column to determine if the table should be moved. If so move the table half the distance to correct the error.
- 10.) The table and gantry are in alignment when both measurements meet the specification.
- 11.) If A and B are not equal, move the table and repeat until measurements meet the specification ($\pm 1\text{mm}$).

4.9 Cradle/Table Parallel Check

Note: The table pictured in the photos in this section appears for demonstration purposes only. Your actual table may vary. Center line marks

- 1.) With the cradle in the home position, rotate the collimator to the 9 o'clock position. Confirm with a level placed on the collimator face plate.

Figure 1-42 First Cradle/Table Parallel Check

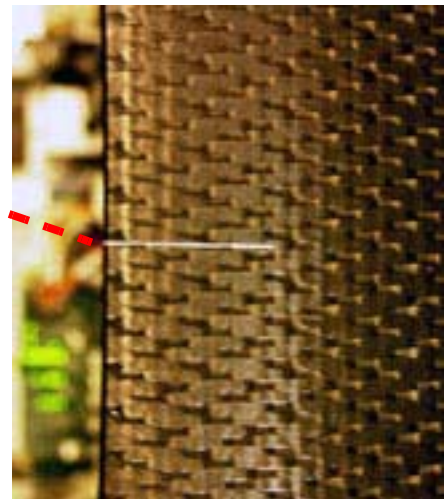
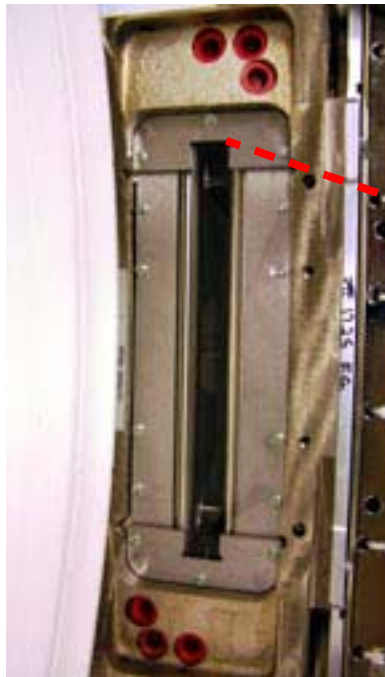


- 2.) Measure the distance from the front top corner of the collimator face plate to the mark at the center front of the cradle. See [Figure 1-42](#), [Figure 1-43](#) and [Figure 1-44](#). Note all measurements in [Table 1-4](#).

Figure 1-43 Collimator Face Plate to Front of Table - Overview

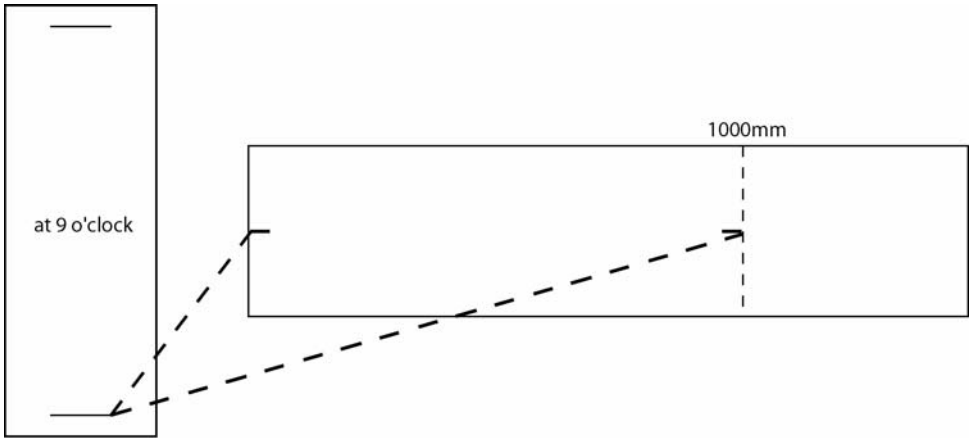


Figure 1-44 Collimator Face Plate to Front of Table - Detail



- 3.) Measure the distance from the same point of the collimator face plate to the center point on the cradle at the 1000mm mark. See [Figure 1-42](#).
- 4.) Rotate the collimator to the 3 o'clock position. Confirm with a level.
- 5.) Repeat [Step 2](#) and [Step 3](#). See [Figure 1-45](#).

Figure 1-45 Second Cradle/Table Parallel Check



- 6.) If needed, move the table using an appropriate tool and re-measure each side until measurements are equal side to side ± 1 mm.

Note: This final adjustment may be slightly different than the placement obtained using the laser.

Table 1-4 Alignment Worksheet

MEASUREMENT	A: 3 O'CLOCK	B: 9 O'CLOCK	DIFFERENCE: A-B
Test #1			
To front of cradle			
To 1000mm mark			
Test #2			
To front of cradle			
To 1000mm mark			
Test #3			
To front of cradle			
To 1000mm mark			

- 7.) Recheck cradle level (front to back and across) and re-level as required.

4.10 Tighten the Lock Rings

- 1.) Re-check bubble levels and loading first. Refer to [Figure 1-23, on page 47](#).
- a.) Re-check that bubbles levels at locations, A through D on the table.
 - b.) Re-check that bubbles levels at both ends of the gantry.
 - c.) Re-check that adjusters at Locations 1, 2, 3, 4, 7, 8 and 9 carry a portion of the load.
 - d.) Re-check that gantry “inside” levelers carry a portion of the load.

CAUTION



Eye protection is required when using a hammer and chisel.

- 2.) Tighten the lock rings at Locations #1 through #9 with the spanner where possible. Use a hammer and chisel to tighten the lock rings only where you can not use the spanner.

4.11 Remove Bottom Gantry Cover - RT Only

Before the gantry mounting holes may be drilled, the bottom cover must be removed from the RT gantry:

- 1.) Remove screws from the cover mounting location (see [Figure 1-46](#)).

Figure 1-46 RT Base Cover Removal



- 2.) Move the cover backward so the gantry holes can be accessed.
- 3.) Follow the procedure for drilling holes (Section [4.12](#), below)
- 4.) After drilling is completed, put the cover back in place, and reattach using the removed screws.

4.12 Drill the Table Anchor Holes

WARNING POTENTIAL FOR PATIENT INJURY.



**IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING
SYSTEM OPERATION.**

4.12.1 Notes to Mechanical Installers

4.12.1.1 Note 1: Basic Anchoring Information

GE provided floor anchors are designed for use ONLY on concrete floors that meet the 102 mm (4 inch) concrete floor requirement. Supplied floor anchors must be installed by a trained contractor, and shall be set to a minimum depth of 76.2 mm (3 in.) at each anchor point. ANY anchors having more than 25.4 mm (1 in.) of thread showing above the nut, when torque is set to 75 N-m (55 ft. lb.), shall have a second anchor installed in the closest adjacent hole. This is because the minimum anchor engagement length in the concrete was not met. The second anchor shall be installed to the standard depth and torque specification. **Do not cut anchor bolts that extend longer than the 25.4 mm (1 in.) limit.**

4.12.1.2 Note 2: Alternate Anchoring

If at least four anchors cannot be set for the gantry, and at least four anchors for the table using the alternate anchor holes, then the installer must inform the PMI that the minimum anchoring cannot be met. Additionally, the customer's structural engineering contractor must be engaged to determine the anchoring method, set the anchors, and certify that their anchoring meets the stated GE minimum load requirement and torque specification.

4.12.1.3 Note 3: Non-Concrete Floors

All other anchoring methods - on floor types other than the concrete minimum - must be determined at the customer's expense by a structural engineering contractor. The anchoring and method must be certified by the customer's contractor to meet the stated GE minimum load requirement and torque specification.

4.12.1.4 Note 4: GE Notification

Mechanical contractors or installers (FEs) should not assume the role of determining acceptable methods to install or anchor equipment on non-102 mm (4 in.) concrete floors. The PMI or appropriate GE contact person shall be notified that the facility's floor type DOES NOT MEET the installation mounting requirement for the installation procedure (described in this Installation Manual), and therefore the table-gantry mounting process CANNOT continue.

4.12.2 Requirements

4.12.2.1 Tools Required

- Standard Install Tool Kit
- Hammer Drill
- ½ in. x 12 in. Drill Bit (Metric equivalent must not be used)
- ½ in. Drill Bushing (shipped in install support kit)
- Vacuum with HEPA or drywall dust filter
- Vacuum Hole Attachment – to clean debris from the holes
- PPE

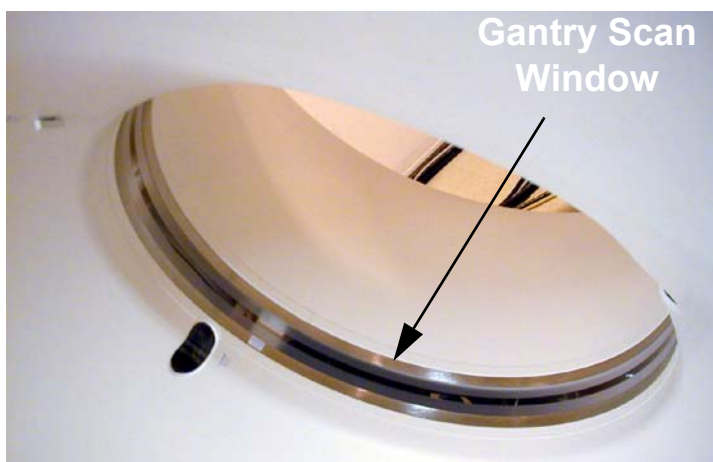
4.12.2.2 Time and Personnel

- 0.5 Hours labor on site
- 2 Engineers

4.12.3 Drilling Procedure

- 1.) Make certain that the gantry scan window is in place, to prevent dust from entering the gantry.

Figure 1-47 Gantry Scan Window

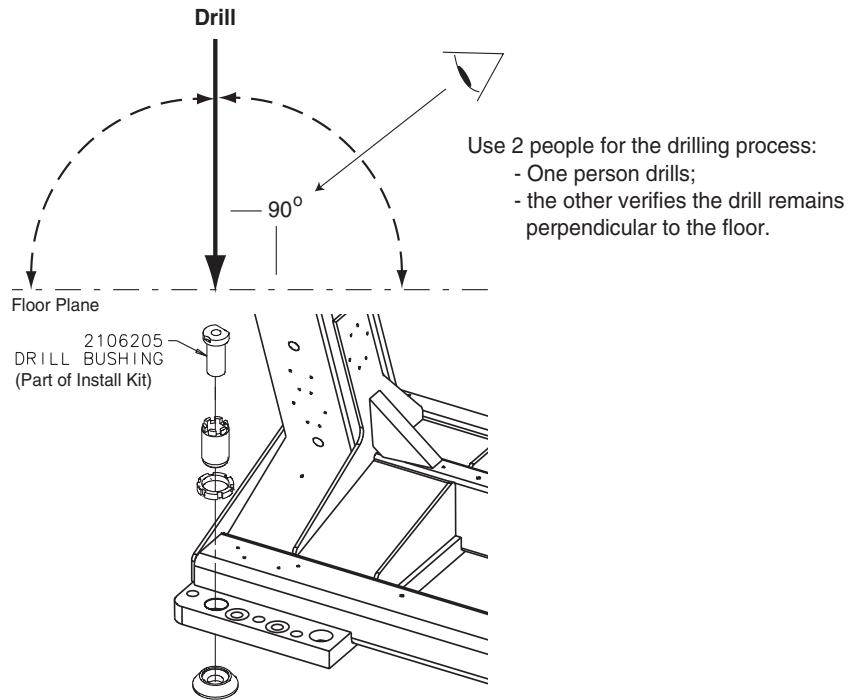


- 2.) Use a piece of tape to mark the drill bit depth of 190 mm (7-11/16 in.) from the tip of the 13 mm (½ in.) masonry drill bit.
- 3.) Use the 13 mm (½ in.) bit to drill all seven (7) anchor holes (refer to [Figure 1-50](#) for locations.)
 - Review [Figure 1-48](#), prior to drilling.

NOTICE
Potential for
Equipment
Damage from
Dust

You must cover all electronic assemblies in the table base prior to drilling to prevent damage due to the dust created during drilling.

Figure 1-48 Drilling Position

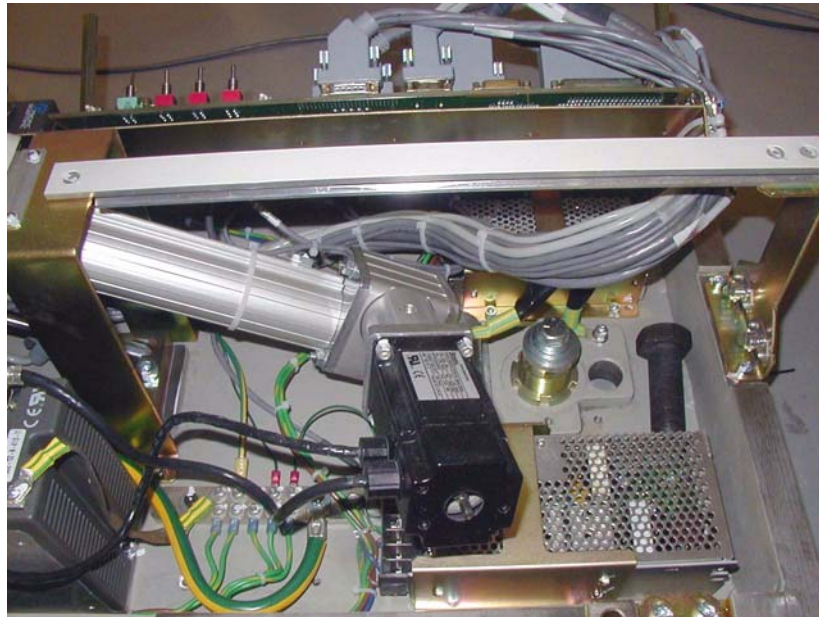


- **Important** - Drill the holes perpendicular to the floor. While one person drills the holes, position a second person to watch the relationship between the drill bit and floor. Make sure the bit remains absolutely perpendicular to the floor throughout the drilling operation. Always use the mechanical guide supplied with the alignment kit.
- Stop drilling every 15 or 20 seconds and clear the hole of debris. This helps to prevent binding of the drill bit.
- a.) Place appropriate protection to prevent damage and dust contamination to electronic assemblies. Refer to [Figure 1-19, on page 44](#), for placement of dust protection materials.
- b.) Drill each hole until the mark on the drill bit is even with the top of the drill bushing. (195 mm/7.7 in.)
- c.) Place the drill bushing inside each adjuster, to keep the hole vertical and centered within the adjuster.
 - * Use the drill bushing to center the anchor holes in adjuster Location #5, #6, and #7, to provide maximum lateral alignment capacity when you center the cradle on isocenter during subsequent system testing.
 - * Take care not to injure yourself on the gantry cover brackets.
 - * Take care to avoid the actuator motor lead when you drill the Location #7 anchor hole. Remove the lead if necessary.

Note:

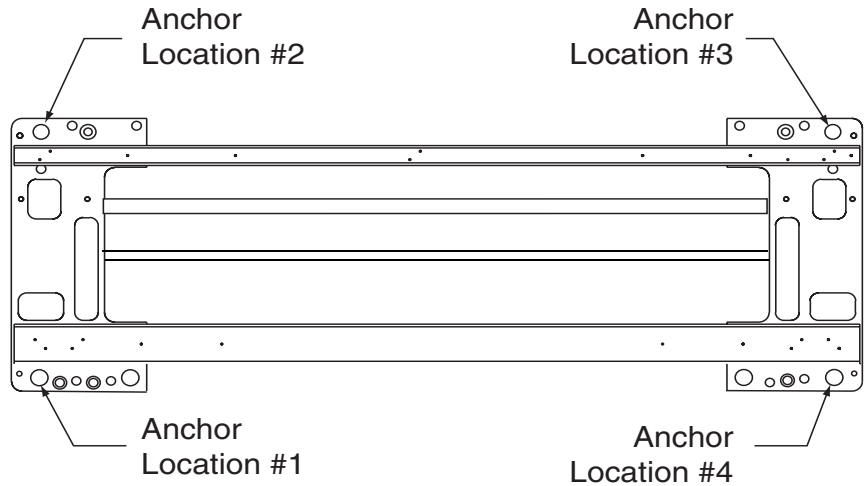
An 18" bit will make drilling anchor location #7 easy, and will prevent component damage (see [Figure 1-49](#)).

Figure 1-49 Table Assembly Showing Actuator Motor and Anchor Location #7

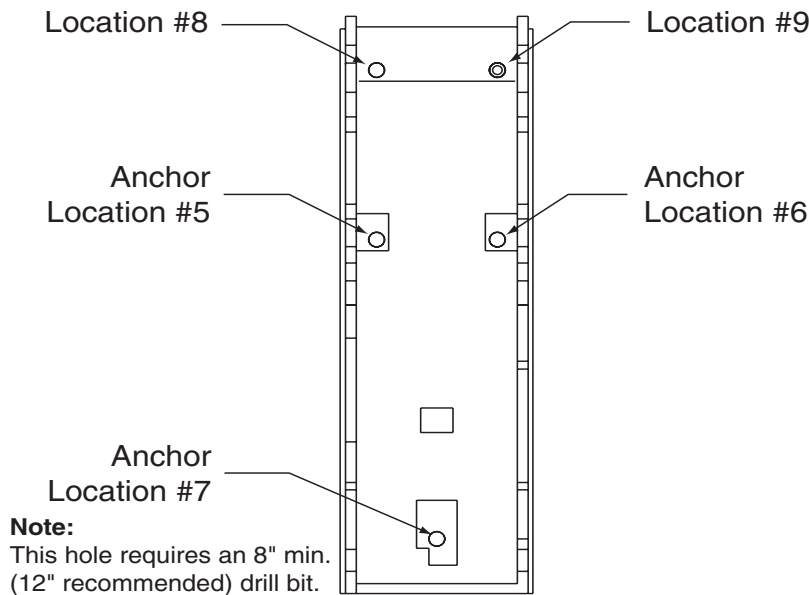


- d.) Stop the drill every 15 or 20 seconds and clear the hole of debris, to prevent binding of the drill bit.
- 4.) Vacuum all debris from the inside of the hole:
Note: A drywall dust filter should be used on the vacuum.
 - a.) Place the funnel tip inside the hole; place the vacuum hose in the funnel.
 - b.) Continue to vacuum while you drill, if you can, to keep gantry and table as free of dust contamination as possible.
 - c.) When you finish clearing the anchor hole, vacuum the debris from the surrounding area.
 - d.) Stop drilling frequently to let the drill bit cool.
 - e.) All holes must be a minimum of 108 mm (4.25 in.) deep (see [Figure 1-54](#)). Re-check all holes.
- 5.) Do not drill anchor holes at Location #8 and #9. These locations help support the weight, but do not anchor the table into place.

Figure 1-50 Anchor Locations, Standard (Except RT)

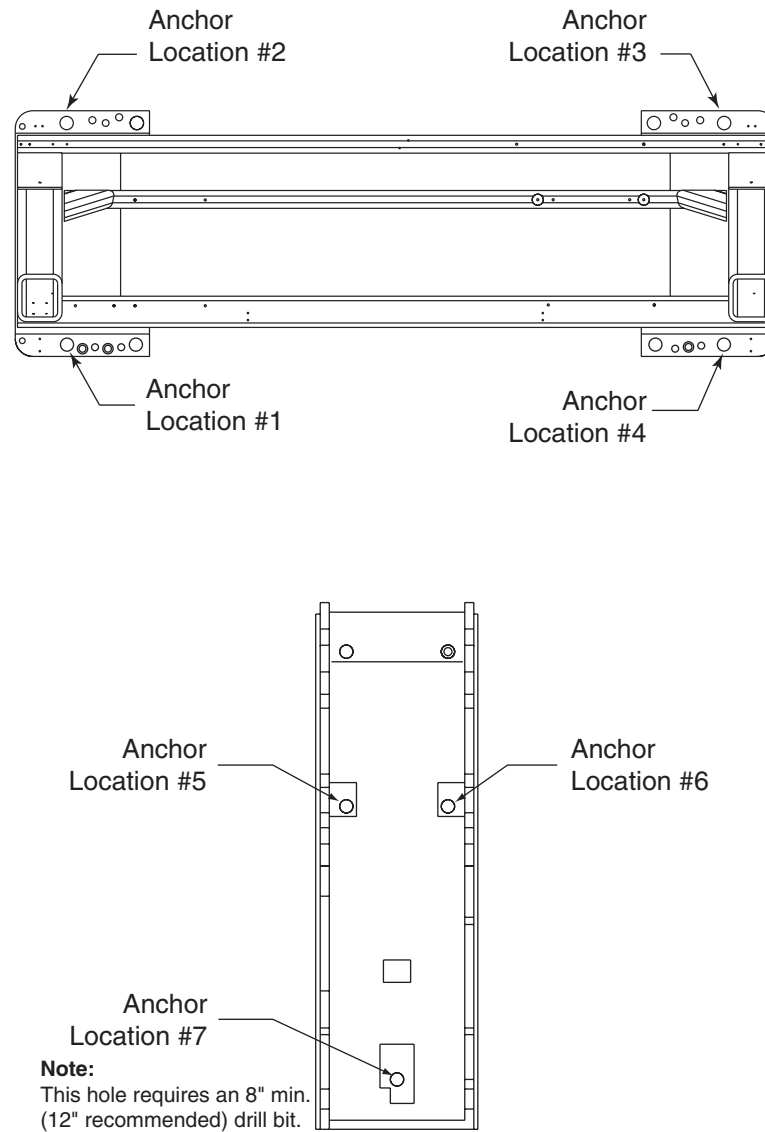


Note: Anchor holes in base are 7/8" (2.3 cm).



Note:
This hole requires an 8" min.
(12" recommended) drill bit.

Figure 1-51 RT Anchor Locations (standard)



- 6.) **For LightSpeed RT only:**
Make certain to replace the cover, and reattach it using the removed screws.

4.13 Gantry & Table Redundant Anchor Holes

If you cannot use one of the adjuster anchor holes due to structural interference, such as reinforcement bars in the concrete, you must use one of the redundant anchor locations, as shown in [Figure 1-52](#).

- The gantry requires a minimum of four (4) anchors, one (1) in each corner.
- The table requires a minimum of three (3) anchors, one (1) at location #5, #6 and #7.

If you must use a redundant anchor hole in the gantry, you must remove the gantry covers. See [Appendix A](#), for gantry cover removal.



POTENTIAL FOR PATIENT INJURY.
IMPROPERLY-SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

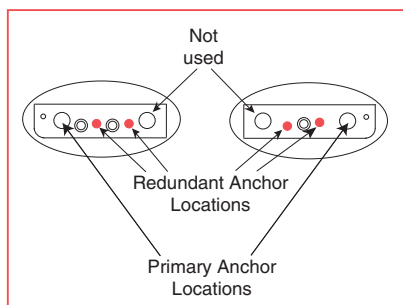
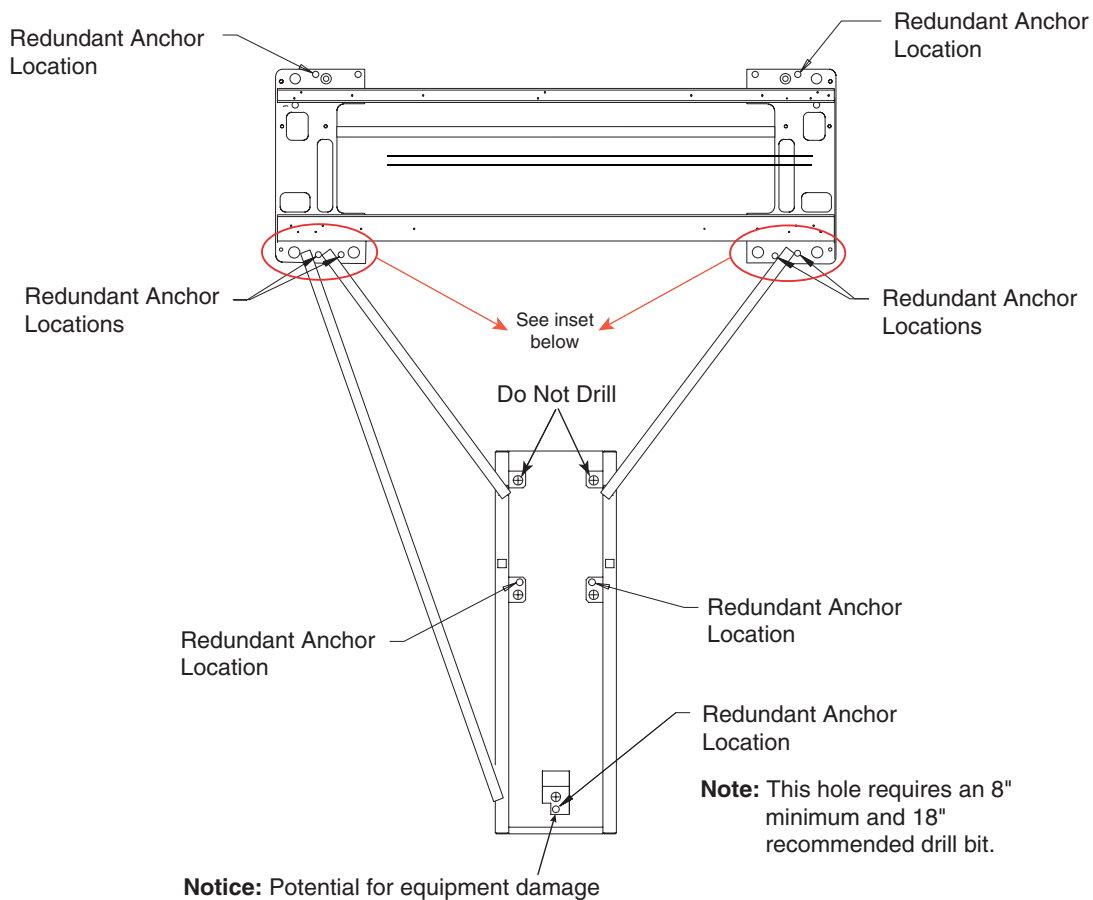
It is the purchaser's responsibility to provide an approved support structure and mounting method for all floor types other than those listed. General Electric is not responsible for any failure of the support structure or method of anchoring, including seismic requirements and/or through-bolting.

Note: GE is not responsible for anchoring methods other than those listed in the pre-installation manual. Provided floor anchors are designed for use ONLY on concrete floors that meet the 102 mm (4 in.) concrete floor requirements.

MOUNTING REQUIREMENTS	GANTRY	TABLE
Minimum Floor Thickness:	101.6 mm (4 in)	101.6 mm (4 in)
Recommended Drilling Depth:	95.25 mm (3-¾ in)	95.25 mm (3-¾ in)
Average Anchor Embedment:	89 mm (3-½ in)	89 mm (3-½ in)
Minimum Anchor Embedment:	76.2 mm (3 in)	76.2 mm (3 in)
Available Alternate Anchor Locations:	Yes	Yes
Shipped Anchor Size:	203.2 mm (8 in)	203.2 mm (8 in)
Alternate Anchoring Methods:	Yes (see notes, above)	Yes (see notes, above)
Floor Levelness Requirement:	8 mm (0.3125 in) over 10 ft	8 mm (0.3125 in) over 10 ft

Table 1-5 Gantry and Table Mounting Requirements

Figure 1-52 Table Base Redundant Anchor Locations



Mark the ½ in. drill bit with a piece of tape 170 mm (6-11/16 in.) from the tip of the bit, and follow the instructions in Section 4.12, on page 59.

4.14 Install the Anchors - 102 mm (4 in.) Concrete Floors Only

NOTICE These instructions apply to installing anchors in 4-inch (or greater) concrete floors **ONLY**. (Refer to the notes in Section [4.12.1](#), on page 59.)

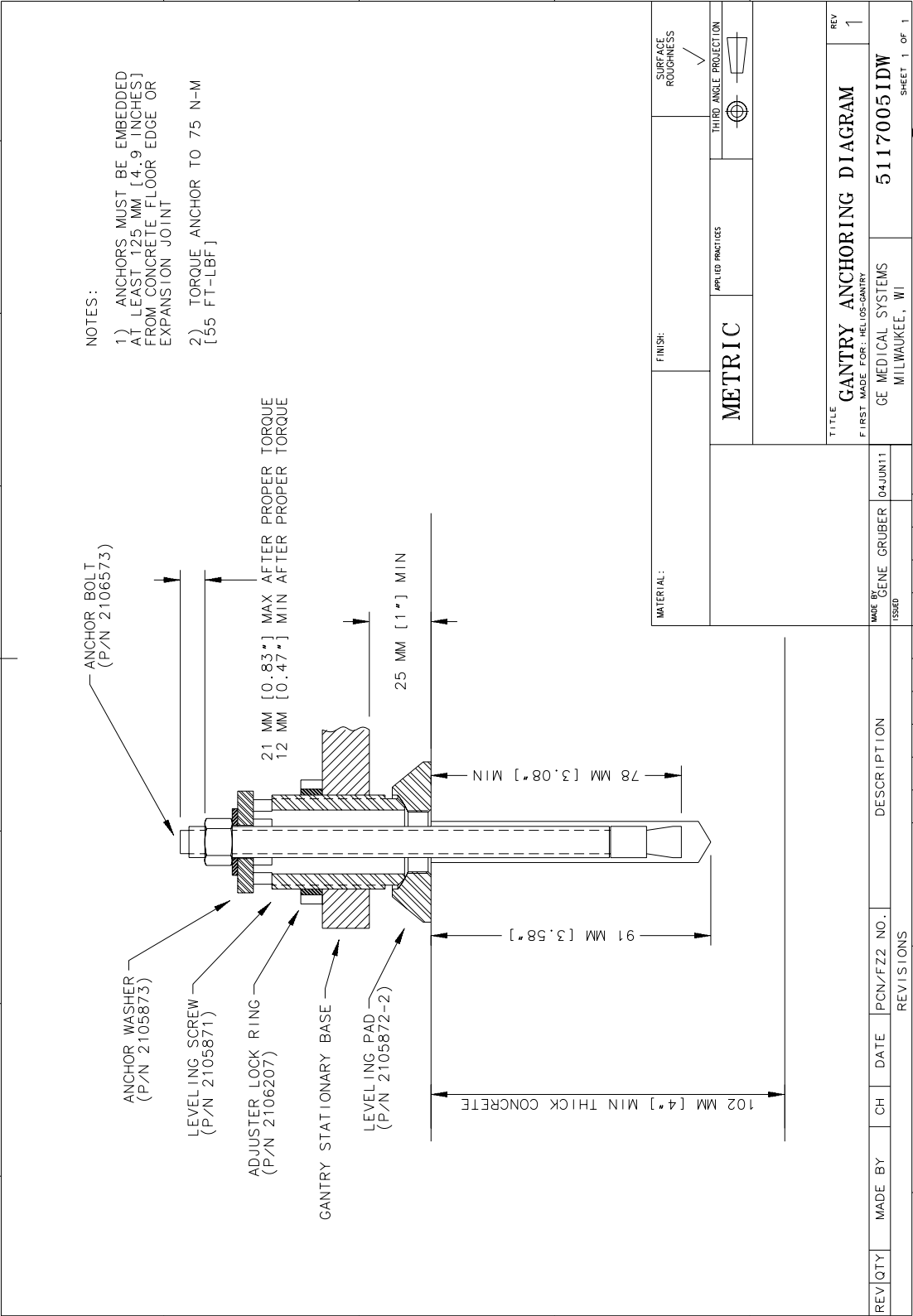
Recommended - Use "Hilti Kwik-Bolt II" anchors for this procedure. P/N 2106573 (13 mm dia. x 203 mm) (½ in. dia. by 8 in. long).

- 1.) Remove the three alignment bars and repack the install kit.
- 2.) Refer to [Figure 1-54](#). Assemble the anchors before you install them:
 - a.) Remove the nut and washer from the anchor.
 - b.) Add a 6.35 mm (¼ in.) thick washer (PN 2105873) under the regular anchor washer
 - c.) Reassemble the anchor washer and nut and position nut so top is flush with threads of anchor before hammering into hole (refer to [Figure 1-54](#)).

Figure 1-53 Measure Table Base Anchor Thread

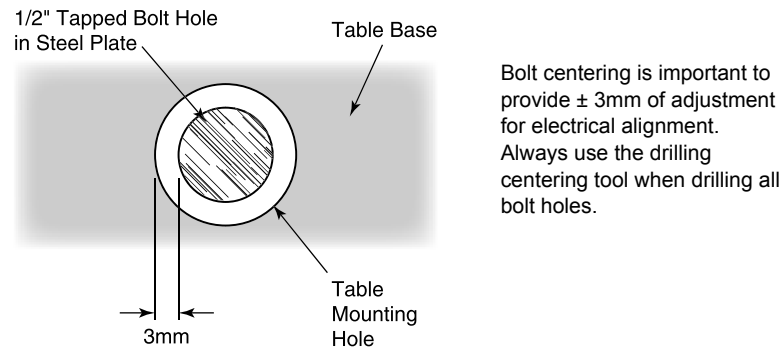


Figure 1-54 Table Base Anchor Assembly



- 3.) Fixed site Installations must be centered under mounting holes in table base, with 3 mm (0.12 in.) of space visible surrounding the tapped bolt hole. [Figure 1-55](#).

Figure 1-55 Center tapped holes under mounting holes in table base



CAUTION



Eye protection is required when using a hammer.

- 4.) Refer to [Figure 1-54](#) for instruction on inserting the anchors in each hole. Using the anchor seating tool, hammer the anchor downward to *completely insert the anchor in the hole*.
- 5.) Repeat Sections [4.8.4.2](#), and [4.9](#).

This procedure is designed to mechanically center the table cradle within the gantry. All steps are important to achieve final electrical x-ray alignment of 1 mm (0.04 in). Alignment bars are still needed to level and position the gantry. Alignment bars must be calibrated yearly.

- 6.) Once alignment has been verified, torque all mounting bolts.
Tighten the Location #1 through #7 anchors and torque to $75 \pm 6 \text{ N-m}$ ($55 \pm 5 \text{ ft.-lb.}$).
- 7.) Replace all the table parts you removed during installation, including covers, ground strap and the center support bar. Check the motor lead.

Note: If you cannot replace the lower table cover because the floor interferes, adjust all of the table and gantry levelers by $\frac{1}{2}$ turn increments to raise the table/gantry until the lower table covers clear the floor. (Refer to [Figure 1-25, on page 48](#), for table base cover clearances.) Return to Sections [4.3, on page 37](#), [4.7, on page 46](#) and [4.10, on page 57](#) to level the gantry, level the table, and tighten the locking rings, respectively.

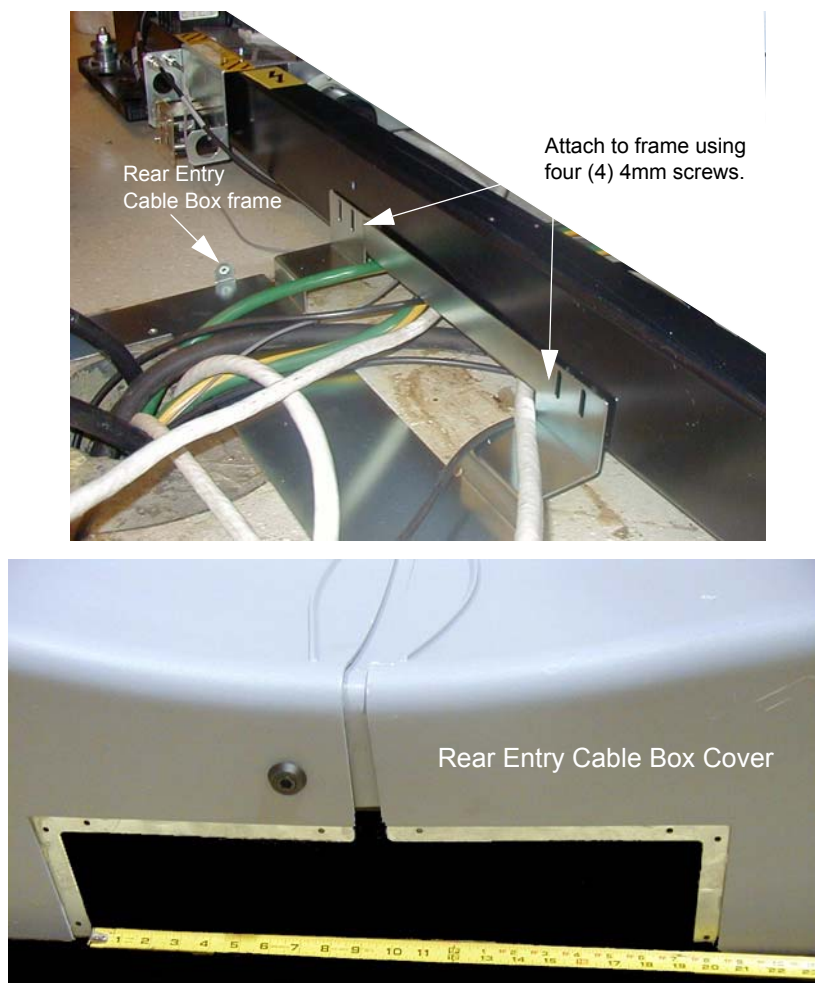
Section 5.0

Rear Entry Cable Box

Use a rear entry cable box (B7850RC) when the cables to the gantry cannot be brought up inside the gantry base. The box is not supplied with the system and must be ordered separately.

The rear entry cable box frame attaches to the gantry base. See [Figure 1-56](#).

Figure 1-56 Rear Entry Cable Box



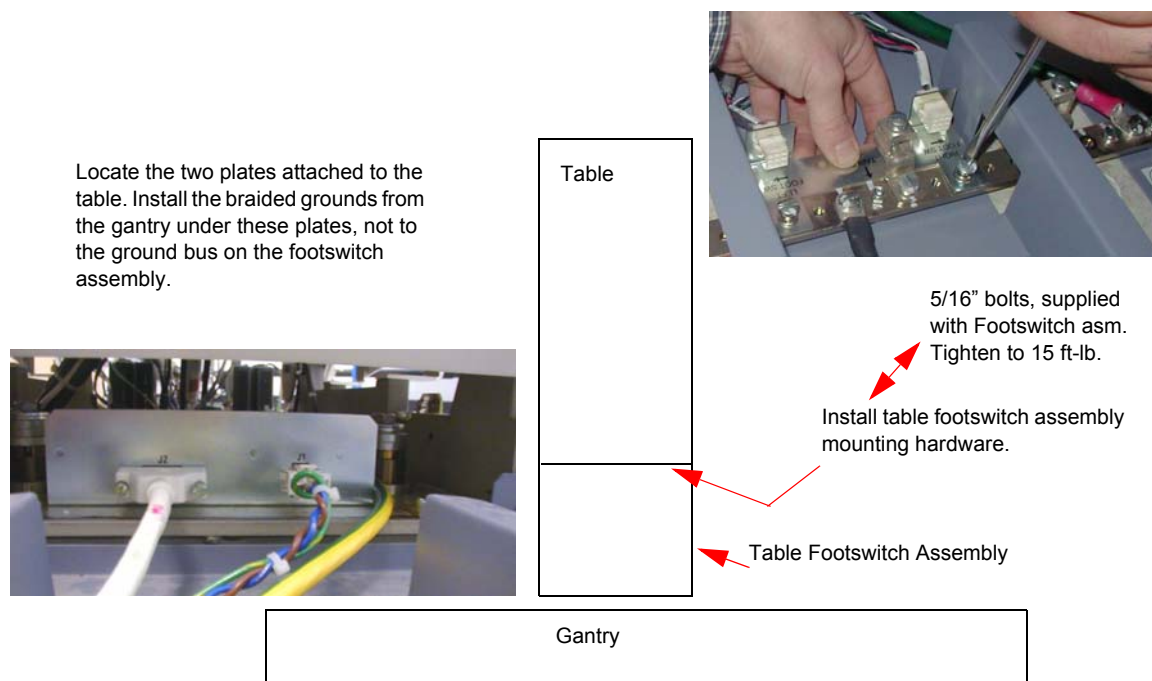
If necessary, cut the filler plate to fit the rear cable box into the cover opening. Use the supplied white molding around the filler plate.

Section 6.0

Install Table Footswitch Assembly

Install the table foot-switch assembly and filler strip as shown in [Figure 1-57](#).

Figure 1-57 Install Table Foot-Switch Assembly

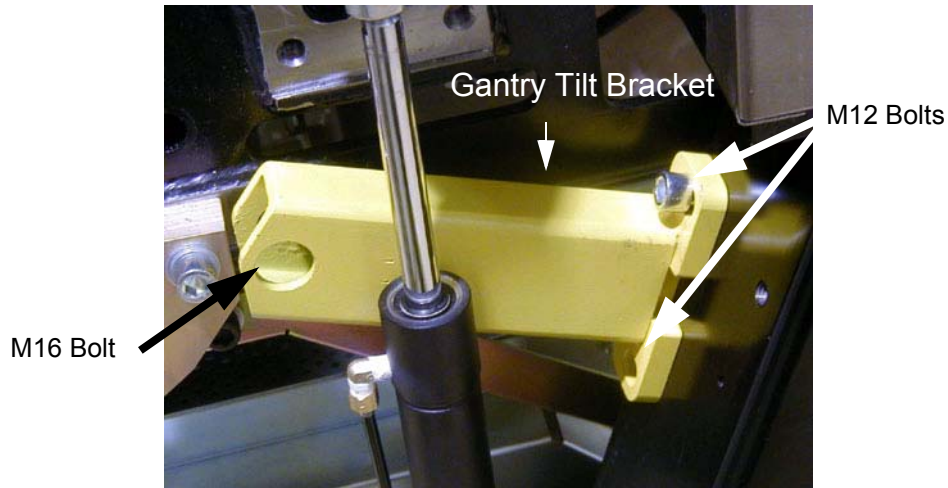


- 1.) Remove the foot switch assembly and locate the hardware.
- 2.) Loosen six (6) hex head screws and remove the footswitch assembly.
- 3.) Place the footswitch cover in a secure place.
- 4.) Locate the 2 ft. velcro strip supplied in the install support kit, and apply two strips to the bottom of the footswitch assembly.
- 5.) Using the two 5/16 in. bolts and blue Loctite (242), attach the footswitch assembly.
- 6.) Remove the 2 in. braided strip and gantry cable that are secured to gantry base, and place them in the cable tray.
- 7.) Remove the three (3) bolts that secure the footswitch bar and set them aside.
- 8.) Take off the footswitch bar and discard it.
- 9.) Locate the new footswitch bar from the install support kit.
- 10.) Install the new footswitch bar on the cover using the original three bolts.

Section 7.0

Remove Gantry Tilt Bracket

Figure 1-58 Gantry Tilt Bracket Removal



- 1.) Remove the M12 bolts using a 10 mm Hex wrench. See [Figure 1-58](#).
- 2.) Loosen the M16 bolt 1-2 turns and check the gantry tilt bracket, it should be loose to the touch. If loose continue with step 4.
- 3.) Check the hydraulic connections for leaks or lack of fluid. You will have to wait until the system can be energized to use the tilt controls to relieve the load on the tilt bracket prior to removal. Do not use force to remove the bracket.
- 4.) If the bracket feels loose, remove the M16 bolt using a 14 mm Hex wrench.
- 5.) Remove the bracket.
- 6.) Close the gantry covers and reinstall the scan window.
- 7.) Store brackets in the service cabinet.

CAUTION



If tilt bracket is not loose, stop and put the M12 bolts back in and tighten tilt bracket back in place. If there is a load on the tilt bracket, removal may cause the gantry to suddenly tilt all the way back due to a possible lack of hydraulic pressure.

Section 8.0

Position the Power Distribution Unit



WARNING

LOCKOUT/TAGOUT IS REQUIRED BEFORE PERFORMING THIS TASK. USE THE SUPPLIED LOTO KIT.

Note:

- 1.) Roll the PDU into position on its permanently-mounted casters. Leave at least 15.5 cm (6 in) between the PDU and back wall to allow cooling air to circulate.
- The customer’s electrical contractor connects the primary incoming power (steps 2 through 5, below).**

Table 1-6 Contractor Connections

Connection or Wall Box	AWG #	Connection From	Connection To PDU	Installed & Checked
TS1	#1	PDB-A	TS1-1	
	#1	PDB-B	TS2-2	
	#1	PDB-C	TS3-3	
	#1/0	GND	N/G (Do NOT connect anything to neutral point.)	

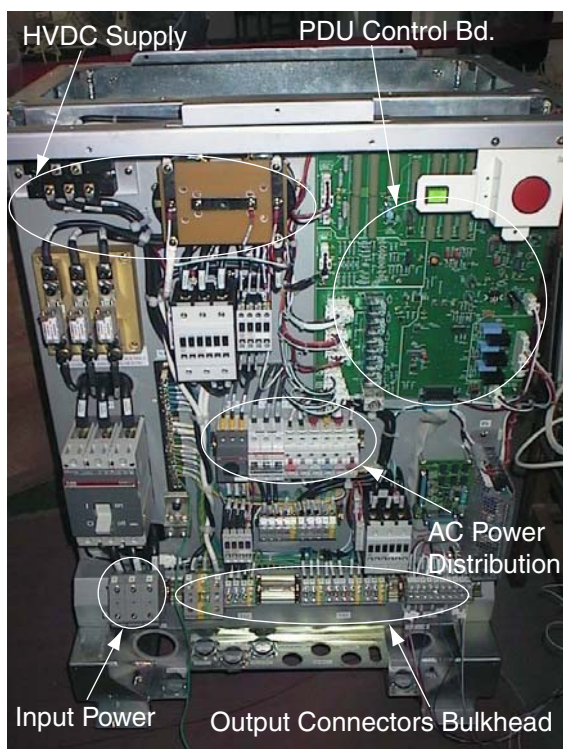
- 2.) Run the main input power conductors and ground though flexible metal conduit (attached between the PDU chassis and room duct-work) so you can move the PDU away from the wall during service.

Figure 1-59 Flexible Conduit for PDU Power



- 3.) Locate the hole cover plate in Box 1 and attach the flexible metal conduit to the PDU.
- 4.) Cut the three-phase and 1/0 ground wire to size.
- 5.) Attach the cables as shown in [Figure 1-59](#) and [Figure 1-60](#).

Figure 1-60 PDU Area Locations



Section 9.0

Install Operator Console

9.1 Unpack Console

Figure 1-61 Console boxed on skid



- 1.) Remove all items from the console.
- 2.) Remove all packing materials and discard them.
- 3.) Place the step-board under the front edge of the skid and step on it to raise the front edge of the skid, as in [Figure 1-62](#).

Figure 1-62 Step-board used to raise front edge of skid



- 4.) Remove the two front cushions from the bottom of the skid. See [Figure 1-63](#)

Figure 1-63 Cushion on bottom of skid



- 5.) Remove the Seismic Brackets from each side of the console. See [Figure 1-64](#). Save the brackets for later use if you need to mount the console to the floor.

Figure 1-64 Removing the Seismic Brackets



- 6.) Lift up on the strap on the front of the step-board ([Figure 1-62](#)) to lower the skid. Remove the step-board.
- 7.) Ensure the console stabilizers are in line with the notched portion at the front of the skid. This allows enough clearance to smoothly roll the console down the ramps. Refer to [Figure 1-65](#).

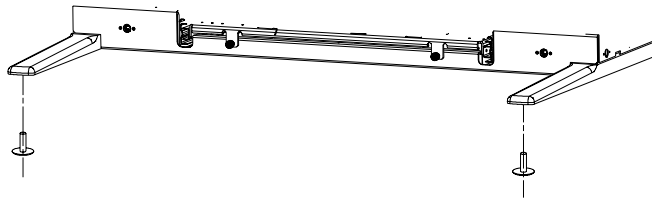
Figure 1-65 Console ready to be unloaded



- 8.) Move the console to the installation location.

- 9.) Attach the two pads to the console stabilizers. See [Figure 1-66](#).

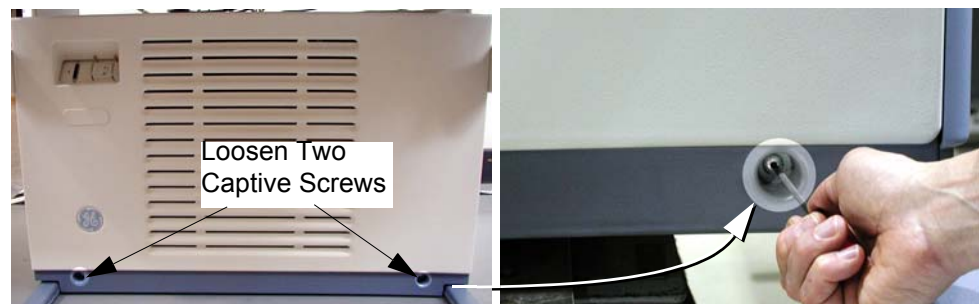
Figure 1-66 Console Stabilizer Pads



- 10.) Screw the pads all the way into the legs

9.2 Remove Console Covers

Figure 1-67 Front Cover Removal - Global Consoles



- 1.) From the service desktop, shut down the system.
- 2.) Turn off the console power.
- 3.) Loosen two captive screws at the bottom of the console.
- 4.) Rotate the bottom of the cover outward and upward until it can be lifted free of the console at the top.

9.3 Adjust Table Top Height and Position Console

- 1.) The console normally arrives with the bottom hole on the monitor top aligned with the fourth bolt from the bottom. See [Figure 1-68](#).

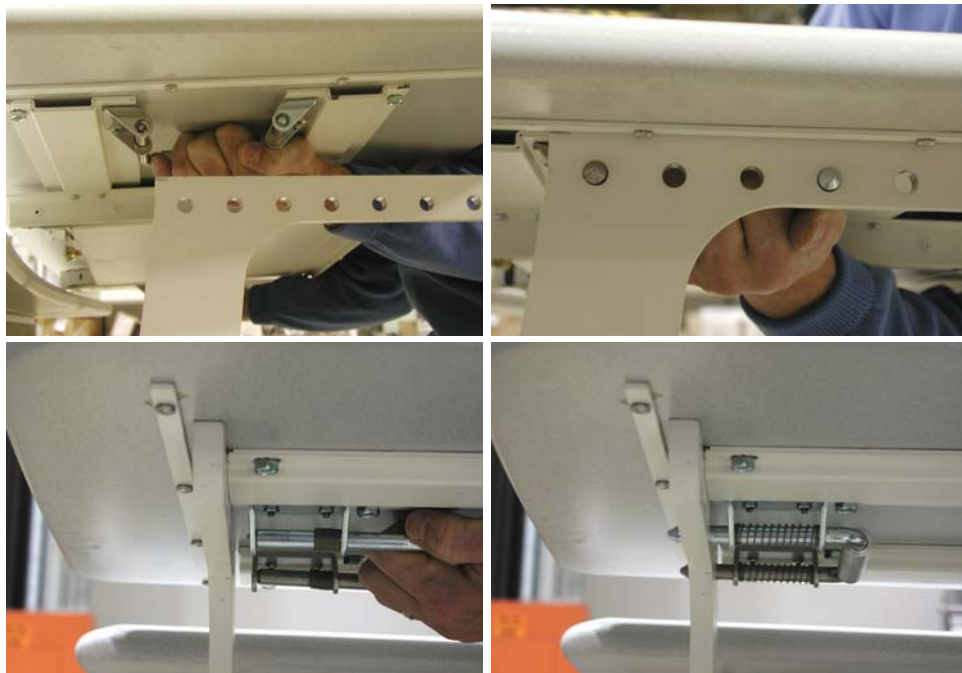
Figure 1-68 Global Console tabletop adjustments



- 2.) For optimum operator comfort, align the bottom hole of the keyboard table top with the fifth hole from the bottom on the console.
Your site may have different requirements and you may have to adjust the monitor top and/or keyboard table top up or down from this position.
 - Always select a table top height that permits the operator to see the patient on the table.
 - Keep the one bolt-hole relationship between the monitor top and the keyboard table top. Fasten the keyboard table top one (1) hole lower than the console table top (Figure 1-68).
- 3.) Install the console side covers. Side covers are left and right specific.
- 4.) Move the console into its final position, maintaining a 15 cm (6 in) minimum distance between the console and the wall.

9.4 Attach Keyboard Table Top

Figure 1-69 Keyboard Table Top Placement



- 1.) Locate the spring-loaded latches at either end of the underside of the keyboard table top.
- 2.) Pull the latches inward (toward center of table top).
- 3.) Position the table on the brackets so that the latches align with the holes.
- 4.) Release the latches, making certain that the pins fully engage the bracket holes (pins should protrude beyond outside edge of bracket).

9.5 LCD Monitor and SCSI Tower Placement

- 1.) Locate and unpack the two LCD monitors.
- 2.) Carefully place the monitors on the console desktop.
- 3.) Locate and unpack the SCSI drive tower.
- 4.) Place the SCSI drive tower on the console desktop. Suggested placement appears in [Figure 1-70](#).

Figure 1-70 Global Console Tabletop Component Placement



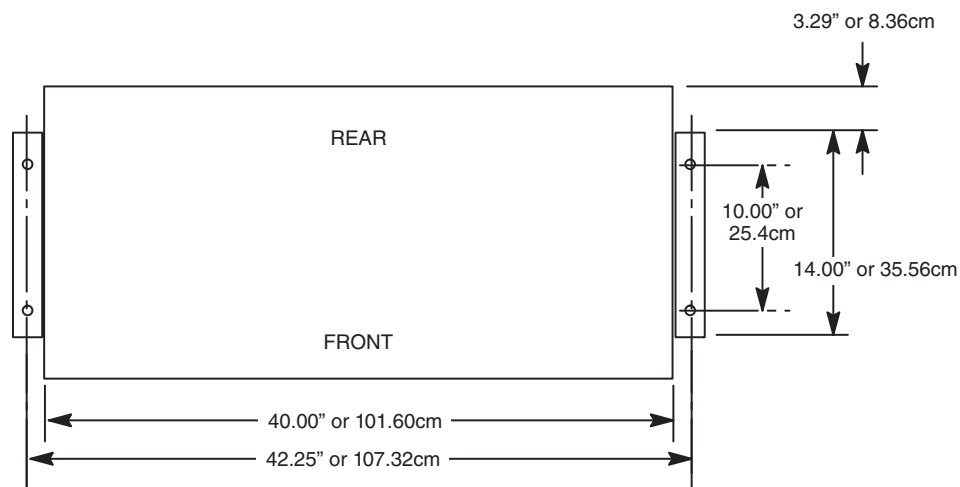
Section 10.0 Seismic Mounting

The Seismic Kit is part number R4390JC.
A seismic kit is also available for the SCSI Tower.

10.1 Console

If site specifications require seismic mounting, use ½ in. bolts to mount the brackets to the floor. Refer to [Figure 1-71](#) for hole placement. The console seismic brackets are shipped with the console. They are used on the console shipping skid.

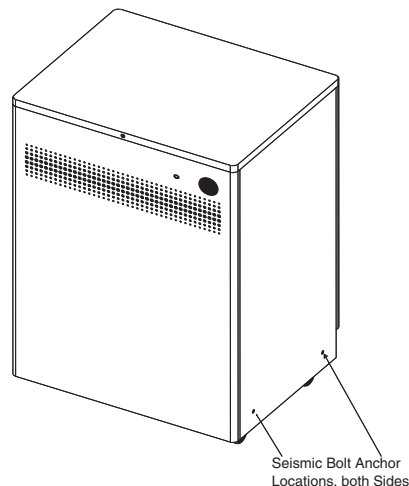
Figure 1-71 Seismic Console Mounting Hole Locations



10.2 Power Distribution Unit

If site specifications require seismic mounting, use the PDU seismic brackets that were shipped with the seismic kit. Refer to [Figure 1-72](#) for hole placement.

Figure 1-72 Seismic PDU Mounting Hole Locations



Chapter 2

Power, Ground & Interconnect Cables



NOTICE Potential for Data Loss and/or Equipment Damage

To prevent potential data loss and equipment damage, please do the following:

- When instructed, record data collected from the procedures in this chapter on GE Form e4879. For more information about this form, see [Section 8.0](#) of Chapter 4.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Introduction

Site use of conduit, floor duct, wall duct, or a raised computer floor, as well as the individual component layout, determines the system cable sequence. If your site has floor or wall ducts that interfere with placement of the table/gantry, it may be important to have the movers unload the cable boxes (8 and 9) first and run those cables while others unload the subsystems.

- Try to run the system cables after the contractor completes the contractor-supplied wiring.
- All ground wires and other contractor wiring should be complete to the point of equipment placement.

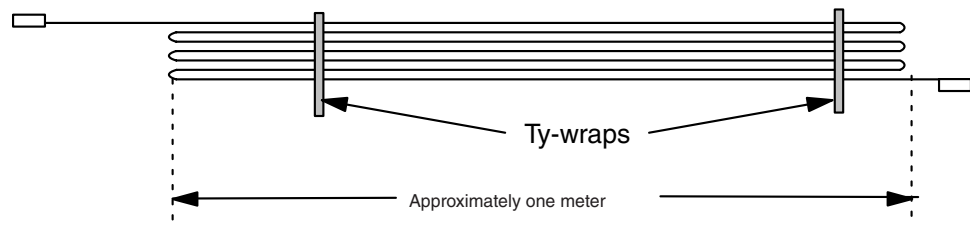


NOTICE Potential for Equipment Damage

Do not store excess cable in the bottom of the PDU or gantry.

When possible, store excess cable length in a serpentine configuration, approximately one meter long. (**Do NOT coil excess cable.**)

Figure 2-1 Excess Cable Storage Configuration



- Keep signal and control cables away from power cables and power wiring. When you lay cables in a raceway, locate the signal cables in a separate section of the raceway, or a separate conduit.
- Check all connections for tightness.
 - Use suitable tools and judgment.
 - Check all visible connections, especially ground connections.
- Check for reasonable cable routing.
 - Take into consideration necessary take-up distances for equipment maintenance, etc.
 - Try to complete as neat a job as possible.

1.1 System Component Identification

Identify all system cables by the system component designators listed in Table 2-1 on page 82. Each end of a system cable has a label, and may have a color near the connector (see Table 2-2 on page 82) to indicate the component and the jack identifier of the component.

Table 2-1 System Component Identifiers

DESIGNATOR	SYSTEM COMPONENT
CT2	Gantry
CT1	Patient Table
PM	Power Distribution Unit
OC	Operator Console (Console Computer)
WL	X-Ray ON Warning Light

1.2 Cable Color Identifiers

The ends of the cables may be marked with a piece of blue, yellow, red, or orange colored tape to help with the cable installation. Table 2-2 on page 82 lists the subcomponent and corresponding color.

Table 2-2 Cable Color Identifiers

SUBCOMPONENT	COLOR
Gantry	Blue
Table	Yellow
PDU	Red
Console Computer	Orange

Table 2-3 LightSpeed 5.X System Interconnect Cables

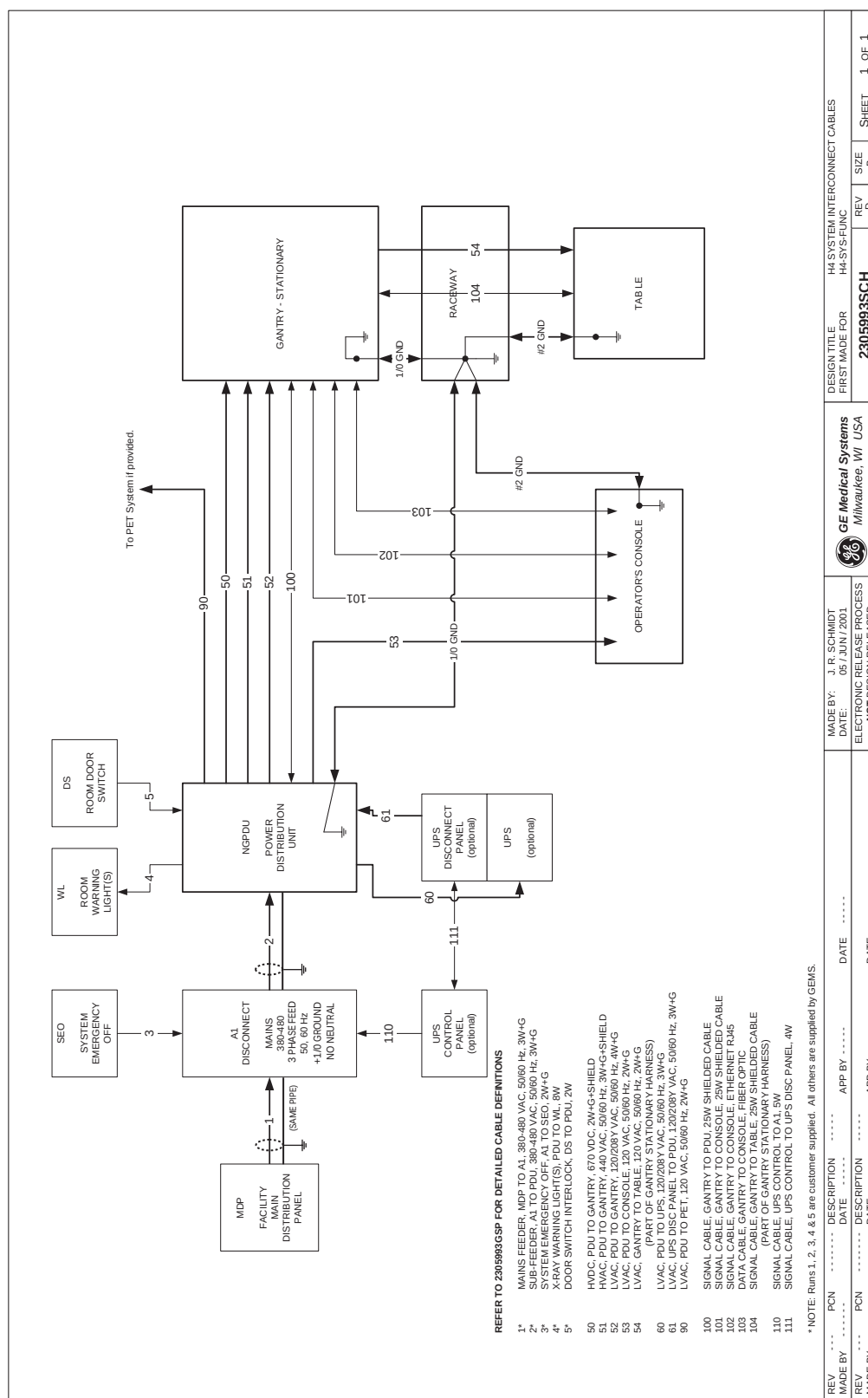
RUN NO.	DESCRIPTION	PART NUMBER	
		Long Cables (Kit 2281840-4)	Short Cables (Kit 2281840-5)
1	Facility MDP to Room Disconnect (A1)	cust. supplied	cust. supplied
2	Room Disconnect (A1) to PDU	cust. supplied	cust. supplied
3	Room Disconnect (A1) to System E-Off	cust. supplied	cust. supplied
4	PDU to Room Warning Light(s)	cust. supplied	cust. supplied
5	PDU to Scan Room Door Switch	cust. supplied	cust. supplied
50	HVDC Power Cable - PDU to Gantry	2343529	2343529-2
51	HVAC Power Cable - PDU to Gantry	2343530	2343530-2
52	LVAC Power Cable - PDU to Gantry	2343528	2343528-2
53	LVAC Power Cable - PDU To Operator's Console	2343531	2343531-2
54	LVAC Power Cable - Gantry to Table	n/a	n/a
55	Ground, PDU to Raceway	2371450	2371450-2
56	Ground, Raceway to Console	2371450-3	2371450-4

Table 2-3 LightSpeed 5.X System Interconnect Cables (Continued)

RUN NO.	DESCRIPTION	PART NUMBER	
		Long Cables (Kit 2281840-4)	Short Cables (Kit 2281840-5)
60	LVAC Power Cable - PDU to Optional UPS	-	-
61	LVAC Power Cable - UPS Disconnect Panel to PDU	-	-
90	LVAC Power Cable - PDU to PET	-	-
100	Signal Cable - Gantry to PDU	2333152	2333152-2
101	Signal Cable - Gantry to Console	2333150	2333150-2
102	Signal Cable (Ethernet) - Gantry to Console	2352714-2	2352714-3
103	Data Cable (Fiber Optic) - Gantry to Console	2117848-2	2117848-7
104	Signal Cable - Gantry to Table	n/a	n/a
110	Signal Cable - UPS Control to Room Disconnect (A1)	-	-
111	Signal Cable - UPS Control to UPS Disconnect Panel	-	-

Section 2.0

System Interconnect Diagram



Section 3.0

Contractor Connections

Table 2-4 Contractor PDU Connections

CONNECTION OR WALL BOX	AWG #	CONNECTION FROM	CONNECTION TO PDU	INSTALLED AND CHECKED
A1	#1	Load - T1	TS-1 L1	
	#1	Load - T2	TS-1 L2	
	#1	Load - T3	TS-1 L3	
	#1/0	GND	TS-1 GND (Do NOT connect anything to neutral point.)	
WL (Warning light) <i>See Figure 2-26, on page 103.</i>	#14	LV Source -1	TS6 1	
	#14	LV Source -2	TS6 2	
	#14	X-Ray ON Light -1	TS6 3	
	#14	X-Ray ON Light -2	TS6 4	
	#14	Sys-ON Light -1	TS6 5	
	#14	Sys-ON Light -2	TS6 6	
	#14	Ready Light -1	TS6 7	
	#14	Ready Light -2	TS6 8	
DS (Scan Room Door Switch)	#14	Door SW-1	TS6 9	
	#14	Door SW-2	TS6 10	

Note:
Add #2 ground wire.

IMPORTANT: Add AWG #2 ground wire from Table frame to Table/Gantry raceway ground bar (as shown in [System Interconnect Diagram](#)).



WARNING



WORK WITH THE ELECTRICAL CONTRACTOR TO BE SURE EXTERNAL POWER SOURCE IS TURNED OFF.

Section 4.0 Console Connections

4.1 SCIM, Keyboard, Trackball & Mouse Installation

- 1.) Route the keyboard cable under the SCIM, as shown in [Figure 2-2](#).

Figure 2-2 SCIM control with keyboard cable routed through SCIM



NOTICE

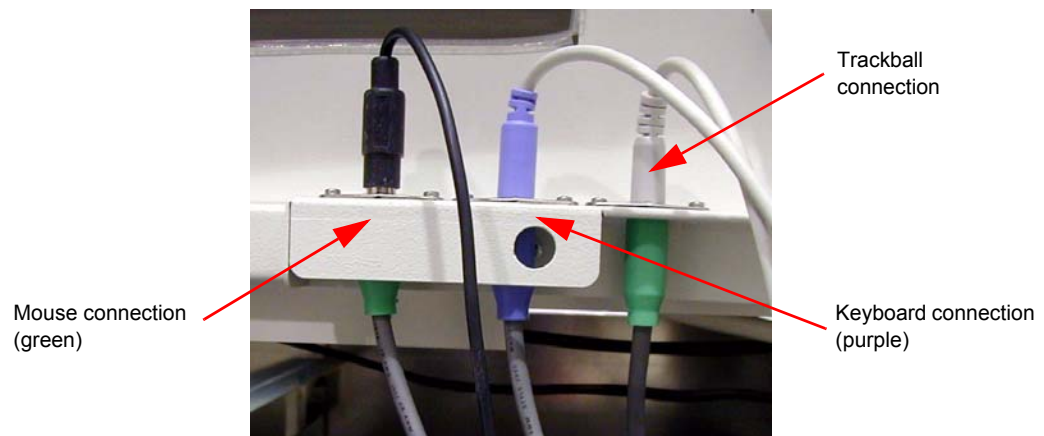


Potential for equipment damage

Never connect a mouse or keyboard with the host computer powered “ON”. Doing so can destroy components within the host computer.

- 2.) The keyboard and mouse connect to the ports on the top of the console front bulkhead. The keyboard connects to the port on the right, the mouse connects to the left port ([Figure 2-3](#)).

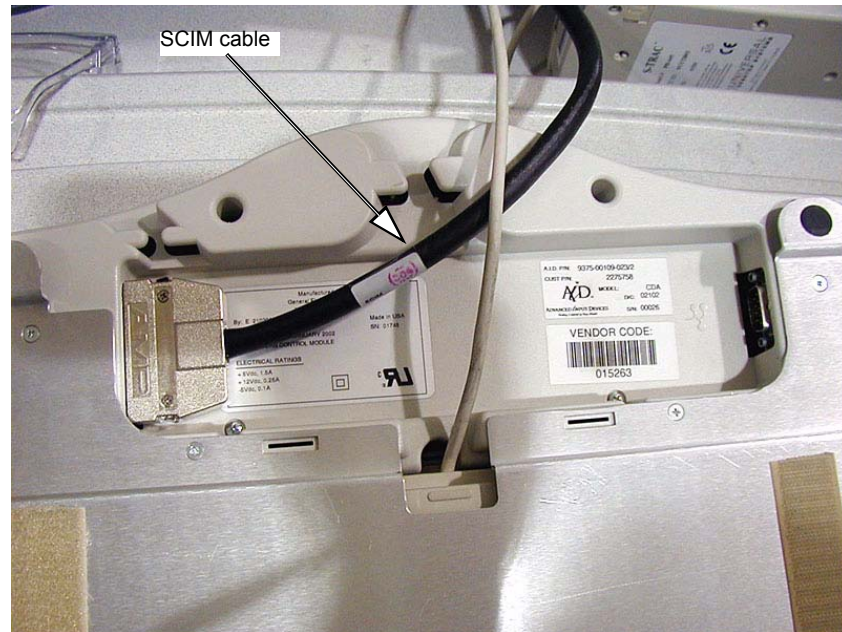
Figure 2-3 Global Console front bulkhead, showing keyboard and mouse connections



- 3.) Connect the SCIM cable to the SCIM as shown in [Figure 2-4](#) (note the cable routing).

Note: Make sure the SCIM connector fits snugly. Some molding may need to be removed to allow the cable to fit snugly.

Figure 2-4 SCIM bottom, showing cables and keyboard mounting bracket



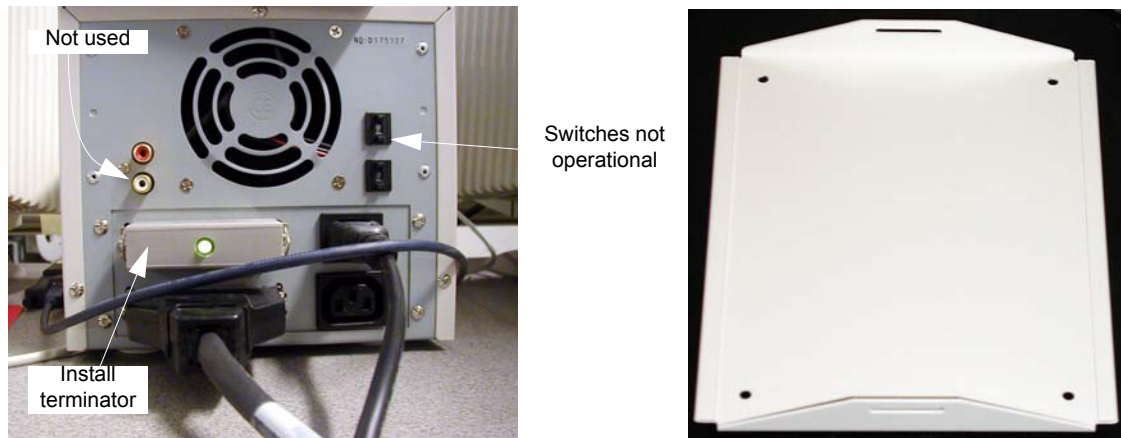
- 4.) Select and install the proper overlay for your system: (1) with Tilt or (2) without tilt. Verify that none of the buttons get caught and stuck under the overlay. Pay close attention to the prescribed tilt button on systems with the tilt feature.
- 5.) The keyboard should attach to the SCIM using the supplied Velcro strip and fit snugly against the SCIM when finished, as shown in [Figure 2-5](#).

Figure 2-5 SCIM connected to the keyboard with the US English tilt overlay installed



4.2 Connecting the SCSI Tower

Figure 2-6 SCSI tower: Connections (left) and Optional Seismic Mounting Bracket (right)



- 1.) Connect the SCSI cable to the rear of the SCSI tower.

Note: **The cable is included in the SCSI Cable Kit and is the same as the DASM cable.**

- 2.) Connect the SCSI terminator to the rear of the SCSI tower.
- 3.) Connect the power cable to the rear of the SCSI tower.

4.3 Connecting the LCD Monitor

With the monitor and computer switched off, connect the video signal cable to the monitor's video input.

NOTICE Equipment Damage Possible



Do not touch the video signal cable connector pins as this might bend them. When connecting the video signal cable, check the alignment of the HD15 connector. Do not force the connector in the wrong way, otherwise the pins might bend.

Connect the following cables:

SCIM:

- SCSI cable
- Template
- Keyboard

Scan Monitor:

- Video cable
- Power cable
- Cable keeper

Display Monitor:

- Video cable
- Power cable
- Cable keeper

Note: A DARC video cable may be shipped with the system. This is only to be connected when troubleshooting. Do not connect during normal installation.

4.4 Power Panel Connections

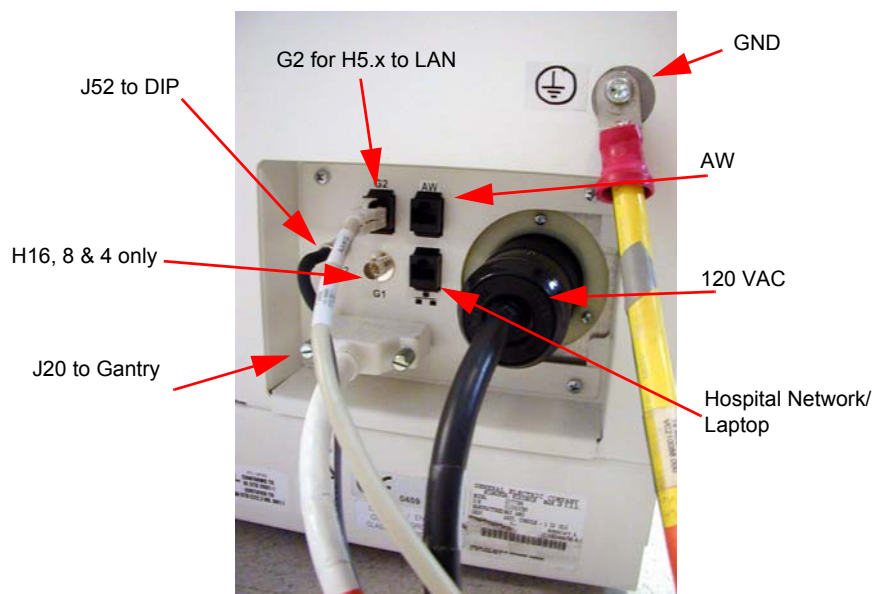
Note: Console power is single phase power. Outlet assigned is not critical.

- 1.) Connect the console power cable to the console power panel.
- 2.) Connect console component power cords as listed in [Table 2-5](#). ("J numbers" increment from top to bottom, left to right)

Table 2-5 Power Panel Outlet Assignments - GC-GRE

J#	DEVICE	J#	DEVICE
J1	Video Splitter	J9	Host Computer
J2	Remote Monitor	J10	Power Monitor
J3	Modem	J11	DARC
J4	IG 3	J12	Power Monitor
J5	IG 1	J13	
J6	IG 2	J14	Power Media Conv
J7	SDDA	J15	Left Fan
J8	DASM	J16	Right Fan

Figure 2-7 Power Connections

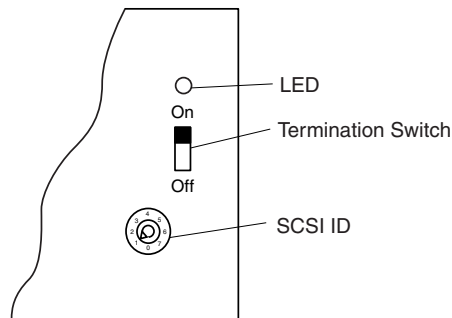


4.5 Installing a DASM (Hardware Option)

If you have a digital DASM to install, complete the following steps to connect it to the computer.

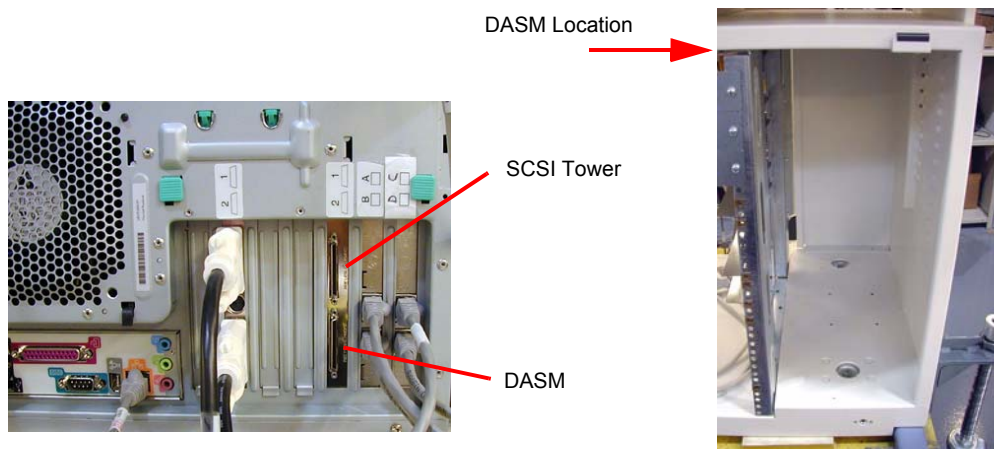
- 1.) Turn the DASM over so you are looking at the bottom.
- 2.) Set the DASM SCSI ID to 1.
- 3.) Set the DASM termination to ON.

Figure 2-8 DASM Bottom



- 4.) Set the RS232/RS422 switch to the RS232 position, for a digital DASM.
- 5.) Refer to [Figure 2-9](#). Attach the SCSI cable supplied with DASM kit to the back of the DASM.

Figure 2-9 Rear view of HP xw800, showing DASM SCSI cable connection



- 6.) Attach the free end of the SCSI cable to the rear of the Host Computer (see [Figure 2-9](#)).
- 7.) Plug DASM power cord into J10 (or any open slot).
- 8.) Attach remaining cable(s) from camera to DASM and install the DASM to the right of the slide-out tray.

Note: See your *Software Load Procedures* manual for instructions on configuring cameras.

4.6 Modem Option

If you have a modem to install, do it now. Place the modem on top of the outlet box and attach it with velcro to the power supply.

Hook up the power, phone, and serial line as shown in drawing 2378862. The drawing is located in the back cover of the console.

Figure 2-10 Modem

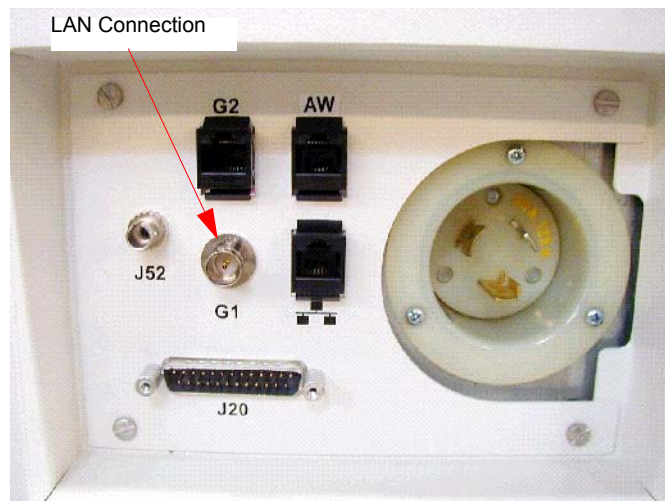


4.7 LAN Connections

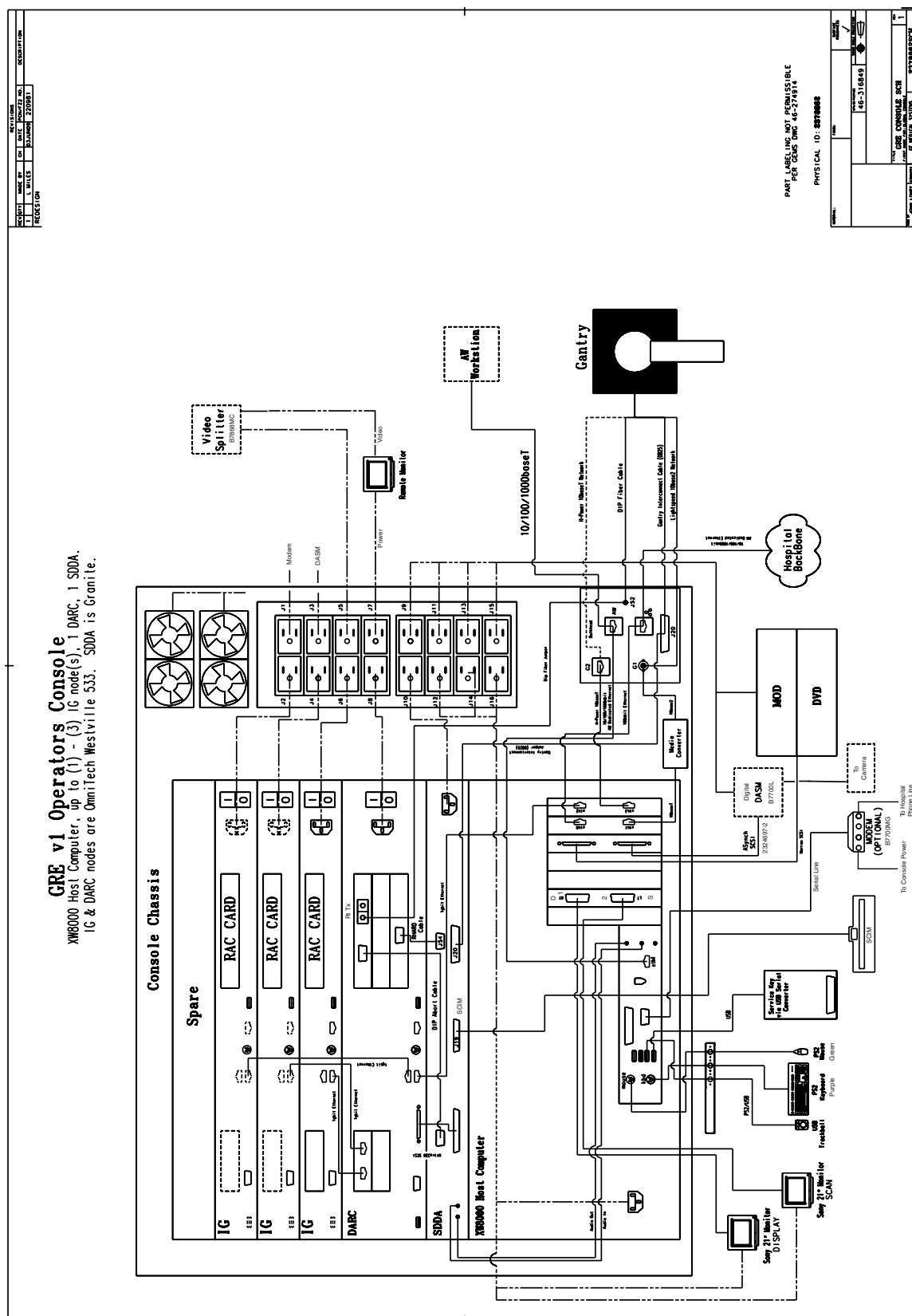
Plug LAN cable (2266887) into G1 of rear bulkhead on console.

Note: G2 is used for LightSpeed Pro¹⁶ only.

Figure 2-11 LAN Connection



4.8 GRE Interconnect



Section 5.0

Install Options

5.1 Install Optional Bar Code Reader

Follow the installation procedures in the Bar Code Reader box.
When finished neatly dress all cables.

5.2 Install Optional Remote Monitor

Follow the installation procedures in the Remote Monitor box.

5.3 Install Cardiac Respiratory Gating Option

Follow the instructions shipped with the option. The options board should be only mounted on the non-motor side of the gantry. Neatly dress all cables along the gantry base so that the base covers fit properly.

5.4 Install Respiratory Gating Option

Follow the instructions shipped with the option. The option is attached to the GOB, and should be only mounted on the non-motor side of the gantry. Neatly dress all cables along the gantry base so that the base covers fit properly.

5.5 Install Injector Option

Follow the instructions shipped with the option.

5.6 Install The IVY Monitor and Stand

Follow the instructions shipped with the Monitor and stand kit. Review the instructions carefully before assembling the stand and accessory basket to avoid repeated steps.

5.7 Customer Accessories (Head Holders and Extender)

Open the boxes and installed the appropriate language warning labels.

The head holders ship with shims installed to assure proper fit. Check that shims are included and a pair is installed. The holder should fit snugly. Follow the procedure in [Section 4.0](#) of Chapter 3 to install Head Holders.

5.8 Install Service Cabinet

The service cabinet you receive may ship disassembled. Assembly takes about 1.5 hours.

- 1.) Assemble the cabinet following the instructions located in the cabinet's shipping box.
- 2.) When you complete assembly, place the cabinet in the location shown on the site print.
- 3.) Place all service materials shipped with the system in the service cabinet.

5.9 Install UPS

Follow the instructions shipped with the UPS option. The option ships with two sets of instructions, a GE set and a Powerware set.

The GE set instructs you to install the connections between the UPS and the PDU and between the UPS and the A1.

Note: A GE A1 Disconnect with UPS controls is required for this option.

The Powerware set instructs you to internally connect the batteries and do a power-up check. Refer to both manuals for additional guidance.

WARNING LOCKOUT/TAGOUT IS REQUIRED WHEN WORKING IN THE A1 DISCONNECT.

Section 6.0

Gantry Cable Connections

Please refer to [System Interconnect Diagram](#), for complete system interconnect details.

Table 2-6 Gantry Cable Connections

TO	FROM	CABLE DESCRIPTION
Gantry Power Pan	PDU	HVDC
Gantry Power Pan	PDU	440VAC
Gantry Power Pan	PDU	120VAC
Gantry Power Pan	Console	Fiber*
Gantry Power Pan	Console	LAN
MSUB (J9)	Console	Control
MSUB (J11)	PDU	Control

**Take extreme care when installing the fiber optic DAS data cable. Do not step on, kink, or sharply bend this fragile DAS cable.*

NOTICE



- 1.) If using a rear cable entry box (B7850RC), install it now, before routing cables to Gantry.

Potential for equipment damage.

Observe correct polarity when connecting the high voltage DC power. Reversing these leads will result in serious equipment damage. The HVDC positive conductors have red insulation and are labeled “ONE.” The HVDC negative conductors have black insulation and are labeled “TWO.” Lead “ONE” must be connected to lead “ONE,” and lead “TWO” must be connected to lead “TWO.”

Observe correct phase rotation when connecting the axial motor power. Phases one, two and three should be connected top to bottom.

- 2.) Install the cables to the Gantry power pan. The power pan is located on the rear of the Gantry at its base. See [Figure 2-12](#), [Figure 2-13](#), and [Figure 2-15](#) for connections.

Note:

The Gantry 120VAC cable may not fit under the Gantry frame. Install this cable before Gantry placement—or remove the power plug—to route it under the Gantry.

Figure 2-12 Gantry Power Pan Connections

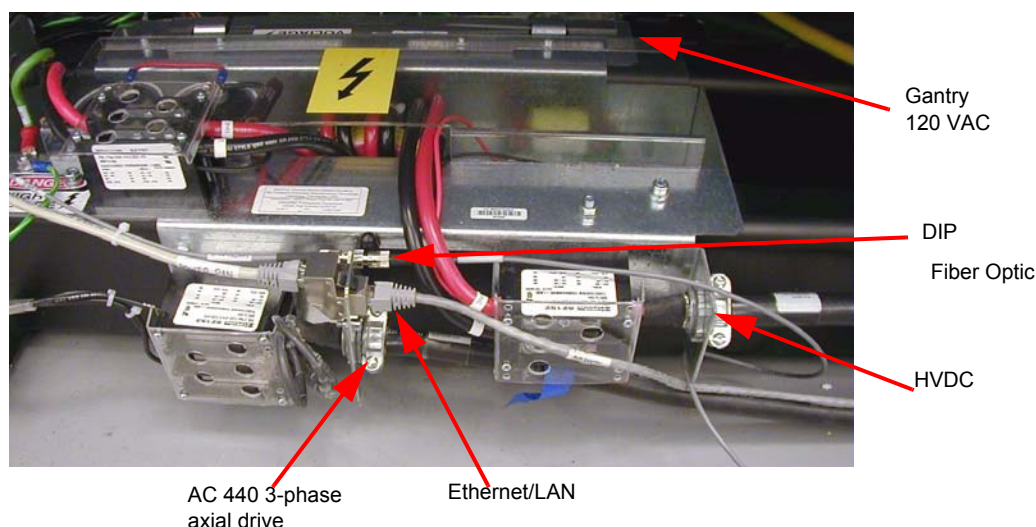
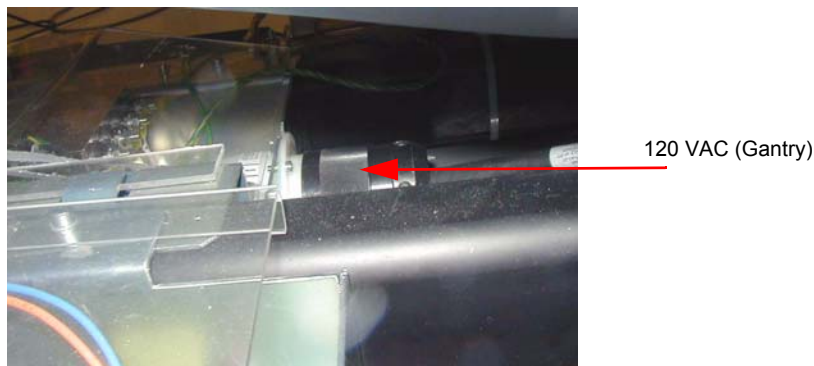


Figure 2-13 Gantry Power Pan



- 3.) If the system has a second CAT5 LAN cable for the cardiac option, run this cable between the console and the gantry.
- 4.) Connect this cable to the gantry base cardiac monitor base plate (GOB board). Different plates are included with the cardiac option kit.

NOTICE



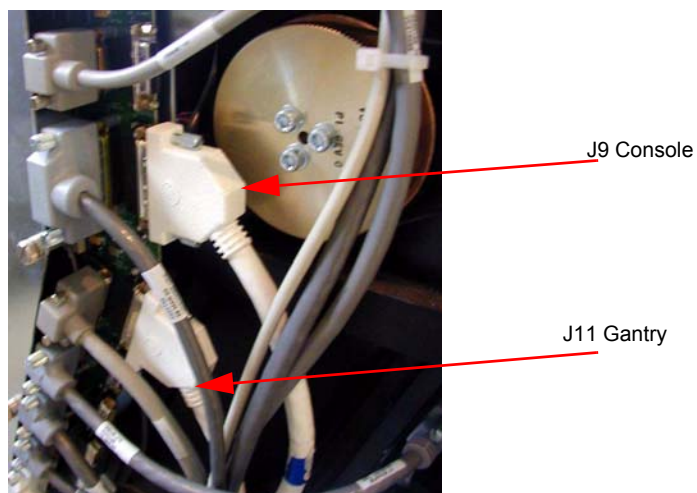
The GOB connectors are not soldered to the board. DO NOT use the cable to pull on the board.

Figure 2-14 Cardiac Option Board



- 5.) Install cables to the Gantry TGPU.

Figure 2-15 TGPU Connections



Section 7.0

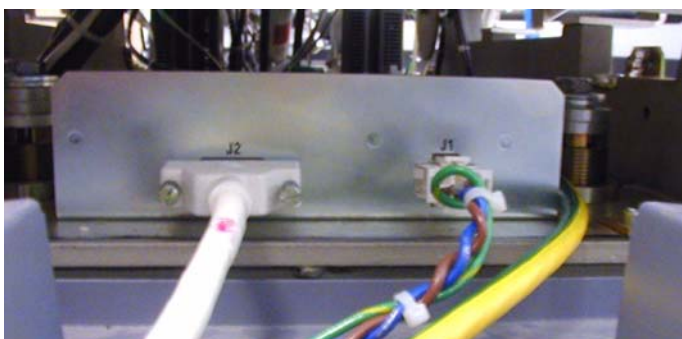
Table Connections

Pull and connect the following cables:

Table 2-7 Cables Connected to Table

J#	CABLE DESCRIPTION
J1	120 VAC
J2	Signal Cable

Figure 2-16 Table Bulkhead Connections



Section 8.0

PDU Cable Connections & Configuration



CAUTION

Do not work in an energized PDU. When working on the PDU, follow this simple rule: Always tag and lock out power to the PDU at the “main” disconnect. Failure to do so can result in electrocution or death.

Do apply power to the PDU until all work has been completed and all PDU covers are in their proper place.

8.1

NGPDU

As seen in [Figure 2-17](#), a number of cables must be installed throughout the PDU. Specific details on each connection can be found in the sub-sections that follow. Use [Figure 2-17](#) for reference. The PDU has been designed to have cables routed into the PDU from its behind and/or from beneath it.

Figure 2-17 PDU Cable Connections - Front

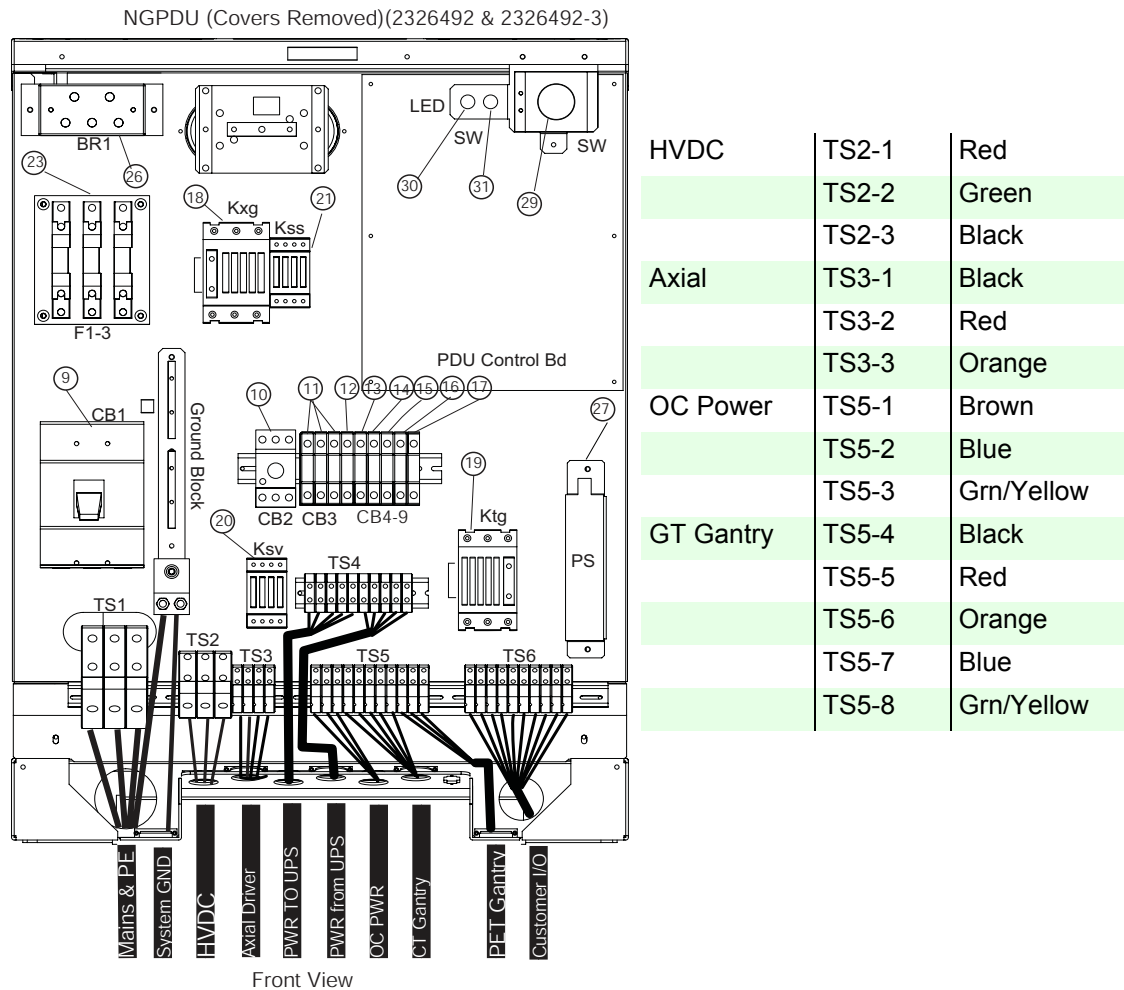
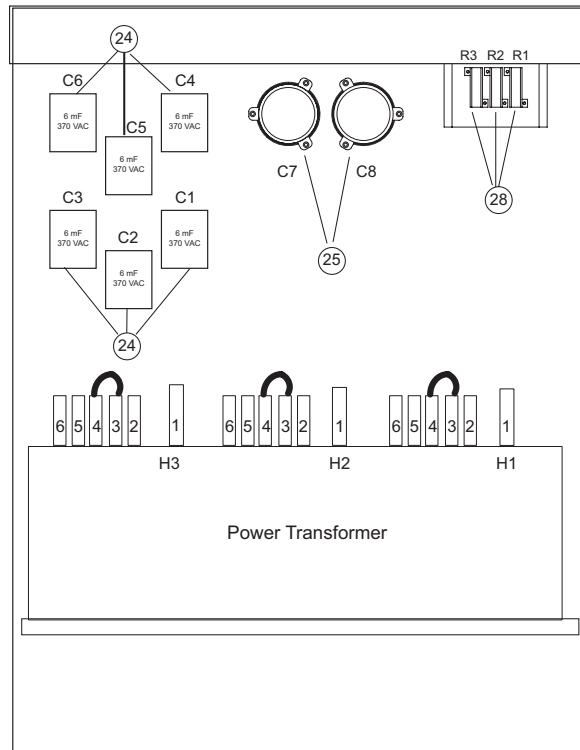


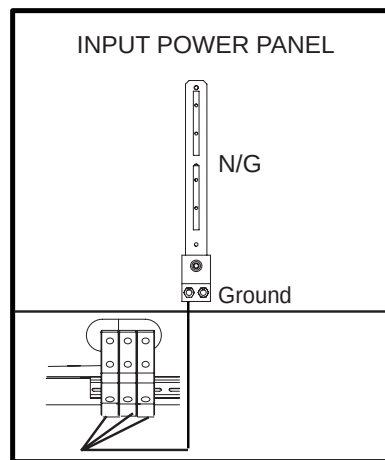
Figure 2-18 PDU Cable Connections - rear



8.1.1 Panel - 380 - 480VAC Mains “TS1” Input Power Connection

- 1.) Remove the TS3 panel front cover.
- 2.) Strip the wires to fit securely on the power block.
- 3.) Observe incoming phases (L1, L2 and L3) and insert bare leads into power block.
- 4.) Insert “vault” ground into PDU “vault” ground lug.
- 5.) Tighten all fasteners securely and replace the TS3 front panel.

Figure 2-19 Input Power Panel Connections

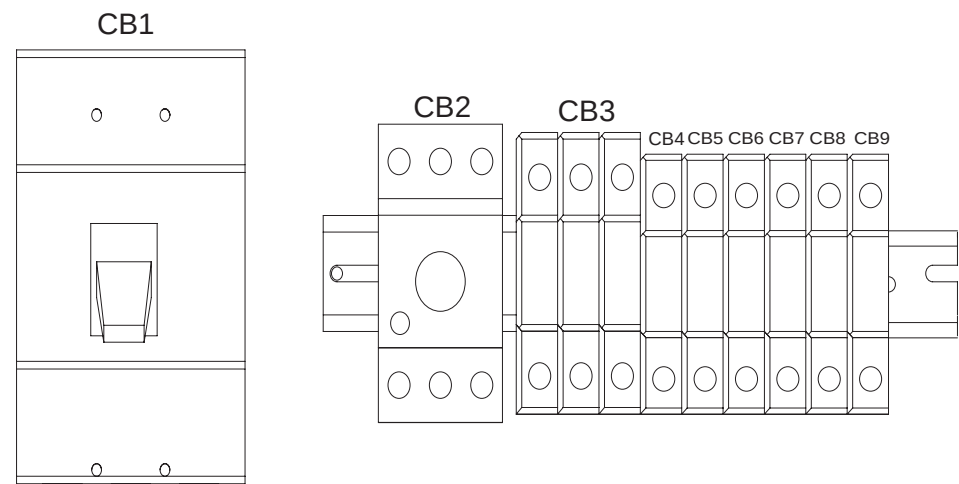


When Mains power is available to the PDU, the “TS1” power light will be illuminated.
(See [Figure 2-17.](#))

8.1.2 Panel - Circuit Breakers

Place the circuit breakers in the “off/down” position during installation, even with Mains incoming power tagged and locked out. After you have completed work on the PDU, you may return the circuit breakers to the “ON” positions.

Figure 2-20 Circuit Breaker Panel



By design, when CB3 is in the “OFF” position, circuit breakers 4, 5, 6, and 7 are switched “OFF”. CB3 is essentially in series with these breakers.

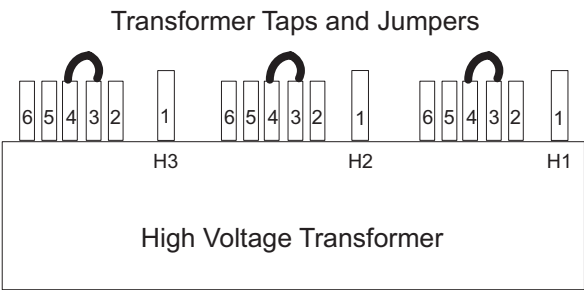
Table 2-8 Panel Circuit Breaker Descriptions

CIRCUIT BREAKER	DESCRIPTION
CB3	Fully Winding Protection (Master power of CB 4, 5, 6, and 7)
CB4	CT Gantry Service Outlets
CB5	CT Gantry rotating loads
CB6	Table & CT Gantry Station Loads
CB7	Operator Console
CB8	PET Gantry
CB9	NGPDU Control Power Supply

8.1.3 Transformer (480VAC) Taps

Verify that the transformer taps are set properly. The transformer taps are set to 480 VAC operation at the factory. The taps should be set as shown in [Figure 2-21](#).

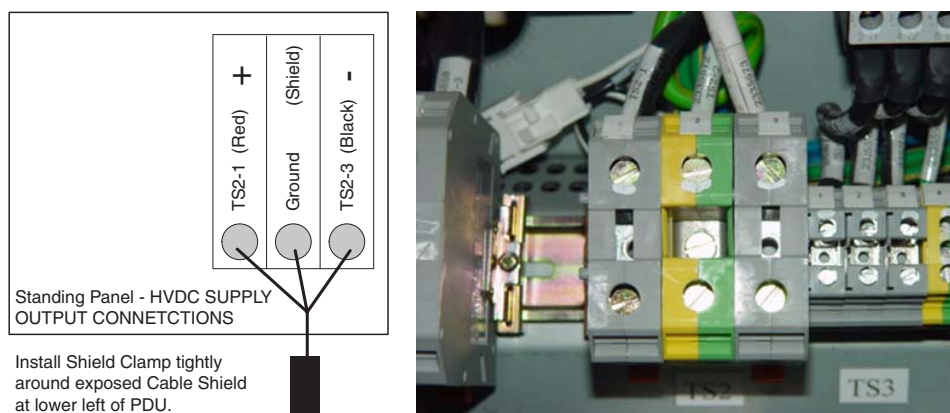
Figure 2-21 PDU Tap Positions for 480 volt operation



8.1.4 HVDC Connection

Connect the internally shielded HVDC cable to TS2 on the standing panel. See [Figure 2-17](#) for the location of the connector and [Figure 2-22](#) for details. Observe polarities and grounds.

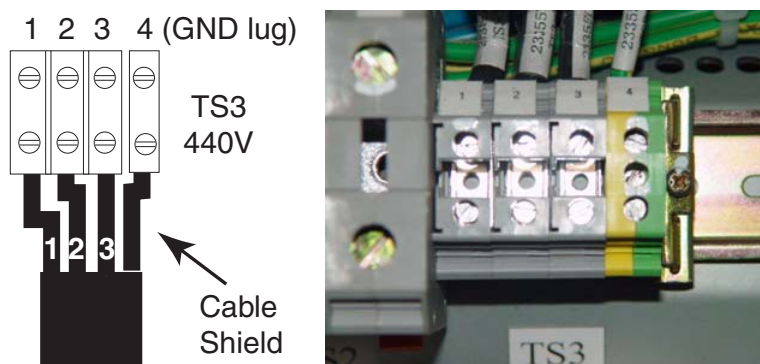
Figure 2-22 HVDC Connection



8.1.5 440V Connection

Connect the internally shielded 440V cable from the Gantry to TS3 on the panel. See [Figure 2-17](#) for the location of the connector and [Figure 2-23](#) for details. Observe the labels on the cable leads for proper identification and orientation.

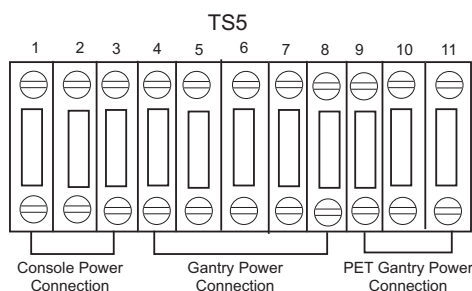
Figure 2-23 440VAC Connection



8.1.6 Gantry & Console Power Connections (120V)

Both Gantry and Console power cables come pre-terminated. Simply plug the Gantry power cable into "J4" and the Console power cable into "J5" as shown in [Figure 2-24](#).

Figure 2-24 Gantry & Console Power Connections



8.1.7 Console Power Cable Re-termination

Power cable re-termination should be used as a last result. Short and long cables are available.

- 1.) Carefully remove the power plug and **record the color of the wires** in [Table 2-9](#). The terminals are labelled X, W and G on the plug.

Table 2-9 Console Power Cable Termination

TERMINAL	X	W	G
Description	Hot	Neutral	Ground
Color			

- 2.) Cut the cable to desired length and dress ends.
- 3.) Re-install the power plug, according to the orientations recorded in [Table 2-10](#).
- 4.) Verify that less than 1 ohm of resistance exists between the following connections:

Table 2-10 Resistance Verification Points

FROM PDU PLUG	TO CONSOLE PLUG END	
CB1 -11 (Black) (J5 Phase X (Brown))	Phase X (Brown)	☐ Check box when complete
A3 Neutral Buss Bar (Blue) (J5 -13 Phase W)	Phase W (Blue) Neutral	☐ Check box when complete
A3 Ground Buss Bar (J5 -22 Ground Green)	Ground (Green) Ground Green screw	☐ Check box when complete

8.1.8 PDU Control Cable

The PDU control cable comes pre-terminated and should not be re-terminated in the field. Excess cable length must be stored. Simply plug the cable into “J2” on the “A4” panel. Secure it by using the fasteners intergrated into cable’s connector shell.

8.1.9 System Ground Connection

Connect the ground wire (green with a yellow strip) from the Table/Gantry raceway ground bus to the system ground lug in the PDU. See [Figure 2-17](#), on page 98, and [Figure 2-25](#), below.

Figure 2-25 PDU System Ground Connection

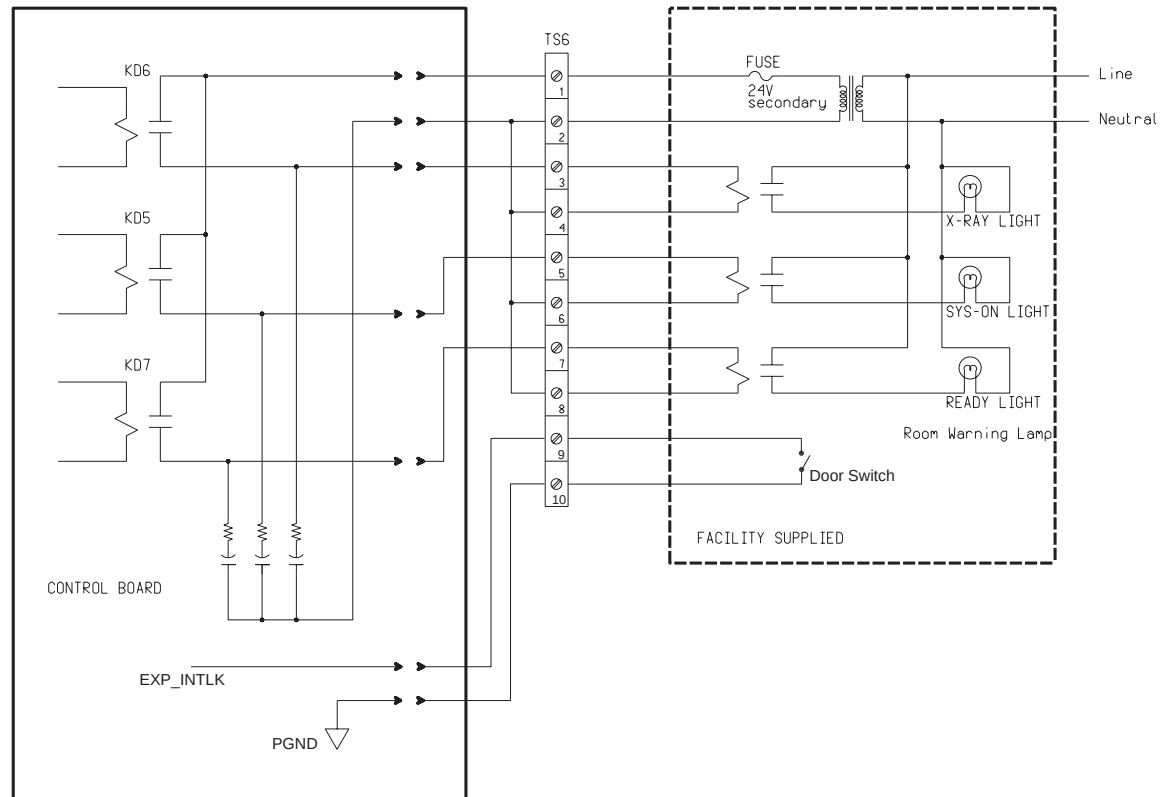


8.1.10 Warning Light & Door Interlock Connections

8.1.10.1 Warning Light Configuration & Connection

- 1.) Warning Light is controlled by signals from the system.
- 2.) This step is site specific. The PDU by default is configured for “no” external warning light connection. If you have external warning lights, see [Figure 2-26](#) for proper connection.

Figure 2-26 Typical TS6 Warning Light & Door Interlock Connections



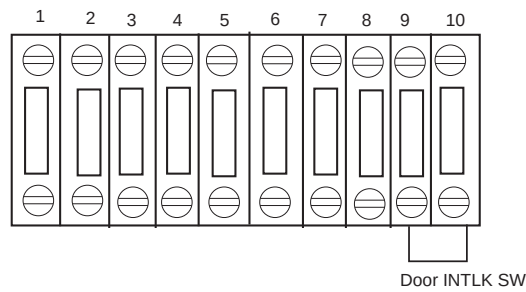
It is recommended that you use the four (4) wire method of adding a X-ray warning light to a room, as shown in [Figure 2-26](#). When using this method, you:

- Minimize EMC interference.
- Increase contact life of the relay used in the PDU.

8.1.10.2 Door Interlock Connections

Door interlocks are used to prevent X-Rays from being generated when the scan room door is open. The Door Interlock circuitry in the PDU is shipped from the factory engaged. This means the system cannot generate X-ray until disengaged. A short must exist between pins 9 & 10 for X-ray to be generated. Using a small piece of wire, short pins 1 and 2 together. See [Figure 2-27](#).

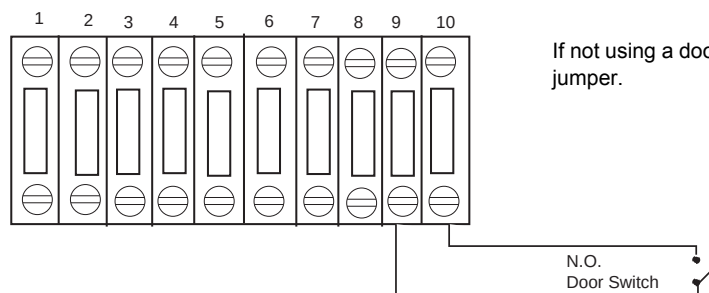
Figure 2-27 Without a Door Interlock



**If jumper is not in place,
exposures will not be made.
Check this jumper if you get
scan interlock errors.**

To use the system with a door interlock, wire a normally open switch between pins 1 & 2 that is attached to the interlock.

Figure 2-28 With a Door Interlock



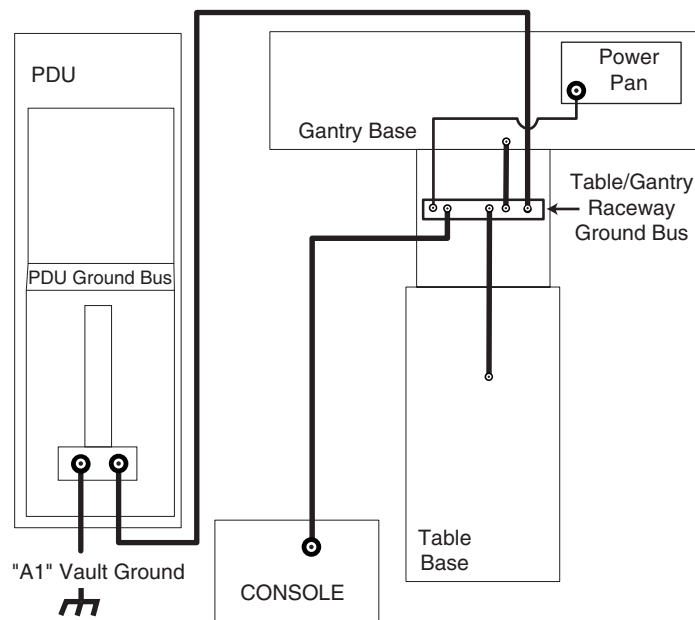
**If not using a door switch, add a
jumper.**

Section 9.0

System Ground Connections

As seen in [Figure 2-29](#), the Table/Gantry raceway ground bus is used to centralize all system grounding. The system ground is tied to vault ground at the PDU, through its chassis.

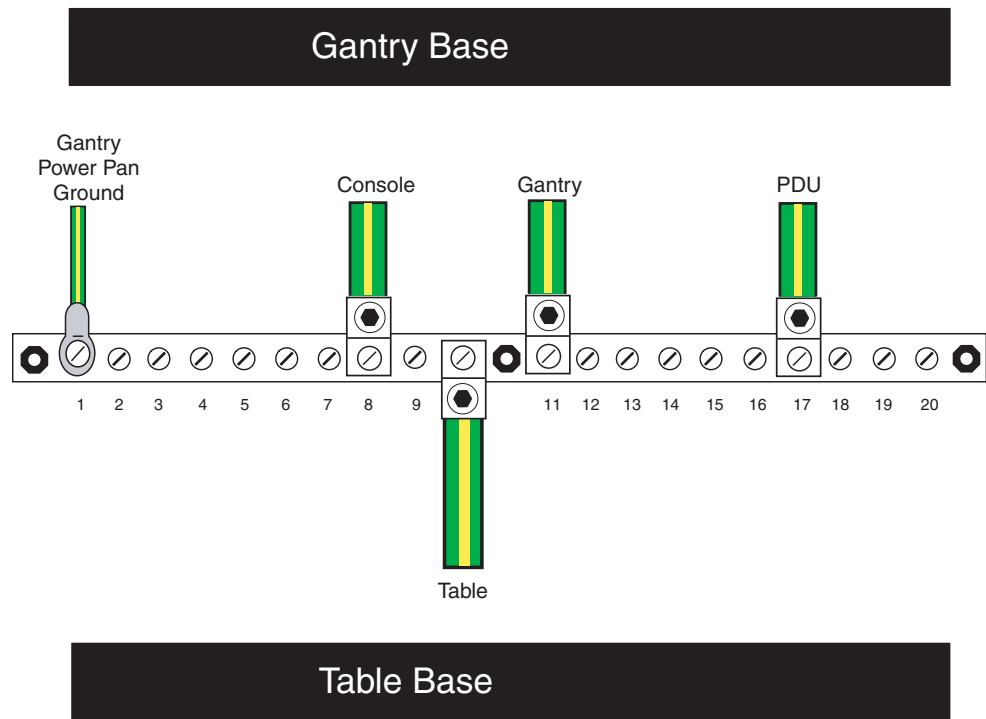
Figure 2-29 CT System Ground Connections



The Gantry is tied to system ground at a number of points. It is important that all of these ground connections are securely made. See [Figure 2-30](#).

Check that the terminals will not be loosened, by moving or swinging the cables rather strongly by hand.

Figure 2-30 Table/Gantry Raceway Bus - Grounds



Various types and sizes of wire are used to ground the system. Please use the type and sizes specified in [Table 2-11](#), below.

Table 2-11 System Ground Connections

AWG #	CONNECTION TO	CONNECTION TO
#1/0	PDU	Power Main
#1/0	Gantry (Power Pan)	Raceway
#2	Console	Raceway
#1/0	Gantry	Raceway
#2	Table (frame)	Raceway
#1/0	PDU	Raceway

Chapter 3

System Continuity & Ground Checks



NOTICE Potential for Data Loss and/or Equipment Damage

To prevent potential data loss and equipment damage, please do the following:

- When instructed, record data collected from the procedures in this chapter on GE Form e4879. For more information about this form, see [Section 8.0](#) of Chapter 4.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

System Continuity & Ground Checks (Mechanical Contractor Task)

Use this section to check cable and ground connections.

1.1 Tools Required

- Digital VOM
- 30 ft of #18 wire
- 600 VAC meter leads

1.2 Procedure

Reference [Figure 3-1: Front View of NGPDU, with Covers Removed on page 109](#) and [Figure 3-2: Gantry Power Pan on page 109](#).



WARNING



USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES; LOCK OUT WALL POWER.

- 1.) Remove all System Power at the A1 Mains Disconnect Panel. Follow Lockout/Tagout procedures.
- 2.) Put the UPS in the Service Position.
- 3.) Remove the PDU A3 input power panel cover.
- 4.) Verify, with a voltmeter, that mains power is disconnected.
- 5.) Verify that less than 1 ohm of resistance exists between the following ground connections:

Table 3-1 Mains Connections to PDU

FROM	TO	
Wall ground connection	PDU Cabinet	☐ Check box when complete

6.) Verify that less than 1 ohm of resistance exists between the following connections:

Table 3-2 Resistance Verification Points

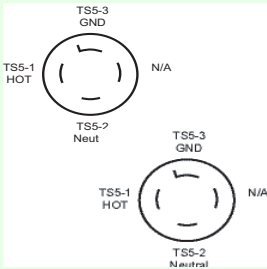
FROM	SIGNAL NAME (COLOR)	TO	
PDU TS2-1	+HVDC (Red)	Gantry HV Power Pan TS1-1	☐ Check box when complete
PDU TS2-2	HVDC Ground (Green/Yellow)	Gantry Power Pan Chassis	☐ Check box when complete
PDU TS2-3	-HVDC (Black)	Gantry HV Power Pan TS1-2	☐ Check box when complete
PDU Ground Bus	HVDC shield	Gantry HVDC cable shield	☐ Check box when complete
PDU TS3-1	Axial drive 440vac (Black)	Gantry HV Power Pan TS2-1	☐ Check box when complete
PDU TS3-2	Axial drive 440vac (Red)	Gantry HV Power Pan TS2-2	☐ Check box when complete
PDU TS3-3	Axial drive 440vac (Orange)	Gantry HV Power Pan TS2-3	☐ Check box when complete
PDU TS3-4	Axial drive ground (Green/Yellow)	Gantry Power Pan Chassis	☐ Check box when complete
PDU Ground Bus	Axial drive shield	Gantry 440 VAC cable shield	☐ Check box when complete
PDU TS5-1	120vac Phase A (Brown)	Console Power Plug: 	☐ Check box when complete
PDU TS5-2	120vac Neutral (Light Blue)		☐ Check box when complete
PDU TS5-3	Ground (Green/Yellow)		☐ Check box when complete
PDU TS5-4	120vac Phase A (Black)	Gantry LV Power Pan A1J1 Filter - L1	☐ Check box when complete
PDU TS5-5	120vac Phase B (Red)	Gantry LV Power Pan A1J1 Filter - L2	☐ Check box when complete
PDU TS5-6	120vac Phase C (Orange)	Gantry LV Power Pan A1J1 Filter - L3	☐ Check box when complete
PDU TS5-7	120vac Neutral (Light Blue)	Gantry LV Power Pan A1J1 Filter - N	☐ Check box when complete
PDU TS5-8	Ground (Green/Yellow)	Gantry Power Pan Chassis A1J1 Filter Ground Stud	☐ Check box when complete

Figure 3-1 Front View of NGPDU, with Covers Removed

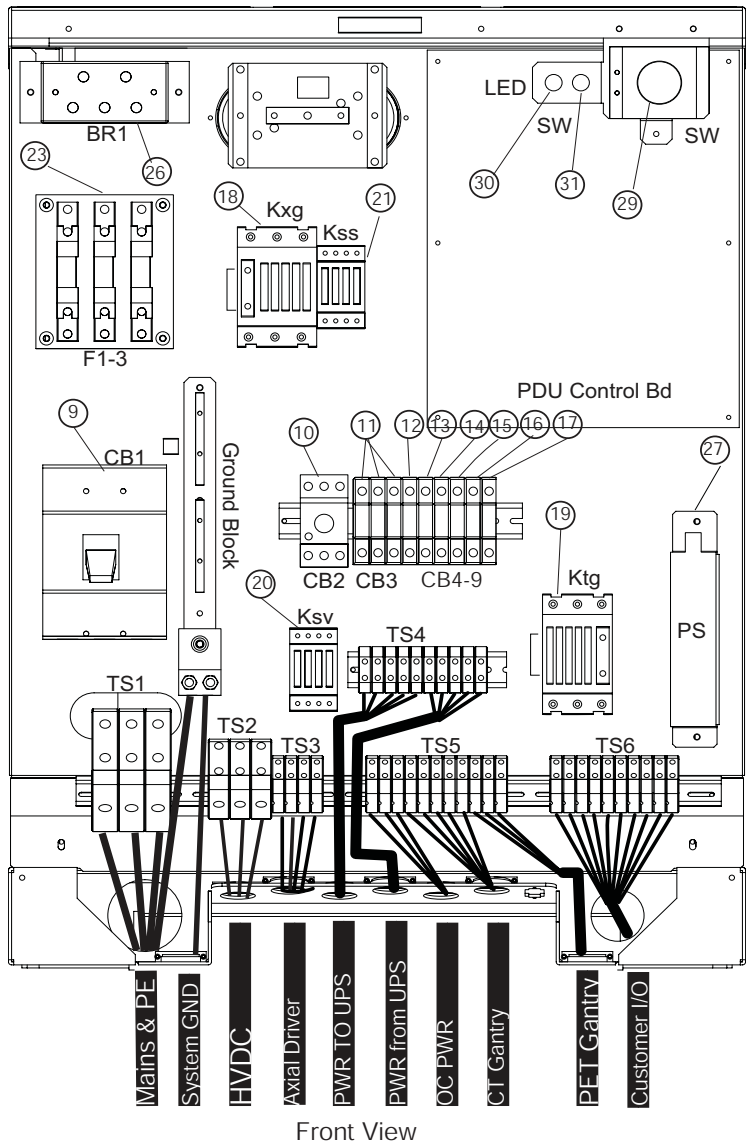
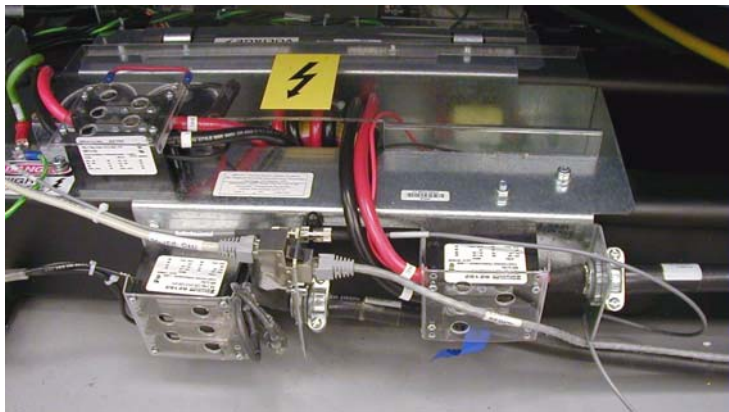


Figure 3-2 Gantry Power Pan





WARNING



TURN OFF ALL PDU CIRCUIT BREAKERS.

- 7.) Set an ohmmeter to the lowest scale. Check between the following points for shorts to ground. Verify no continuity exists between the following points:

Table 3-3 No Continuity Verification Points

FROM PDU	TO A1 BREAKER BOX	
TS2-1 (+HVDC) (Red)	vault ground	☐ Check box when complete
TS2-3 (-HVDC) (Black)	vault ground	☐ Check box when complete
TS3-1 (440vac output) (Black)	vault ground	☐ Check box when complete
TS3-2 (440vac output) (Red)	vault ground	☐ Check box when complete
TS3-3 (440vac output) (Orange)	vault ground	☐ Check box when complete

- 8.) Leave the metal cover off the PDU A3 input power panel until you complete the checks in the next section.

Section 2.0

Site Ground Continuity Check

- 1.) Use an ohmmeter to verify the presence of **less than 1.0 ohm of resistance** between each of the following points:

Table 3-4 Resistance Verification - Site Ground

FROM	TO	
PDU Ground Bus	Vault Ground	☐ Check box when complete
PDU Ground Bus	Table/Gantry raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Table Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console Chassis	☐ Check box when complete
All Display or Computing Options (if any)	Operator Console Chassis	☐ Check box when complete

- 2.) Install remaining covers on the:

- Gantry
- Table
- Raceway
- Console
- PDU

Section 3.0 Seismic Mounting

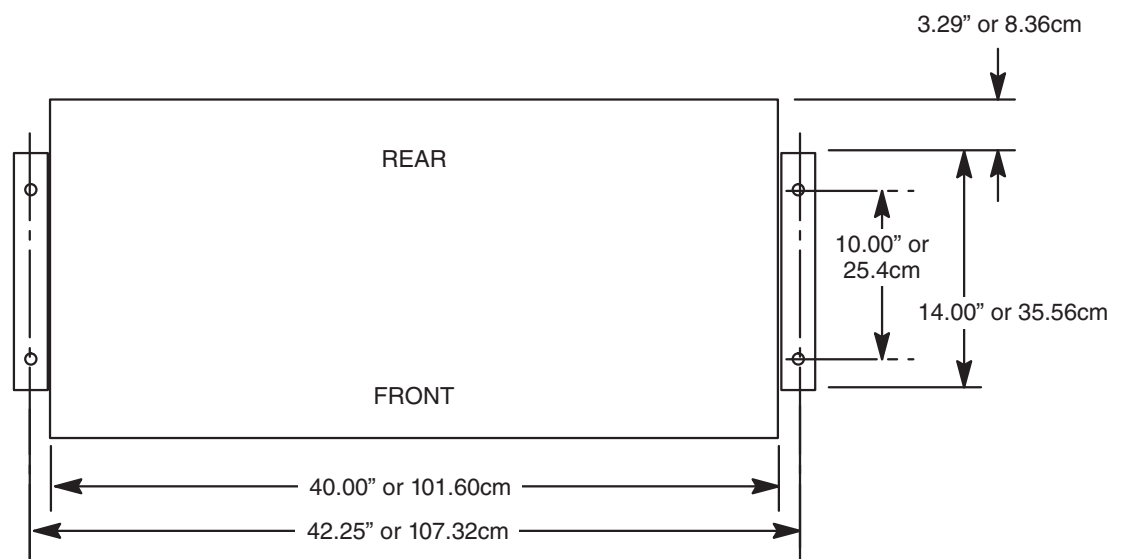
Before proceeding with seismic mounting for any of the components in this section, be sure to allow sufficient space to unbolt and move the component from its mounted location for service.

- All four mounting bolts may need to be removed.
- If removing the component requires lifting, use an appropriate-sized pry bar to lift each corner of the component.
- Two installers may be required to safely complete the task.

3.1 Console

If site specifications require seismic mounting to the floor, use 1/2 in. bolts, or those supplied in the seismic mounting kit, to mount the seismic brackets that shipped with the console. Refer to [Figure 3-3](#) for mounting hole locations, and mount the console so it can be easily removed for service.

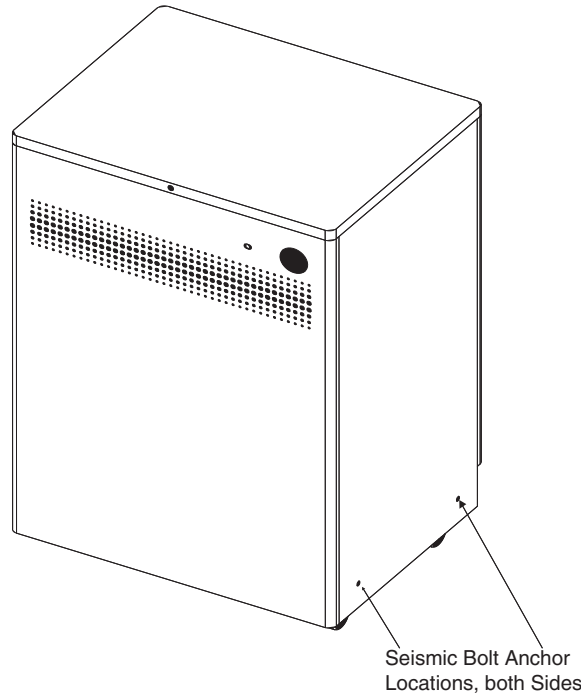
Figure 3-3 Seismic Console Mounting Hole Locations



3.2 NGPDU

CAUTION The PDU is very heavy and may present a crush hazard if proper precaution and tools are not used. If site specifications require seismic mounting, use 1/2 in. bolts, or those supplied in the seismic mounting kit, and the seismic brackets and anchors that shipped with the NGPDU. Refer to [Figure 3-4](#) for mounting hole locations, and mount the PDU so it can be easily removed for service.

Figure 3-4 Seismic PDU Mounting Hole Locations



3.3 Uninterruptible Power Supply (UPS)

CAUTION The UPS is very heavy and may present a crush hazard if proper precaution and tools are not used. If the site has a UPS (shown in [Figure 3-5](#)), refer to the *2-Phase UPS Installation* manual, *Direction 5174051-100*. If site specifications require seismic mounting, use 1/2 in. bolts, or those supplied in the seismic mounting kit, and the seismic brackets that shipped with the UPS. Refer to the manufacturer's installation instructions to locate the seismic mounting holes, and mount the UPS so it is easily removed for service.

Figure 3-5 UPS



Section 4.0

Axial Head Holder Shim Installation

4.1 Overview

This procedure applies to the following Table Types:

GT 2000	All head holders
GT 1700	All head holders
HP 1600	All head holders
PET Tables	All head holders

4.2 Requirements

Table 3-5 Personnel Requirements

REQUIRED PERSONS	PRELIMINARY REQS	PROCEDURE	FINALIZATION
1	10 mins	15 mins	5 mins

Table 3-6 Tools and Test Equipment

ITEM	QTY	EFFECTIVITY	PART#	MANUFACTURER
Standard FE Tool Kit	1	-	-	-

Table 3-7 Replacement Parts

ITEM	QTY	EFFECTIVITY	PART#	MANUFACTURER
Shim Kit	1	-	-	-

NOTICE Understand and Follow All General Table Safety Procedures.

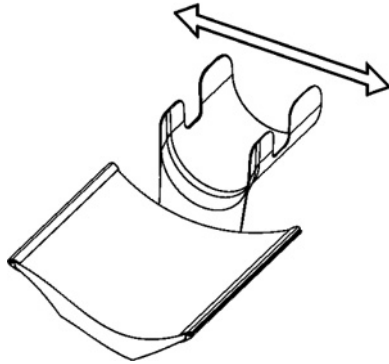


4.3 Required Conditions

Check head holder for a tight fit. If the head holder fit is loose, follow this procedure.

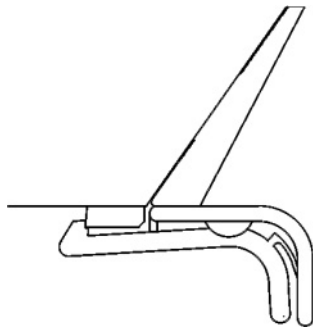
Introduction:

- Some Axial Head Holders have a large free-play in the horizontal direction which could potentially lead to motion and therefore image artifacts.
- Installation of the 2327335 rubber shim kit can minimize this motion.

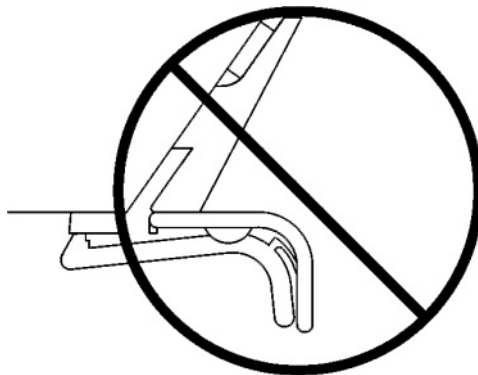


Notes before Selecting Shim Thickness:

- While selecting the best shim size, do not attach the rubber shim yet using the adhesive on the back. It is best to use a piece of tape to hold on the shim in order to see if the size is correct.
- Selecting a shim size that is too thick may result in:
 - Difficulty latching the head holder properly. The head holder must latch so that a patient is not injured.
 - Damage to the plastic latch or the plastic screws that secure it.



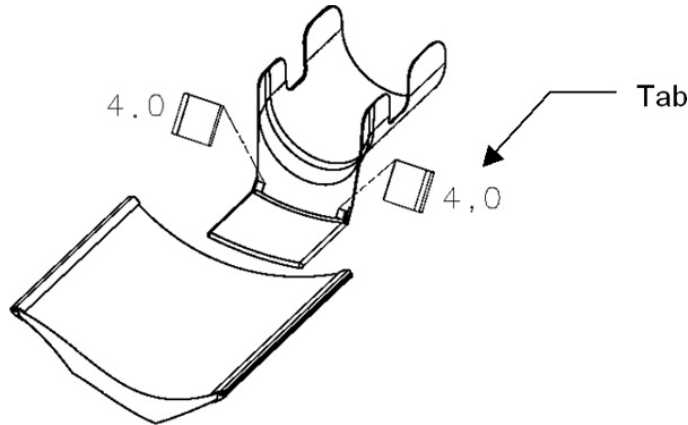
Correct - Head Holder is latched onto first step of plastic latch mechanism (The head holder does not need to be latched onto the second step)



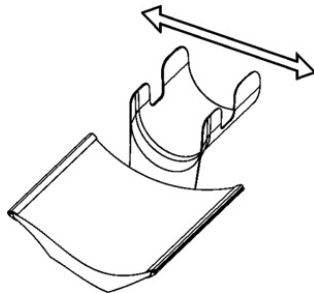
Wrong - Head Holder is NOT latched after installing shims

4.4 Procedure

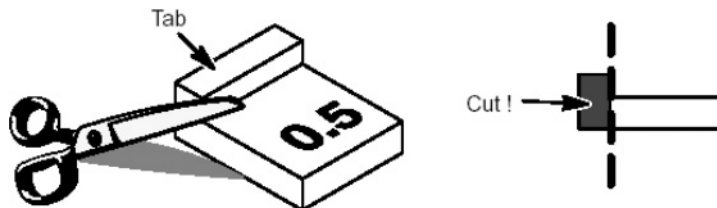
- 1.) First place the two 4.0mm shims (thickest size) onto both edges of the head holder as shown (use a piece of tape to temporarily secure them)
 - The shim must be placed with the tab facing out
 - The thickness is printed on the shim



- 2.) Insert the head holder into the cradle
- 3.) Check if the head holder is latched onto the cradle at the first step of the plastic latch mechanism. (The head holder does not need to be latched onto the second step)
- 4.) Check if the head holder has a small free-play in the horizontal direction



- 5.) If the rubber is too thick, repeat steps 1-4 using a thinner shim (3.5, 3.0...0.5mm) until the head holder is latched (without excessive force) and fits securely in the cradle.
If the thinnest shim (0.5mm) is too tight, the tab can be cut off to reduce the thickness



- 6.) Clean off the surfaces where the shims will mount using alcohol.
- 7.) Peel off the paper from the back of the selected shims and attach with the tabs facing out. Hold each shim with your fingers for a few seconds to attach it to the head holder.

4.5 Finalization

Review latching of head holder with customer after installation.

Section 5.0

Mechanical Installation Completion Section

Complete the *Mechanical Installation* section of the GE e4879 and turn it over to your site FE. The electronic file for the checklist is found on the LightSpeed 5.X Service Information CD-ROM. The electronic checklist is also located in the downloadable forms directory of the CD-ROM.

Appendix A

Removal & Installation of Covers

Section 1.0 Gantry Side Covers

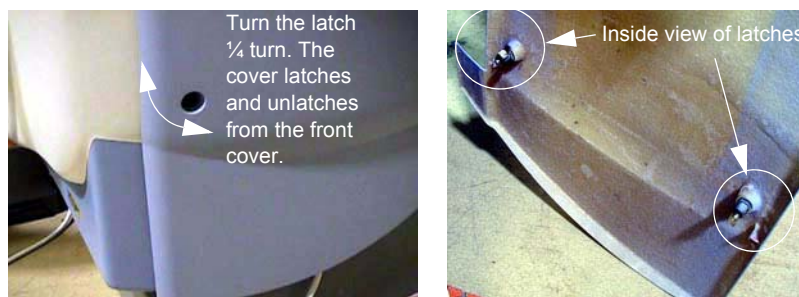
1.1 Side Cover Removal

CAUTION



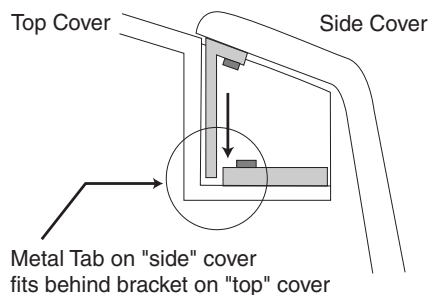
- 1.) Lower Table to home (lowest) position.
Potential for injury if covers removed and power is left "ON".
Always remove the right side cover first, and turn OFF power at the Service Switch Panel.
- 2.) Use an 8mm Hex wrench to unlatch the side cover from the front cover. See [Figure A-1](#).

Figure A-1 Side Cover Latches



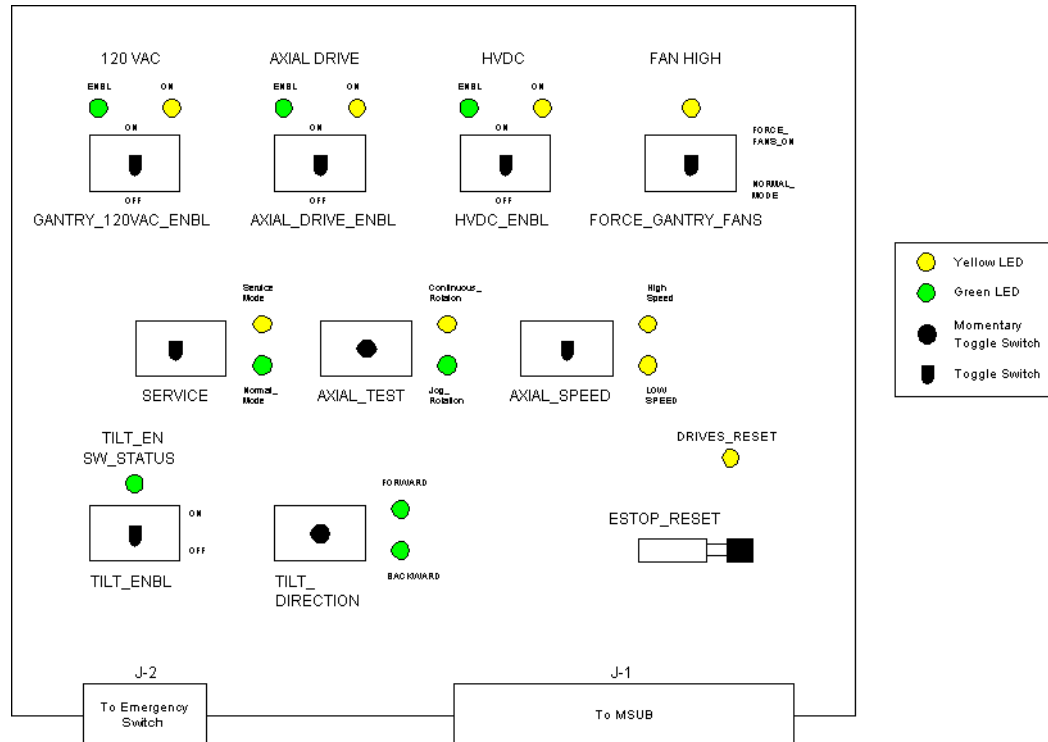
- 3.) Remove the right side cover by lifting it upward to release the two (2) latches, located on the top edge of the cover. Once removed, the MSUB/TGP left side should be exposed.

Figure A-2 Side and Top Cover Clasp



- 4.) Turn OFF the **120 VAC**, **AXIAL DRIVE** and **HVDC** power switches on the Gantry service switch panel (see [Figure A-3](#)).

Figure A-3 Gantry Service Switches



- 5.) Repeat steps 1-3 for the left side cover.

1.2 Side Cover Installation

- 1.) To install a side cover, place it over the top cover and let the two (2) side cover latches slide behind the metal tabs, located on the top cover. See [Figure A-2](#).
- 2.) Use Hex wrench to secure the side cover to front cover by turning the bolts a quarter turn. See [Figure A-1](#).

Section 2.0 Gantry Top Covers

2.1 Top Cover Removal

CAUTION Potential for Shock.



Voltage may be present. Potential for injury if covers removed and power is left ON. Always remove the right side cover first, and turn OFF power at the service switches.

NOTICE



- 1.) Remove the Gantry right side cover (see Gantry Side Covers Removal and Re-install).
Always turn OFF the HVDC before the 120 VAC. Turning OFF 120 VAC power before HVDC power can result in equipment damage.
 - 2.) Turn OFF the three (3) main power switches (HVDC, 120VAC, and Axial Drive) on the Service Switch Panel (SSP). See [Figure A-3](#).
 - 3.) Disconnect the power cable that connects the fans on the top cover to the Gantry.
- Note: The cooling fans are now mounted on the top covers.
- 4.) Take the end of the top cover nearest to the side cover and tilt upwards.
 - 5.) Slide the cover down to disengage the tab from the mounting bracket. See [Figure A-4](#).

Figure A-4 Top Cover Tabs and Bracket



- 6.) Lift the cover clear and repeat the above steps for the other cover.

2.2 Top Cover Installation

The top cover consists of two (2) pieces. Install the front and rear Gantry covers, if not already installed. See [Section 3.0 on page 120](#), and [Section 4.0 on page 129](#).

- 1.) Take one of the top covers and align the tabs on the cover with its associated bracket. Lift and slide the cover into place.
- 2.) Take the other top cover and align the tabs on the cover with its associated bracket. Lift and slide the cover into place.
- 3.) Connect the cable from the fans to the Gantry.
Always turn ON the 120 VAC before the HVDC. Turning ON HVDC power before 120 VAC power can result in equipment damage.
- 4.) Turn on the three (3) power switches.
- 5.) Ensure fans work properly.
- 6.) Re-install the Gantry side covers.

NOTICE



Section 3.0

Gantry Front Cover

NOTICE Potential for cover damage.



Front and rear cover removal and installation can be safely accomplished by one (1) person using the dollies provided with the system. Failure to use these dollies will significantly increase the likelihood of damage to the covers. Do not lean covers against walls.

3.1 Front Cover Dolly Setup

DANGER



DO NOT USE DOLLIES ON UNEVEN SURFACES SUCH AS STEPS OR ELEVATOR THRESHOLDS. THE DOLLIES ARE DESIGNED TO BE USED ON FLAT LEVEL FLOORS WITHIN THE SCANNING SUITE ONLY. MISUSE CAN RESULT IN PERSONAL INJURY OR DAMAGE TO COVERS OR OTHER FACILITY ITEMS.

CAUTION



Rotating arms on the stand are supposed to be stiff. If they fall freely, tighten the tensioning nuts. Loose rotating arms will reduce the stability of the dollies when supporting the front cover. Do not lubricate.

- 1.) Arrange Dolly sections for assembly. The base and post can be assembled only one way. Refer to [Figure A-5](#) and [Figure A-6](#).

Figure A-5 Front Cover Dolly in Storage Mode

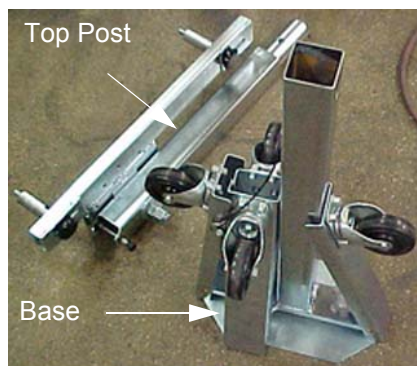
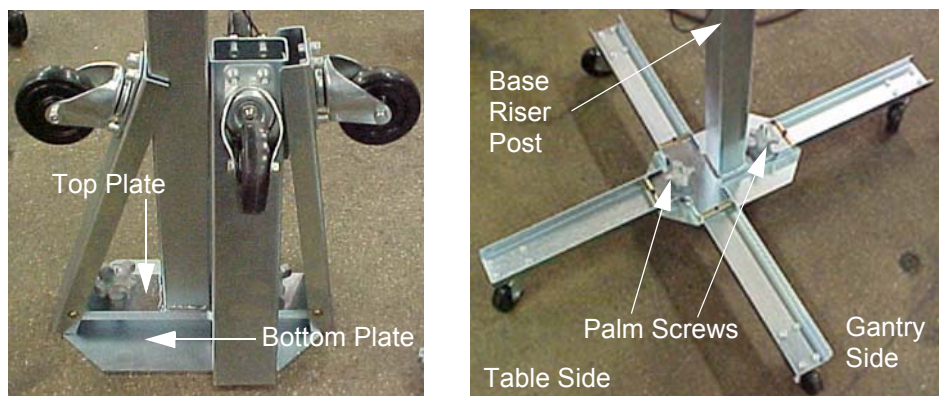


Figure A-6 Front Cover Dolly Base Assembly



- The base uses two (2) palm screws to clamp the four (4) legs in the open or usage mode.
 - The base also uses the same palm screws to prevent the legs from falling in storage mode.
 - The top post can be inserted in either base and is keyed for proper engagement.
 - The top post locking pin prevents the sections from separating during usage.
- 2.) Unfold the base legs by loosening both palm screws to the top of their travel.
 - 3.) Carefully unfold the legs so that the castors touch the floor.
 - 4.) Tighten the palm screws to clamp the legs between the base top and bottom plates.

Note: Lifting the base by the riser post while leaving the castors on the floor will ease palm screw tightening. Reference [Figure A-6](#).

WARNING



ENSURE BOTH PALM SCREWS ARE TIGHTENED SECURELY AND THE LEGS ARE CLAMPED TIGHTLY BETWEEN THE BASE TOP AND BOTTOM PLATES. FAILURE TO DO SO WILL RESULT IN INSTABILITY DURING FRONT COVER HANDLING.

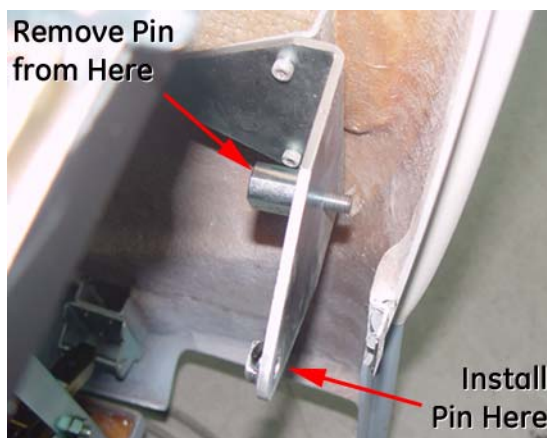
- 5.) Insert top post into the base riser post. Align the key for complete engagement.
- 6.) Insert top post locking pin to secure both top and bottom sections.
- 7.) Reverse above steps to disassemble.

Note: For base storage only one (1) palm screw needs to be tightened. This will engage the bottom base plate and the leg ends preventing the legs from unfolding during transport and storage.

3.2 Removal

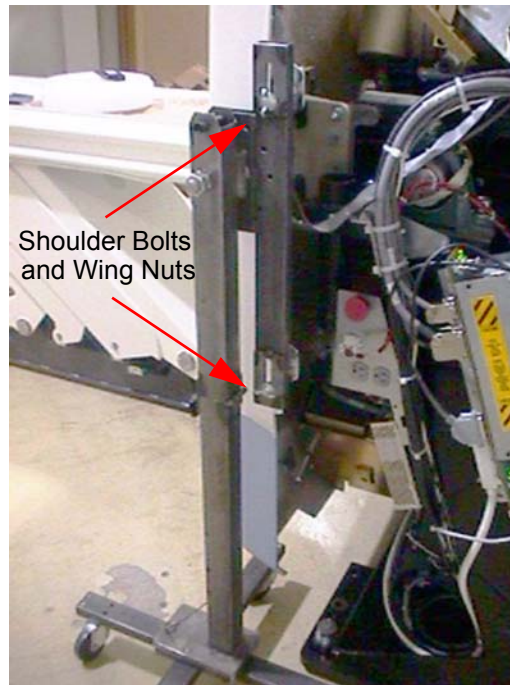
- 1.) Position the Table at its lowest position.
- 2.) Remove Gantry side and top covers, if you have not already done so. See [Section 1.0 on page 117](#). Make sure that the three (3) power switches have been turned off. See [Figure A-3](#).
- 3.) Assemble the front cover dolly.
 - a.) **For RT Only:**
Change the extension pin location on Gantry front cover bracket. See [Figure A-7](#).

Figure A-7 Extension Pin Location (RT Only)



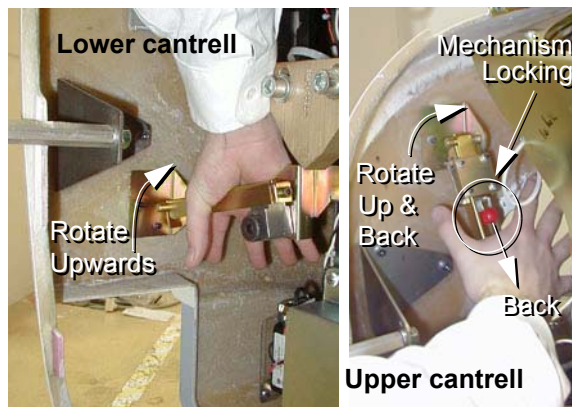
- b.) Tighten the two (2) shoulder bolts to the Gantry securely. This will make cover installation easier. See [Figure A-8](#).

Figure A-8 Front Side Dolly



- c.) Attach side dolly to the shoulder bolts and secure assembly with two (2) wing nuts.
d.) Repeat steps a and b to assemble the other side dolly.
4.) Detach front cover J3 and J2 and front cover BKHD J1 cables.
5.) Remove front cover
a.) Disengage upper and lower cantrell brackets on both sides of the cover.
i.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers. See [Figure A-9](#).

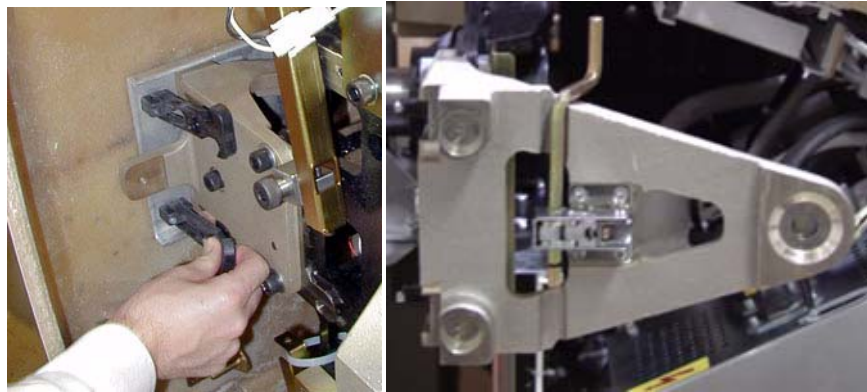
Figure A-9 Releasing cover brackets



- ii.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking mechanism and allow the cantrell to be rotated upwards with steady and firm pressure.

- b.) Disengage the rubber retaining straps on both sides. You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.
- c.) Also lift and rotate cover locking arm to unlocked position.

Figure A-10 Rubber retaining straps and Cover Locking mechanism



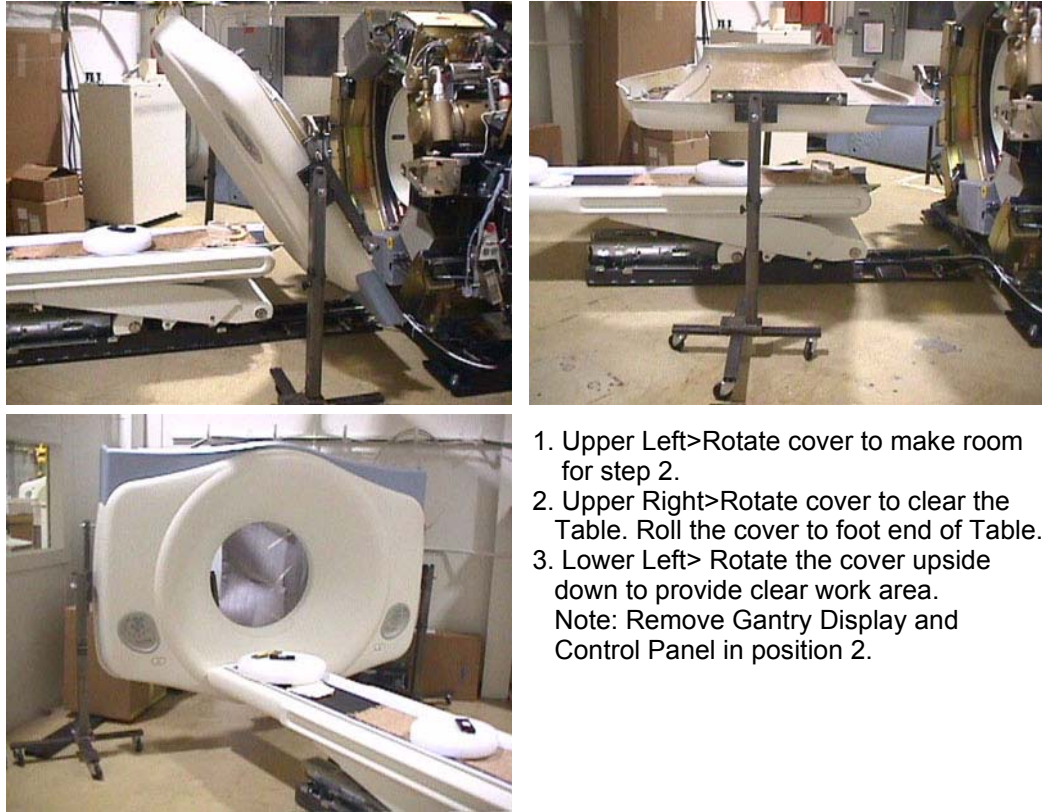
- 6.) Rotate front cover away from Gantry.
 - a.) Move front cover away from Gantry giving enough space (about 5 feet) between front cover and Gantry.
 - b.) Pull the locking pin and rotate front cover away from Gantry. Place locking pin in one of the side dolly perforations. See [Figure A-11](#).

Figure A-11 releasing Front Cover Dolly Hinge



- 7.) Rotate the cover horizontally and move it back and over the Table to a safe location. Once in a safe location, you may over-rotate the cover full vertically but upside down.

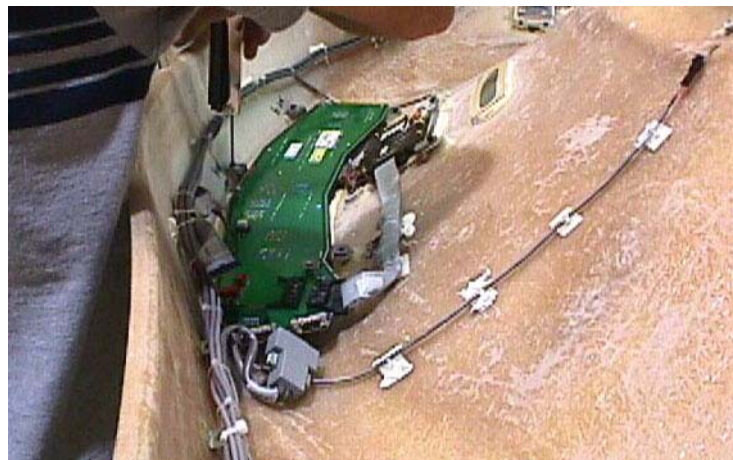
Figure A-12 Front Cover Removal Sequence



1. Upper Left>Rotate cover to make room for step 2.
2. Upper Right>Rotate cover to clear the Table. Roll the cover to foot end of Table.
3. Lower Left> Rotate the cover upside down to provide clear work area.
Note: Remove Gantry Display and Control Panel in position 2.

- 8.) Remove the Gantry display and one (1) of the cover's control assemblies, and place them into the service positions.
- a.) Remove the Gantry display and place it into its service position.
- * The Gantry display is held in place with (5) thumb screws. Use a flat-blade screwdriver to remove the Display. Reference [Figure A-13](#).

Figure A-13 Gantry Display Removal



- * Place the Display in the bracket on the right side of the Gantry. See [Figure A-14](#).

Figure A-14 Gantry Display Service Mounting Location



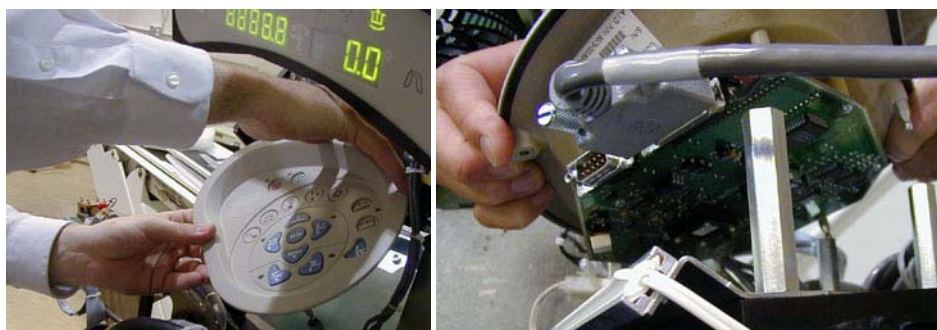
- * Disconnect the cabling at the right rear Gantry cover. Only (1) cable will connect to the Gantry Display. Connect the cable taken from the rear cover to the display.
- b.) Remove a Gantry control and place it into its service position.
- i.) Press on each ball stud until the panel is released. Keep one hand on the control panel at all times to prevent it from dropping to the floor.

Figure A-15 Gantry control panel removal



- ii.) Align the ball studs with their associated receivers and snap into place.

Figure A-16 Control panel service position



- iii.) Connect cable to terminator located on the cantrell arm. See [Figure A-17](#).
- Note:** There are three (3) cables, each of which is unique. The ribbon cable is not used in the Service configuration. The other 2 cables will only fit in the terminator or the control panel, not both.

Figure A-17 Gantry Service Mode Cable Terminator



3.3 Installation

- 1.) Remove the Gantry display and control assembly from their service positions and re-attach them to the Gantry cover.
 - a.) Disconnect cables from Display and Gantry Control Panels.
 - b.) Install Gantry Display in front cover. Secure the 5 thumbscrews. With a flat-blade screw driver gently tighten past finger tight.
 - c.) Install the Gantry control panel making sure the ball studs are secure within the receivers.
 - d.) Re-attach cables.

NOTICE



Potential for front cover damage.

When you rotate the Gantry back to its vertical position, make sure not to scratch the front cover with the edge of the Table cradle.

- 2.) Rotate Gantry back to its vertical position.
- 3.) Attach the front cover.
 - a.) Align the studs on both sides of the front cover with each associated receiver. Receiver is located on the Gantry frame.

Figure A-18 Cover stud and Mounting bracket receiver



- b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps. Then insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.
- 4.) Re-attach upper and lower cantrell brackets on both sides.

- a.) Remove upper Cantrell brackets from service position and rotate them into position over their associated retaining pins. See [Figure A-19](#).

Figure A-19 Service position of upper and lower cantrell brackets

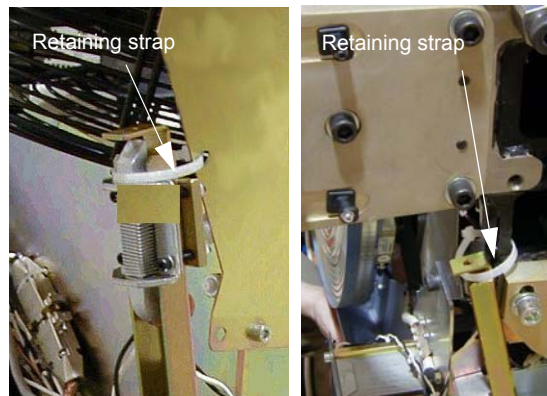
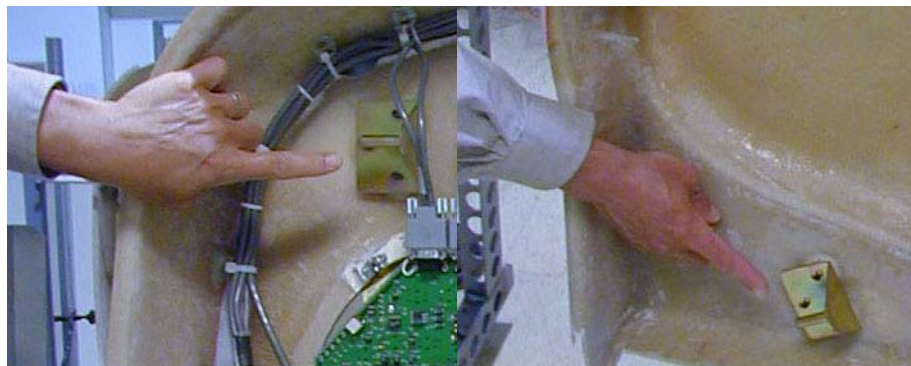


Figure A-20 Cover retaining pins (top and bottom)



Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides. See [Figure A-21](#).

Figure A-21 Locking the cover brackets into place

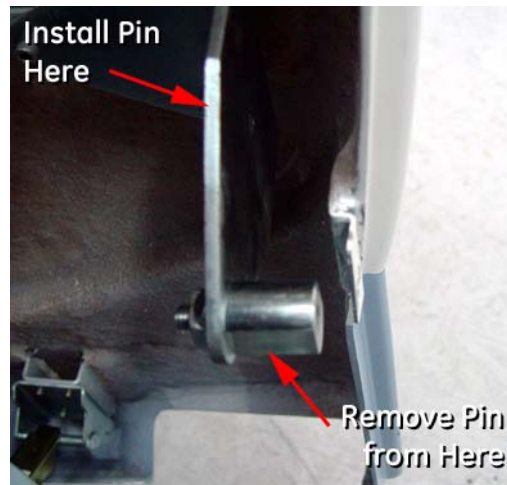


- b.) Remove lower cantrell brackets from service position (see [Figure A-19](#)), and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place. See [Figure A-21](#).

Note: Incorrect adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not require adjustment during normal use.

- 5.) **For Pro16, LS16, Ultra or Plus Only:**
Remove dolly, disassemble and store safely away for later use.
- 6.) **For RT Only:**
Remove dolly. Change the extension pin location on the Gantry front cover bracket (see Figure A-22). Disassemble and store safely away for later use.

Figure A-22 Remove Dolly (RT Only)



- 7.) Re-attach cables to cover.

Section 4.0

Gantry Rear Cover

4.1 Removal

- 1.) Assemble the rear cover dolly.
 - a.) Tighten the two (2) shoulder bolts to the rear cover.

Figure A-23 One side of the Rear cover dolly



- b.) Fit side dolly through the shoulder bolts and secure assembly with two (2) wing nuts. See [Figure A-23](#).
 - c.) Repeat steps a and b for the other side dolly.

CAUTION



Potential for injury if covers removed and power is left "ON".

- 2.) Disconnect cables on the right side of the rear cover.
- 3.) Remove rear cover.
 - a.) Disengage upper and lower cantrell brackets on both sides of the rear cover.
 - 1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers. See [Figure A-9](#).
 - 2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking mechanism and allow the cantrell to be rotated upwards with steady and firm pressure.
 - b.) Disengage the rubber retaining straps on both sides.

4.2 Installation

- 1.) Position cover in back of Gantry
- 2.) Attach the rear cover
 - a.) Align the studs on both sides of the rear cover with the receivers located on the Gantry frame.
 - b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps. Then insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

Note: You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.

- 3.) Reattach upper and lower cantrell brackets on both sides.
 - a.) Remove upper cantrell brackets from service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides.
 - b.) Remove lower cantrell brackets from service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place.

Note: Adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not re-quire adjust during normal use.

- 4.) Remove dolly, disassemble and store safely away.
- 5.) Re-attach cables to cover.
- 6.) Re-install the mylar (scan) window. Carefully bend the scan window and place it into the channel (groove) provided in the covers.

Figure A-24 Installing the mylar window



Section 5.0

Gantry Cover Alignment Adjustment Guide

5.1 Overview

This section explains the adjustment capability available for the Gantry covers and guidelines for performing adjustments.

5.2 Tools Required

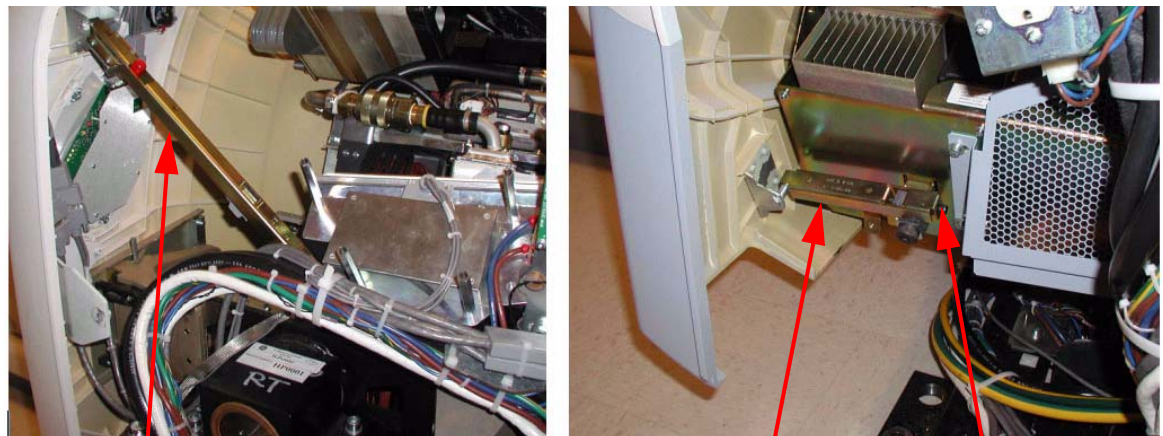
3mm, 5mm, 8mm, Hex wrenches

5.3 Adjustment Guide

5.3.1 Front/Rear Cover adjustment

The front and rear covers are held in a fixed vertical position by the cover locking mechanisms at the vertical middle of each cover on each side. The only adjustment capability is found on each of the cover cantrells (see [Figure A-25](#), one near each corner of the front and rear covers. Adjustments are best done at zero tilt.

Figure A-25 Cantrells



Top Cantrell same
for all top latches

Bottom Cantrell same
for all bottom latches

Adjustment screw on
end of all brackets

- 1.) Remove the Gantry side covers using a 8mm hex wrench to release the bottom latches. Refer to [Section 1.0 on page 117](#), for further details.
- 2.) Disable Gantry rotation, 120VAC and HVDC via the service switch panel.
- 3.) Disconnect the fan power via the white molex connector (one for each top cover). Remove the top covers by lifting up and pulling to the side of the Gantry.
- 4.) Remove the scan window to avoid creases or kinks during cover movements.
- 5.) Using a 5mm hex wrench, adjust the bolt on the Gantry end of the cover cantrells (see [Figure A-25](#)) to lengthen or shorten the cantrell arm that will push or pull the cover corner. The distance between the inside edges of the front and rear covers will be typically 27" + 1/8" (689mm). Distance should remain the same as measured along the inside edges from top to bottom with both covers hanging level as seen with a level on the inside vertical edge.

5.3.2 Scan Window

Position the scan window in the opening to verify that it seats all the way around with no gaps. The scan window should seat easily if the front and rear covers have been adjusted properly. If not, go back and readjust such that the scan window seats properly. There is no easy solution other than adjusting and looking for fit.

5.3.3 Top Cover Adjustment

The top covers rest in a top bracket that is mounted to the stationary Gantry frame and subsequently just rest on top of the front/rear covers. There are also two alignment fingers on the outer edge of the top covers that need to sit in brackets on the front/rear covers.

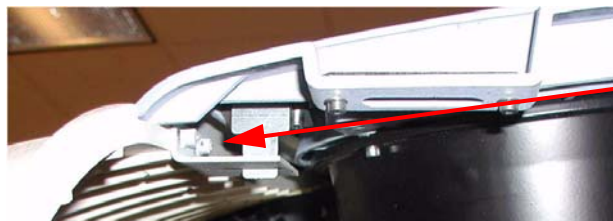
Figure A-26 Top Cover Bracket



Screws on both sides
allow adjustment up/
down only

- 1.) Set the top covers back on the Gantry and look for proper fit. The top covers should rest on the front/rear covers all along the edges. If they do then skip the rest of this section.
- 2.) If the top covers do not rest on the Gantry at the top then lift them back off and use a 5mm hex wrench to loosen the bolts that hold the top cover bracket for adjusting vertical alignment to allow the top covers to rest on the front and rear covers.
- 3.) The brackets on the front/rear covers for the fingers on the outer edge of the top covers are adjustable using a 3mm hex wrench and sliding the bracket till it lines up with the top cover finger. (See [Figure A-27](#))

Figure A-27 Top Cover Fingers



Adjustment bolts for
aligning bracket to
top cover finger

- 4.) Verify the top cover tapered edges fit over the lip along the top edge of the front/rear covers. If not, go back to Section 5.3.1 and readjust the cantrells as necessary.
- 5.) When done adjusting the top covers, reconnect the fan power molex connections.

5.3.4 Side Covers

The side covers just hang from the top covers by two fingers that are not adjustable. When set into the top covers, the side covers should hang straight down with the tapered edges fitting over top of the lip of the front/rear covers.

5.3.5 Finish

- 1.) Re-enable Gantry rotation via the Axial Enable switch on the service switch panel.
- 2.) Replace the side covers and latch using a 8mm hex wrench on the side cover bottom latches.
- 3.) Run an axial scan to ensure proper system operation prior to turning system back over to the customer.

Section 6.0

Gantry Scan Window

6.1 Remove Scan Window

- 1.) Grab the window at the top and pull firmly downward.
- 2.) Continue to pull until the top of the scan window makes contact with the bottom portion of the scan window.
- 3.) Hold the top and bottom portions of the scan window together, grasp both sides of the scan window, move them together and lightly pull upward, until you can free the window from between the front and rear covers.

Figure A-28 Scan Window Removal



6.2 Install Scan Window

- 1.) Install the front and rear covers.
- 2.) Deform the scan window, as shown in [Figure A-29](#), and nest the scan window at the bottom of the opening between the front and rear covers, ([Figure A-30](#)) with the rivets in the 6 o'clock position installation position. Remember the rivets must be in the 12 o'clock position when the mylar window is fully installed.

Figure A-29 Install Scan Window

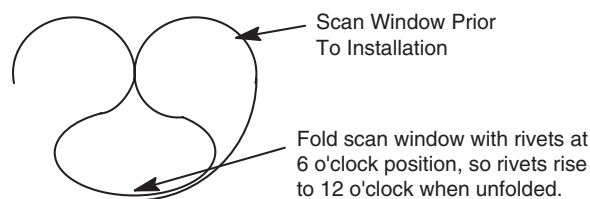
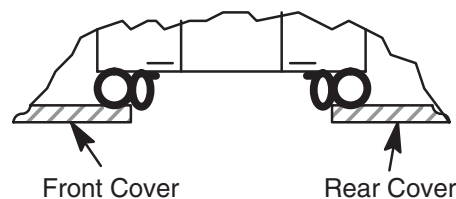


Figure A-30 Scan Window Nested Between Front and Rear Cover



- 3.) After you complete the initial seating of scan window, let the window slowly unfold, and work both sides of the window into position, starting at the bottom and finishing at the top.
- 4.) Make sure you position the window with the rivets at the 12 o'clock position, and the mylar window slit at either the 3 or 9 o'clock position.

Section 7.0

Gantry Base Covers

Refer to [Figure A-31](#) for the following assembly sequence.

Note: Tighten means torque to 2.3 Nm

- 1.) Assemble two (2) brackets (item 11) to front cover (item 5), using four (4) hardware items 1, 2 and 3. Align leg of brackets parallel to front edge of cover and tighten. Position cover on Gantry base, with bracket slots aligned to Gantry holes. Center cover left to right, and attach with four (4) hardware items 1, 2 and 3, as shown, and tighten.
- 2.) Assemble two (2) brackets (item 12) and two (2) brackets (item 13) to Gantry base, using eight (8) hardware items 1, 2 and 3. Finger tighten hardware with brackets moved outward to end of slots. Install side covers (items 6 & 7) on base, pushing brackets (items 12 & 13) inward until properly aligned with front cover. Remove side covers, tighten fasteners and replace side covers, using two (2) hardware items 1, 2 and 3 and one item 4 on each cover, and tighten.
- 3.) Assemble last bracket (item 13) loosely to Gantry base with two (2) hardware items 1, 2 and 3. Assemble latch (item 10) to rear cover (item 8), oriented as shown. Install rear cover (item 8) to base, properly aligned to side cover (item 6). Attach rear cover to bracket with hardware items 1, 2, 3 and 4, tightening all fasteners. Lock latch (item 10).
- 4.) Assemble two (2) latches (item 10) to rear cover (item 9), oriented as shown. Place cover on Gantry base, aligned to covers 7 and 8. Lock both latches.

Figure A-31 Gantry Base Covers (except RT)

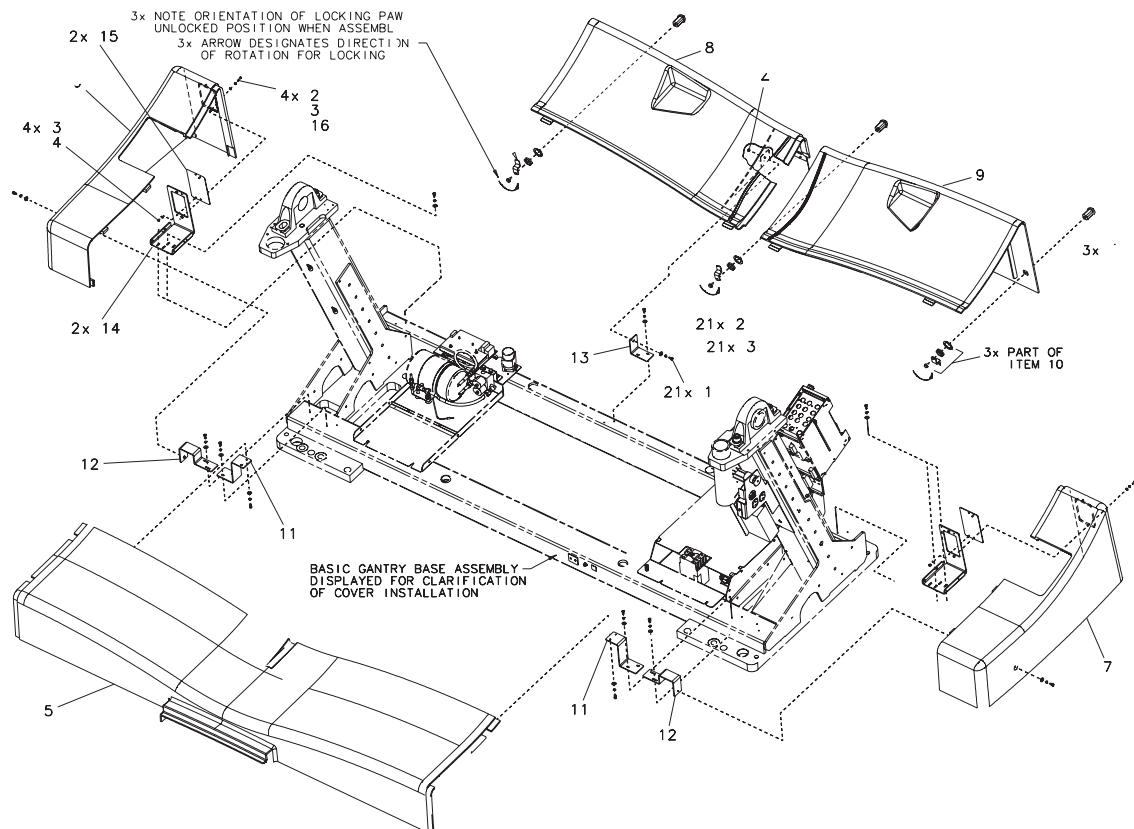
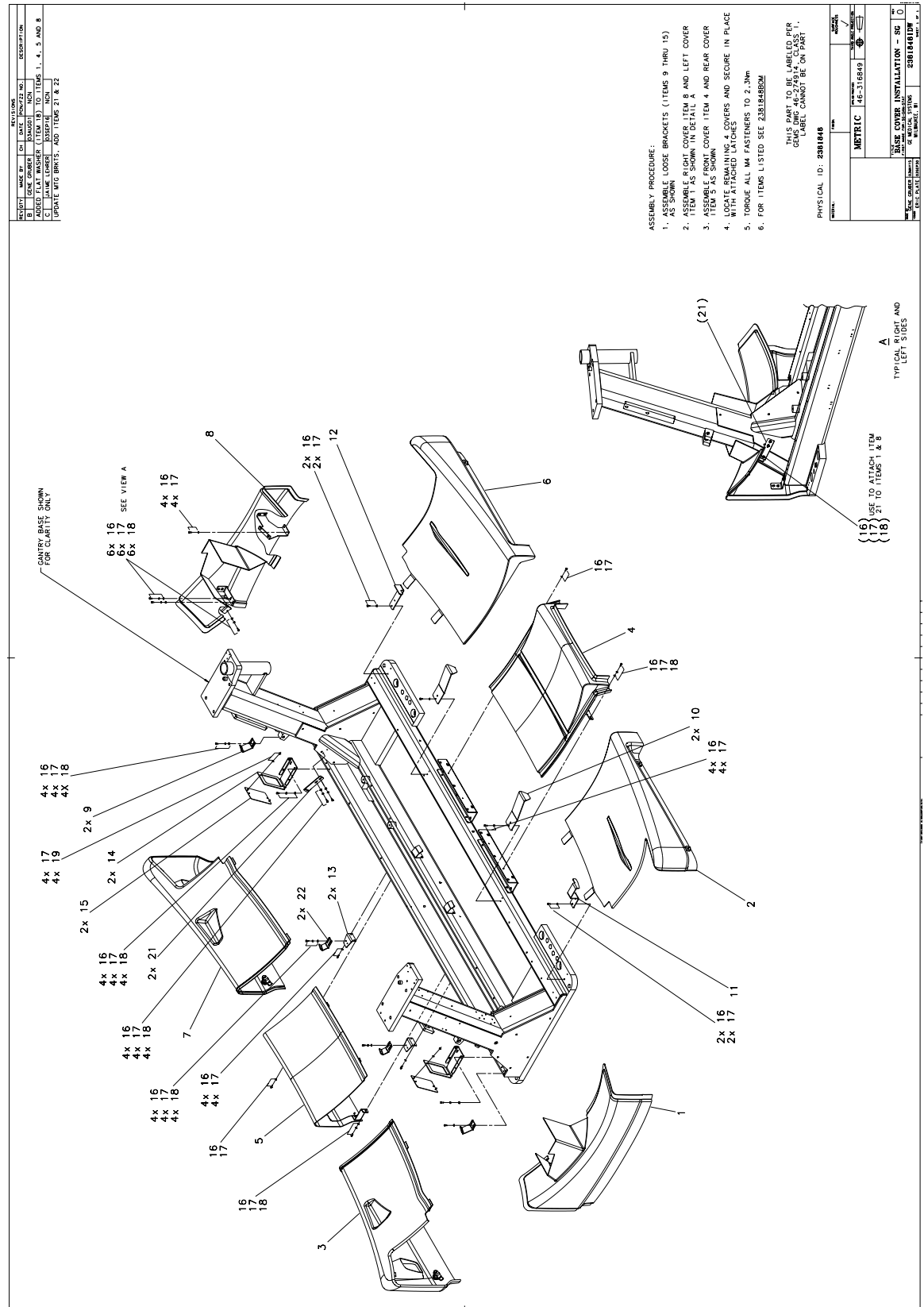


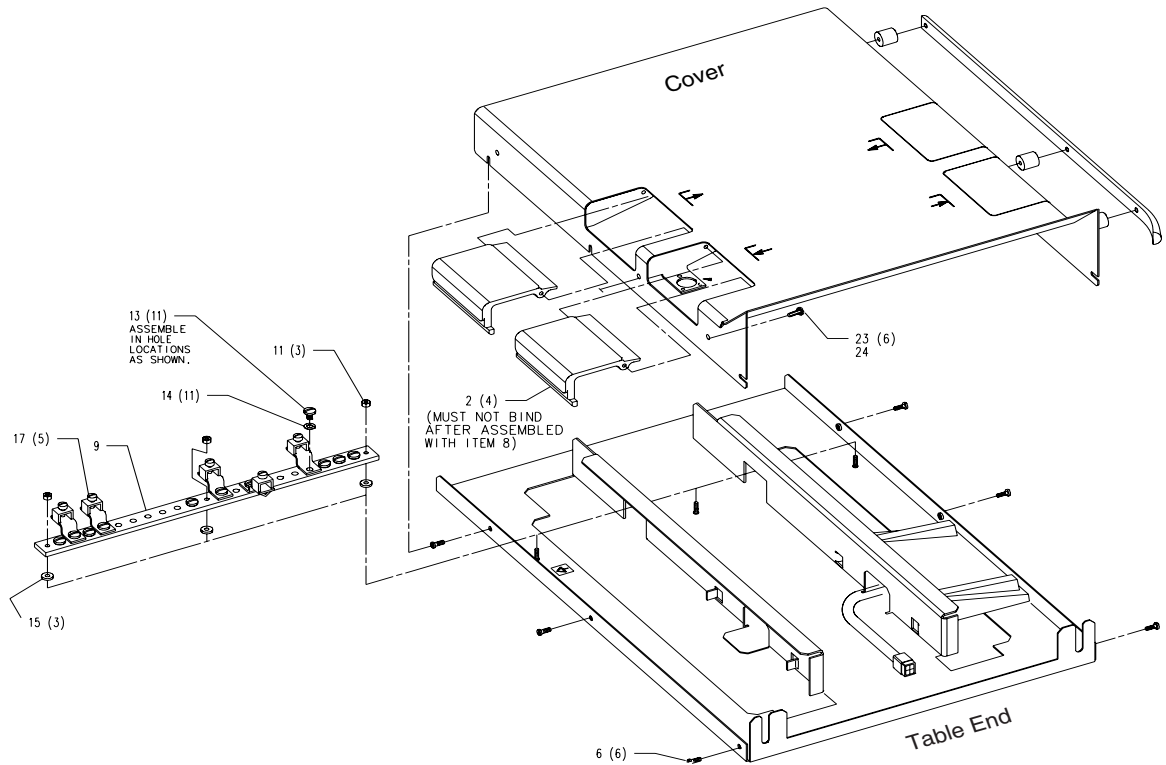
Figure A-32 Gantry Base Covers - RT



Section 8.0

Footswitch Covers

Figure A-33 Table Footswitch Assembly



Appendix B

Pictorial Representation of Required Tools

Use the following guide as a reference, if you are unsure of a tool listed in [Section 2.6, on page 28](#).

Table B-1 Required Tools

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Adapter		Sears Industrial: $\frac{3}{8}$ " to $\frac{1}{2}$ " (9-4258)
Ball-Peen Hammer		Sears Industrial: 1lb/2lb (9-38465)
Canned Air		Miller Stephenson: Aero Duster (MS-222N)
Clamp on Amp Meter		Sears Industrial: 9-WTAD105
Combination Wrench Set		Sears Industrial: U.S. Standard & Metric (9-44048)
Cordless Screwdriver		Sears Industrial: 9-MU65401
Deep Well Socket		Sears Industrial: $\frac{3}{4}$ " X $\frac{3}{8}$ " (included with 9-34496)
Dental Pick		
Diagonal Cutting Pliers		Sears Industrial: Small (9-45077)

* Part Numbers given for reference only. GE Healthcare does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Drill		Sears Industrial: $\frac{3}{8}$ " or $\frac{1}{2}$ " (9-27859)
Drill Adapter		Sears Industrial: 3" X $\frac{3}{8}$ " (9-APSZ24)
Drill Bit Set		Sears Industrial: U.S. Standard (9-66084)
DVM		Sears Industrial: 9-82028 Sears Industrial: 9-FL873
Extension for Ratchet Wrench		Sears Industrial: 3" X $\frac{1}{2}$ " (9-44133)
Gloves		Sears Industrial: Large (9-40502)
Hammer Drill		Sears Industrial: $\frac{1}{2}$ " (9-27205)
Hex Bit Set		Sears Industrial: $\frac{1}{4}$ " (9-SK45508)
Hex Key (Allen Wrench) Set		Sears Industrial: U.S. Standard (9-46284)

* Part Numbers given for reference only. GE Healthcare does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Level		Sears Industrial: 4' (9-39856)
Masonry Bit		
Open-End Wrench (Thin or Standard Tappet)		Snap-on: 10mm (SRSM10) & 21mm (LTAM2124)
Ratchet Wrench		Sears Industrial: $\frac{3}{8}$ " (9-43175)
Reciprocating Saw with Blades		Sears Industrial: 9-MU650921
Safety Glasses		Sears Industrial: 9-18650
Safety Shoes		
Screwdriver Set		Sears Industrial: Phillips & Straight (9-41505)
Socket Set		Sears Industrial: Standard $\frac{3}{8}$ " (9-34496)
Sockets		Sears Industrial: 1 $\frac{1}{8}$ " X $\frac{1}{2}$ " (9-47516)

* Part Numbers given for reference only. GE Healthcare does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Step Ladder		Sears Industrial: 6' (9-WN6006)
Tongue & Groove Pliers		Sears Industrial: Large (9-CL440)
Torpedo Level		Sears Industrial: 9" (9-39829)
Torque Wrench		Sears Industrial: $\frac{3}{8}$ " (9-WR3470)
Universal Joint		Sears Industrial: $\frac{3}{8}$ " (9-4435)
Vacuum Cleaner		Sears Industrial: 8 Gal (9-17780)
* Part Numbers given for reference only. GE Healthcare does not endorse any tool brand name.		

AppendixC

Regulatory Clearance Quick Reference Guide

Section 1.0

Regulatory Code Description

Egress: 29 CFR 1910 Subpart E (OSHA) and NFPA 101 (Life Safety Code) define the minimum requirements for means of egress. The requirement most applicable to equipment installation and room layout is minimum width of exit access. Under OSHA 1910.37(f)(6), the minimum width of exit access shall in no case be less than 28 in. from any potentially occupied point in the room.

Under NFPA 101 (2006 edition) 7.3.4.1, the minimum width of any means of egress is 36 in. However, NFPA allows this to be reduced to 28 in. around furniture or equipment, provided that a 36 in. clearance would otherwise be available without moving permanent walls.

Electrical Clearance: 29 CFR 1910 Subpart S (OSHA) and NFPA 70E (Standard for Electrical Safety in the Workplace) define minimum clearance requirements for the workspace around electrical equipment. Under both OSHA 1910.303(g)(1) and NFPA 70E (2004 edition) 400.15, a minimum clear space of 36 in. depth (with minimum 30 in. width and 78 in. height) must be provided in front of electrical equipment with parts operating at 600 volts or below and likely to require examination, adjustment, servicing, or maintenance while energized.

This safety clearance requirement applies to all GEHC equipment. Although 36 in. is the minimum clearance for most installations, the standards require an increased minimum clearance distance where parts operate above 150 volts (but still below 600 volts) under the following circumstances:

- If the wall or surface directly facing the electrical equipment is grounded (e.g. brick, concrete, or tile) or includes grounded protrusions (such as medical gas ports, metal door or window frames, water sources and metallic sink structures, metallic cabinetry, electrical disconnects or emergency off panels, air conditioners or vents), then a 42 in. clearance depth is required.
- If the possibility exists of exposed and unguarded live parts on both sides of the workspace (for example if a power distribution unit were positioned on the wall directly facing the GEHC equipment), then a 48 in. clearance depth is required.

Section 2.0

Terms and Definitions

Egress: The path of exit from within any room. U.S. regulations require a minimum of 28 in. (711.2 mm) of continuous and unobstructed space, including trip hazards along the path of exit.

Workspace: The dimensional box required for safe inspection or service of energized equipment. It consists of depth, width, and height. The depth dimension is measured perpendicular to the direction of access. The U.S. regulation minimum is 36 in. (914.4 mm), but additional conditions can increase the minimum dimension requirement. GEHC defines this as the envelope of the component superstructure with the external covers in place.

Service Access Width: The width of the workspace in front of the equipment. A minimum of 30 in. (762 mm), or the width of the equipment, whichever is greater.

Head Clearance: The height dimension of the workspace. The height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). 78 in. (1981.2 mm), or the height of the equipment, whichever is greater.

Grounded Wall: Any wall that can be electrically conductive to earth ground. Masonry, concrete, and tile are considered conductive. Additional commonly found aspects of a wall should also be considered grounded. This is not an all-inclusive list:

- Medical gas ports and plates
- Metal doors and window frames
- Water sources and metallic sink structures
- Metallic wall-mounted cabinetry
- A1 main disconnect panel
- Equipment Emergency Off panels
- Industrial equipment (such as air conditioners and vents)
- Expansion joints
- Surface raceway
- Exposed wall conduits
- Floor outlets boxes

The following are not considered as grounded elements of a common wall:

- Standard wall outlet
- Light switches
- Telephones
- Communication wall jacks
- Ceiling tile grids

Section 3.0

Regulatory Minimum Working Clearance by Major Subsystem

Requirements apply to equipment operating at 600V or less, where examination, adjustment, servicing, or maintenance is likely to be performed while live parts are exposed.

Direction of Service Access is defined as perpendicular to the surface of the equipment being serviced.

- Required regulatory clearance distances must be maintained and may not be used for storage. This includes normal system operation as well as service inspection or maintenance..

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access: front of console	914 mm (36 in.)	There are no exposed live part hazards with the cover in place. If the console is placed under a counter, the front edge of the console must be even with the vertical edge of the console workspace. Note: This component is typically serviced from the front with access to the rear.
Service access width: Front of console	762 mm (30 in.)	This is the width of the workspace in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head clearance	1981.2 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981.2 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 3-8 Console Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of NGPDU)	914.4 mm (36 in.)*	There are no exposed live part hazards with the cover in place. This component is typically serviced from the front with access to the rear. *If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) is required on both sides of the workspace with the operator between. *If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.

Table 3-9 NGPDU Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Head Clearance	1981 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 3-9 NGPDU Subsystem

- For the gantry and table, distances are measured from the enclosure, not the finish covers.

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (All Sides)	914 mm (36 in.)	If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) on both sides of workspace with the operator between is required. If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.

Table 3-10 Gantry Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Table Head or Foot)	914 mm (36 in.)	There are no exposed live parts hazards with the cover in place. This component is typically serviced from all four sides. this is the width of the workspace on each side of the equipment. A minimum of 914.4 mm (36 in.), or the width of the equipment, whichever is greater, is required.
Direction of Service Access (Table Sides)	914 mm (36 in.)*	*This distance can be reduced to 711 mm (28 in.) provided a written and signed approval is obtained by the local team from the local AHJ (Authority Having Jurisdiction). The signed document must be on file with GE.

Table 3-11 Table Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Table Foot)	711 mm (28 in.)	For the front gantry cover removal, a minimum of 457 mm (18 in.) is allowed only if an unobstructed egress space of 711 mm (28 in.) is maintained around the equipment for room exit. This also means no trip hazards exist along the path of egress.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.

Table 3-11 Table Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of UPS)	914.4 mm (36 in.)*	There are no exposed live part hazards with the cover in place. This component is typically serviced from the front with access to the rear. *If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) is required on both sides of the workspace with the operator between. *If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Right side and length of UPS)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head Clearance	1981 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 3-12 UPS Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of A1 Disconnect)	914.4 mm (36 in.)*	There are no exposed live part hazards with the cover in place. This component is typically serviced from the front with access to the rear. *If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) is required on both sides of the workspace with the operator between. *If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Right side and length of A1 Disconnect)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head Clearance	1981 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 3-13 A1 Disconnect Subsystem

Section 4.0

Minimum Room Size (Limited Access)

The CT Gantry Left Side Limited Access Initiative provides the capability to reduce the minimum room size for CT Systems while still meeting all installation requirements and specifications. This adds left side flexibility, allowing the CT system to be sited in rooms with widths 22 inches (559 mm) smaller than the current minimum room width.

In the minimum room configuration, left-side access and egress may be restricted for sites with less than 28 inches (711 mm) of left side clearance. Refer to your site's installation print for your room's detail.

4.1 Regulatory Caution

Site prints are required for all system installations including relocation and moves. CT room layout as shown on your site print shall meet all regulatory requirements as described in the installation manual. Additional room components such as cabinets reduce room size. Equipment not shown on the site print may void the caution statement, making the room non-compliant. Actual site measurements before installation will be taken to determine room size and compliance.

For minimum room size sites, where the Gantry distance between the wall and side cover is between 14 inches (356 mm) and 27 inches (686 mm) the left side cannot be considered or used for egress. In this configuration, egress around the left side of the Gantry will be restricted. Unobstructed egress route shall be provided around the Table foot end or behind the back of the Gantry. If mill-work is required or will be installed later, consider room placement locations that allows for regulatory compliance and clearances.

4.2 Operational Caution

In this minimum room layout (14 in. to 27 in. or 356 mm to 686 mm) the customer should consider workflow, customer access for patient care, and critical-care operations space requirements. Additionally, there may be limited equipment access on the Gantry left side when loading patients or when positioning patient equipment in the room between the Gantry and the wall.

Detailed customer installation tasks are detailed in the product PIM Chapters 1-4.

4.3 Suggested Room Size

This room configuration offers the most flexibility for future upgrades. It has sufficient workspace and space to add millwork, while meeting all regulatory requirements. This room would be compatible with most two-step future installations.

4.4 Recommended Room Size

This room configuration allows for some future upgrades. It has sufficient workspace but limited space to add millwork and meet all regulatory requirements. This room may be compatible with some two-step future installations.

4.5 Minimum Room Size

This room configuration allows for no future upgrades. It has limited workspace and no in-room millwork, but meets all regulatory requirements. This room is not compatible with two-step future installations.

Section 5.0 System Specifications

System	Recommended Room Size	Typical Room Size	Minimum Room Size
Pro16 with HP 1600	4420 mm x 7772 mm (14 ft. 6 in. x 25 ft 6 in.) ¹	4115 mm x 5867 mm (13 ft, 6 in x 19 ft, 3 in) ¹	3556 mm x 5867 mm (11 ft, 8 in x 19 ft 3 in) ²
RT (4-slice) with HP 1600	4420 mm x 6096 mm (14 ft, 6 in x 20 ft) ¹	4267 mm x 5867 mm (14 ft x 19 ft, 3 in) ¹	3708 mm x 5867 mm (12 ft, 2 in x 19 ft, 3 in) ²

1 Same Regulatory requirements apply.

2 Same Regulatory requirements apply, with the addition of no energized left side service.No energized left side service is standard on all limited access compatible systems.

Table 3-14

5.0.1 Rooms w/ Less Than 711 mm (28 in) Egress Clearance at Table Foot End

Egress requires a clear unobstructed route out of the room, either around the back of the gantry or around the foot end of the table. If your egress route is not around the foot end of the table, maintain maintain 457 mm (18 in) of clearance between the foot end of the table and the wall. Additionally, there needs to be a continuous width of 3200 mm (126 in)—1600 mm (63 in) on each side of the Table center-line—to any obstruction so that the front cover can be removed. This may limit placement of millwork within the room. Refer to the Pre-Installation manual for more details on service clearances.

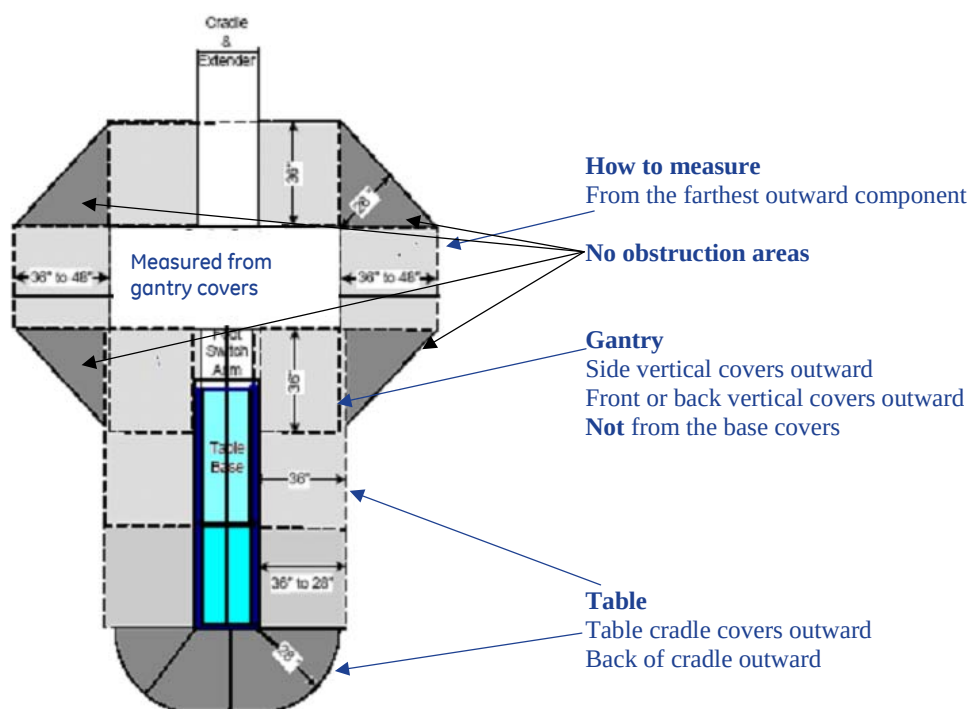
Cautions

For a Pro 16: Rooms smaller than 3556 mm x 5867 mm (11 ft., 8 in. x 19 ft., 3 in.) may require construction to meet the minimum requirements. The design center or your Project Manager of Installation may have additional recommendations for your room size.

For an RT (4-slice): Rooms smaller than 3327 mm x 5867 mm (10 ft., 11 in. x 19 ft., 3 in.) may require construction to meet the minimum requirements.

Section 6.0

How to Measure

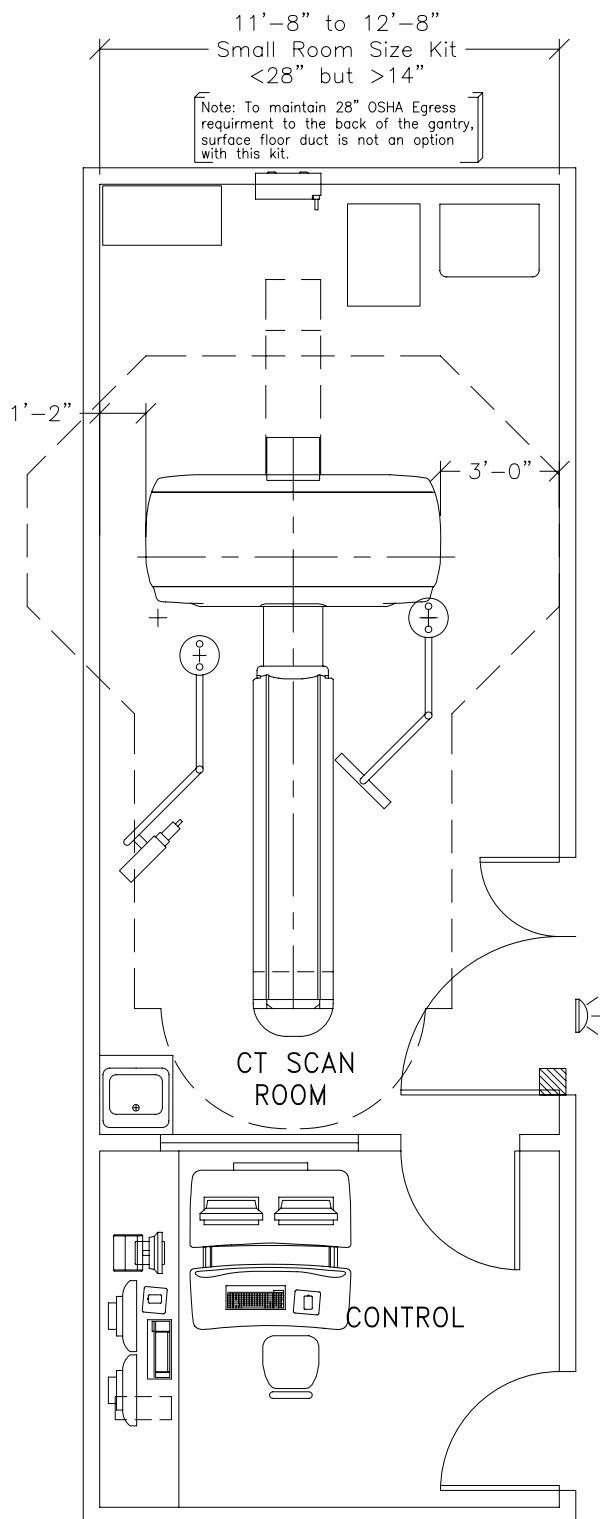


Section 7.0

Minimum Room Size & Requirement Layouts

Room A - Less than 28" (711mm) but greater than 14" (356mm) measured from the covers to the left sidewall.

In this configuration service, egress and workspace are compromised around the Gantry.



English to Metric Conversions:

12'-8" = 3861mm

11'-8" = 3556mm

3'-0" = 914mm

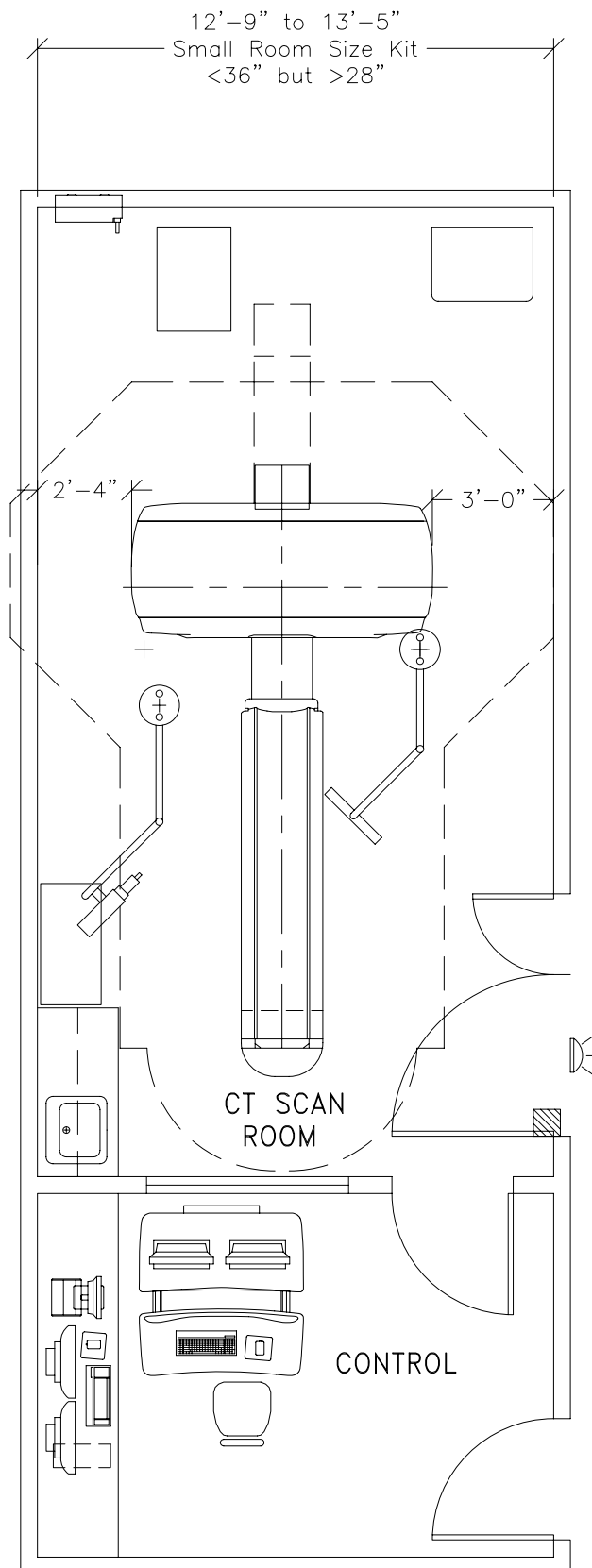
28" = 711mm

1'-2" = 356mm

14" = 356mm

Room B - Less than 36" (914 mm) but greater than 28" (711mm) measured from the covers to the left sidewall.

In this configuration service, egress and workspace are acceptable around the Gantry.



English to Metric Conversions:

13'-5" = 4089mm

12'-9" = 3886mm

3'-0" = 914mm

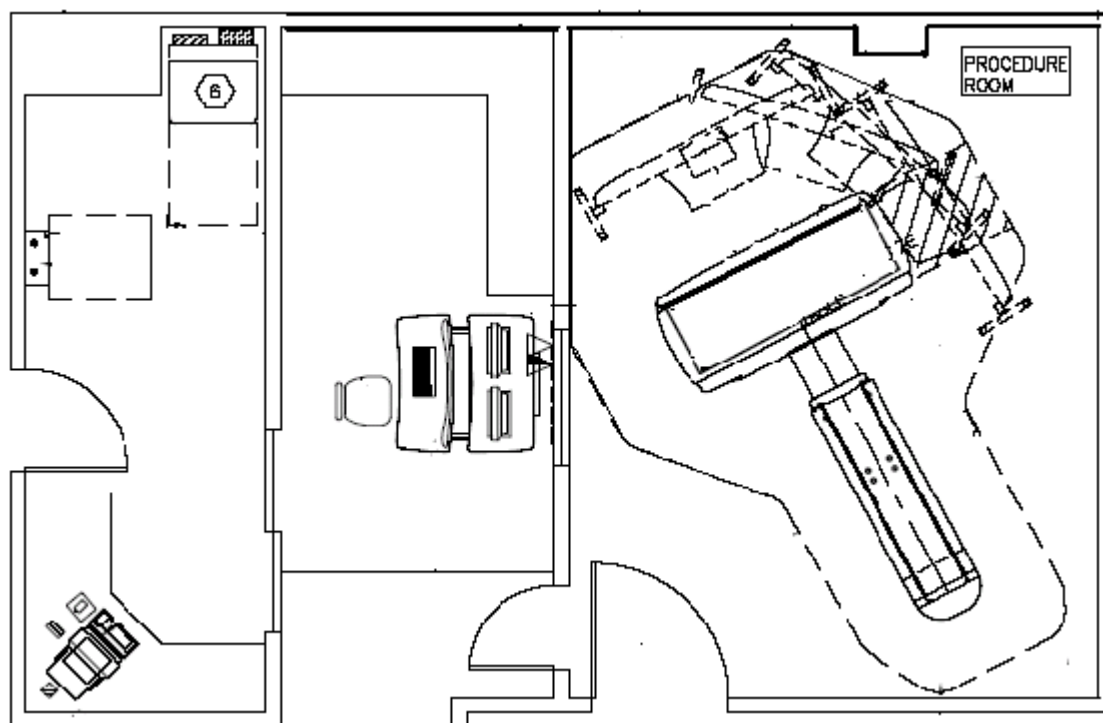
36" = 914mm

2'-4" = 711mm

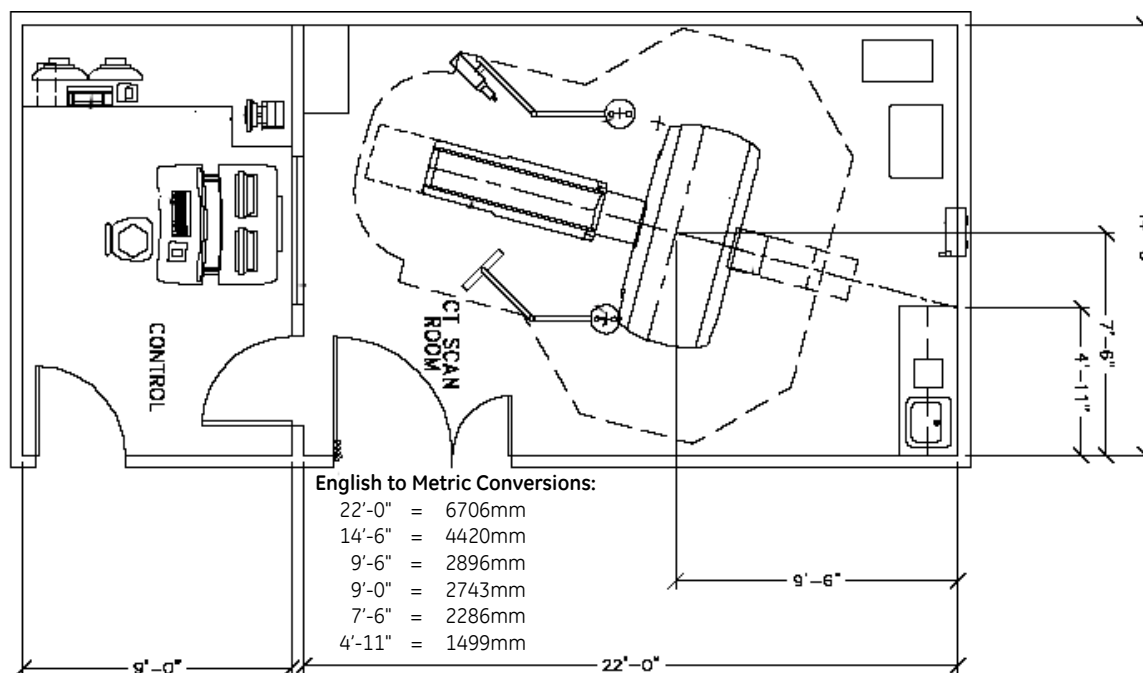
28" = 711mm

Section 8.0

Recommended Room Size & Requirement Layouts



Note: Your room layout may meet the recommended or suggested room requirements but look different than the room shown above. Contact your sales person to have a detail room layout completed for your site.



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